

Impedance Analysis of Porous-Material-Functionalized RF Sensors toward Intelligent E-Nose and E-Tongue for Multidisciplinary Monitoring

Published as part of ACS Applied Materials & Interfaces special issue "Porous Materials for the Adsorption, Degradation, and Sensing of Toxic Chemicals".

Zhiqiang Ma, Yancheng Bi, Chenyang Gao, Mengyao Fu, Qingqing Zeng, Lili Shi, Gang Wu, Wei Li, Dibo Hou, Hongjian Zhang, and Yunqi Cao*



Cite This: <https://doi.org/10.1021/acsami.6c08905>



Read Online

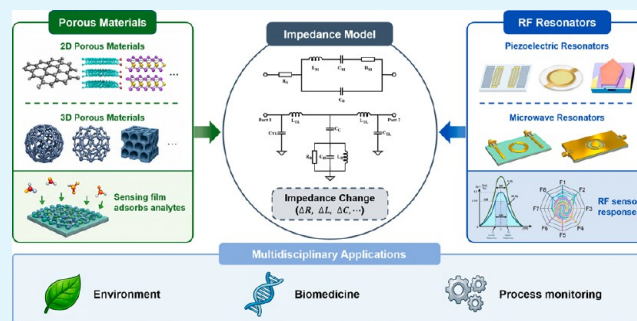
ACCESS |

Metrics & More

Article Recommendations

ABSTRACT: Electronic nose (e-nose) and electronic tongue (e-tongue) systems have emerged as powerful analytical tools for the rapid and specific detection of analytes. However, their performance is often impeded by technical bottlenecks inherent in traditional sensing technologies, particularly limited signal output dimensionality, short lifespan, and difficulties in miniaturization or integration. Radio frequency (RF) sensors, with their inherent advantages of multiparameter signal output, long-term stability, ease of integration, and wireless compatibility, offer a promising platform for e-nose and e-tongue development. In this context, porous materials can act as sensing films to adsorb target analytes, thereby modulating the impedance characteristics of RF sensors and facilitating robust detection. Notably, emerging porous materials, distinguished by their ultrahigh specific surface areas, tunable pore structures, and customizable surface chemistry, induce more rapid, pronounced, and targeted variations in the complex impedance of RF sensors during analyte interaction, which significantly enhance critical sensing metrics, including sensitivity, selectivity, and response kinetics. This review systematically summarizes the current research status of RF biochemical sensors integrated with porous materials, covering fundamental principles, material characteristics, and impedance-driven sensor analysis. Particularly, guided by the equivalent impedance model of RF sensors, we systematically propose design paradigms and performance optimization strategies. Furthermore, we analyze current advances and application scenarios in environmental monitoring, biomedical analysis, as well as process and quality control. Finally, we discuss future directions and ongoing challenges, providing a roadmap for next-generation intelligent e-nose and e-tongue systems.

KEYWORDS: RF sensors, E-nose, E-tongue, porous materials, lumped impedance models, multi-field applications



1. INTRODUCTION

The electronic nose (e-nose) and electronic tongue (e-tongue) are bioinspired sensing systems that mimic human olfaction and taste mechanisms.^{1,2} Both rely on sensor arrays and advanced data processing technologies to detect and quantify target analytes in gases or liquids. Their ability for rapid response and in situ analysis has made them increasingly valuable across various fields, including environmental monitoring,^{3,4} biomedical diagnostics,^{5,6} industrial and food quality assessment.^{7,8} As detection scenarios grow more complex, these systems face rising demands for enhanced performance, including greater sensitivity, accuracy, and broader detection ranges.^{1,9} Addressing these demands fundamentally depends on the collective performance and

information-gathering capacity of their sensor arrays. Traditionally, enhancing the discriminatory power of an e-nose or e-tongue dictates expanding the array by integrating a larger number of individual sensor nodes to capture a broader range of chemical features. However, blindly scaling up the array size not only impedes integration into portable devices due to bulky footprints and high power consumption,¹⁰ but also

Received: May 1, 2026

Revised: June 5, 2026

Accepted: June 8, 2026

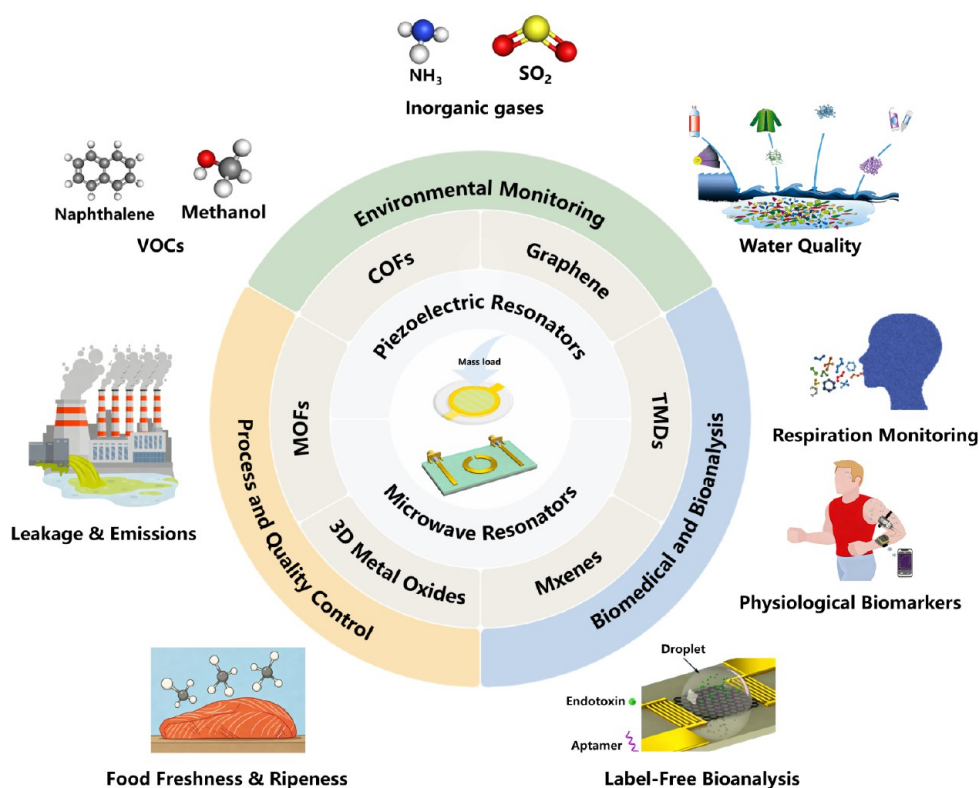


Figure 1. Summary of application scenarios of RF sensors based on porous materials.

exponentially increases the computational burden for pattern recognition algorithms via data redundancy and compounded cross-sensitivity.¹¹ Moreover, this reliance on array expansion inherently exposes and exacerbates the fundamental limitations of the individual biochemical sensors commonly employed. For instance, maintaining high sensitivity in metal oxide semiconductor (MOS) sensors generally relies on elevated operating temperatures;¹² electrochemical sensors suffer from issues like electrode aging, surface contamination, baseline drift, and limited long-term stability;¹³ optical sensing requires complex optical components and precise alignment, which complicates miniaturization and chip-level integration.¹⁴ Moreover, from the perspective of signal acquisition and feature extraction, most conventional sensor nodes within these arrays rely on relatively simple, single-dimensional electrical signal outputs, such as static chemical resistance changes^{15,16} or basic electrical responses (e.g., current, potential, or steady-state impedance).^{17,18} This necessitates a paradigm shift in e-nose and e-tongue development: transitioning from simply expanding array architectures to empowering individual sensor nodes with multidimensional signal generation capabilities.

In contrast, RF sensing technologies, spanning from a few MHz to tens of GHz,¹⁹ present a highly promising solution to the challenges aforementioned. Piezoelectric and microwave resonators, the two primary types of RF sensor components, transduce changes in the intrinsic properties of the measured substance into shifts in the resonant state of the device, enabling precise measurements through electrical signals. These sensors operate at room temperature, offer high stability, exhibit strong anti-interference capabilities, and are naturally compatible with CMOS technology and wireless sensing systems.^{20–22} More importantly, RF sensors provide sub-

stantial potential for expanding feature dimensions. On one hand, their output response stems from the intrinsic electro-mechanical or electromagnetic resonance characteristics of the device. The electromechanical response is primarily influenced by the multidimensional mechanical properties (mass load, viscoelasticity) of the measured substance at the sensing interface,²³ while the electromagnetic response is mainly determined by the complex permittivity (dielectric constant, dielectric loss) of the substance.²⁴ These properties are fully captured in the device's frequency response through the corresponding physical coupling mechanisms. On the other hand, unlike traditional sensors that yield single-point steady-state outputs, the frequency-swept operation captures a continuous spectrum across a defined frequency range. This enables the extraction of not only localized parameters at the resonance peak—such as resonance frequency, amplitude, and quality factor—but also the distributed response characteristics at every individual frequency point across the sweep, yielding a comprehensive complex impedance profile. Furthermore, employing strategies such as harmonic,²⁵ multiresonance,²⁶ and broadband measurement²⁷ expands the available sensing feature space, thereby furnishing richer informational support for the precise discrimination of complex mixtures and the simultaneous detection of multiple components in e-nose and e-tongue systems.

Since bare RF resonators intrinsically lack the affinity required to trap or accumulate target analytes, incorporating sensitive material films has become a common strategy to enhance their performance in e-nose and e-tongue applications. Among the numerous candidate materials, porous materials stand out owing not only to their excellent adsorption properties but also to their high compatibility

with RF device, as elaborated below, making them ideal for the sensitive film in RF sensors.

First, the porous structure inherently improves both the sensitivity and response kinetics of the RF sensor. Unlike traditional dense inorganic or polymer films—where enhancing adsorption capacity necessitates an increase in film volume,²⁸ which often introduces severe viscoelastic or dielectric losses and dampens the high-frequency resonance performance²⁹—the high specific surface area of porous materials supplies abundant active adsorption sites.^{30–32} Simultaneously, the highly interconnected pore network establishes low-resistance diffusion pathways for molecular transport. This structural advantage bypasses the sluggish bulk permeation and surface-only adsorption mechanisms typical of nonporous dense materials, thereby effectively shortening both response and recovery times.³³

Second, the favorable mechanical and morphological characteristics of porous materials ensure that sensor response is not solely dependent on analyte-induced changes in mass loading or permittivity. Instead, the entire porous film undergoes significant changes in physical and chemical properties, such as adsorption-induced viscoelastic evolution,³⁴ interfacial polarization enhancement,³⁵ and dielectric relaxation of molecules confined within the pores.³⁶ These phenomena produce synchronous modulation of multiple parameters such as resonance frequency, amplitude, quality factor, and phase, thereby fully realizing the intrinsic multidimensional sensing potential of RF devices.

Finally, the rich material engineering possibilities afforded by porous materials offer extensive latitude for optimizing RF sensing performance. For instance, composite porous materials that combine high adsorption capacity with high electrical conductivity can be integrated into piezoelectric sensors to simultaneously amplify the mass loading effect and the conductivity change,³⁷ whereas porous/metal oxide heterojunctions can be constructed to serve not only in enhancing adsorption capacity but also in augmenting the dielectric response in microwave sensors.³⁸ Furthermore, the selectivity of various RF sensors can also be improved by utilizing the pore size sieving effect^{39,40} or modifying the surface chemical environment.^{41–46}

To date, owing to the favorable compatibility between porous materials and RF sensors, RF sensors based on porous materials have been successfully deployed across a variety of fields of gas/liquid component detection, as illustrated in Figure 1.

To enable RF sensors to serve more effectively as core sensing units in e-noses and e-tongues for increasingly complex detection tasks, a deeper understanding of their sensing mechanisms and feature dimensions is essential. In particular, systematic investigation of the underlying mechanisms and the extraction of high-dimensional signal features are urgently needed. Sensor impedance models, or equivalent circuit models, provides a powerful framework for this purpose, as it links measurable electrical responses to their physical origins and enables the decoupling of multiple sensing contributions.^{47,48} Whether manifesting as piezoelectric acoustic vibrations and electromagnetic oscillations in physical space, or as capacitance and inductance within electrical networks, the physical essence of resonance fundamentally lies in the periodic storage and conversion of energy within a system. Therefore, the equivalent circuit model can essentially be regarded as a simulation of physical systems sharing similar

energy conversion patterns. Within this framework, the overall sensor response can be decomposed into distinct components associated with the porous sensitive film and represented by lumped-element impedances, which include interfacial effects such as mass loading and viscoelastic changes,⁴⁷ as well as variations in intrinsic electrical properties including complex permittivity and conductivity.⁴⁸ Such analysis establishes a solid theoretical and technical foundation for achieving high selectivity and reliable multicomponent recognition in electronic nose and electronic tongue systems.

Nevertheless, existing reviews on RF sensing have largely focused on device configuration or the development of sensitive materials within relatively narrow scopes, while systematic, mechanism-oriented studies based on impedance analysis remain limited. To address this gap, this study adopts impedance analysis of RF sensors as its central organizing principle, systematically examining the operating principles and core equivalent circuit models of the two mainstream categories of RF sensor components—piezoelectric resonators and planar microwave resonators—along with the structural characteristics of two- and three-dimensional porous sensitive materials and their adaptive integration with RF sensing platforms. Building on this foundation, a comprehensive methodology is proposed for the full-process design and performance optimization of porous material-based RF sensors guided by impedance analysis, which is illustrated through representative applications in environmental monitoring, healthcare, and process and quality assessment. Finally, by integrating insights from both RF device and materials design, this review highlights key development trends and remaining challenges, providing guidance for both fundamental research and practical implementation in RF sensing systems.

2. PRINCIPLE OF RF SENSORS AND THEIR IMPEDANCE MODELS

RF sensors are generally constructed with two indispensable functional modules: RF transducer devices and functional sensitive materials, where the RF transducer serves as the core signal transduction unit, and the sensitive material acts as the specific recognition element for target analytes. The fundamental sensing mechanism of RF sensors is to convert the interactions between target analytes and sensitive materials into quantifiable, high-resolution electrical signal changes.

To achieve high-sensitivity, high-resolution, and high-stability sensing detection, the vast majority of RF sensors adopt a resonant transduction architecture.^{22,49} According to the difference in resonant transduction working mechanisms, the widely reported RF resonant transducers in current research are mainly divided into two categories: piezoelectric resonators and microwave resonators, each exhibiting distinct resonant characteristics and matched application scenarios. To provide a comprehensive theoretical framework, this section sequentially elaborates on the fundamental operating principles and equivalent impedance models of both piezoelectric and microwave resonators. Subsequently, we conclude with a comparative analysis of their primary characteristics, aiming to guide the rational selection of specific sensor platforms for targeted sensing applications.

2.1. Piezoelectric Resonators

Piezoelectric resonators achieve reversible electro-mechanical energy conversion via the piezoelectric effect. They can be categorized by energy propagation mode and structure: surface

acoustic wave (SAW) resonators, in which acoustic waves propagate along the surface of the material, bulk acoustic wave (BAW) devices—such as the quartz crystal microbalance (QCM) and the film bulk acoustic resonator (FBAR)—in which acoustic waves propagate through the bulk of the material.²¹ To clarify the working mechanism of RF sensors based on piezoelectric resonators, the core structural characteristics of the three most widely used devices (SAW, QCM and FBAR)⁵⁰ are schematically depicted in Figure 2.

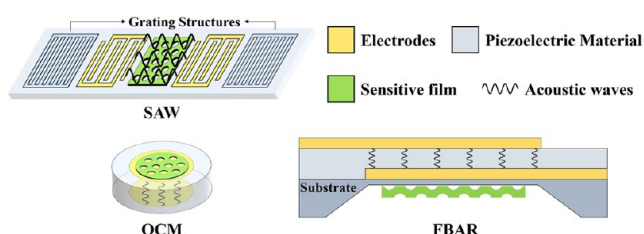


Figure 2. Schematic diagrams of RF sensor structures based on typical piezoelectric resonators: SAW, QCM and FBAR.

2.1.1. SAW Resonators. SAW resonators utilize acoustic waves that propagate along the surface of a piezoelectric substrate, with energy confined within a depth of approximately a few wavelengths.⁵¹ These waves are typically generated and detected by interdigital transducers (IDTs) patterned on the substrate. Benefiting from their simple planar structure, SAW devices feature straightforward fabrication processes and low manufacturing costs.⁵² Applying an alternating electric field (with a frequency range typically between 100 MHz and 1.5 GHz⁵³) to the IDT induces mechanical deformation via the inverse piezoelectric effect, launching SAWs. These SAWs interact with the substrate and sensitive film as they travel. Upon reaching another IDT or reflecting off grating structures, the SAWs induce mechanical strain in the piezoelectric material, which is converted back into an electrical signal via the direct piezoelectric effect, allowing for the detection or processing of the original perturbation.⁵¹ The resonant frequency f of SAW sensor is proportional to the SAW propagation velocity v , which is sensitive to perturbations induced by the sensitive film, and inversely proportional to the SAW wavelength λ , which is directly determined by the geometric parameters of the IDT.⁵⁴ Specifically, the propagation of SAWs is significantly influenced by multiple physical parameters of the sensitive film, including mass loading, viscosity, and electrical conductivity, which can be quantitatively described by the following expression:^{55,56}

$$\frac{\Delta f}{f_0} = \frac{\Delta v}{v_0} = -c_m f_0 \Delta \left(\frac{m}{A} \right) - \left[\frac{K^2}{2} \right] \Delta \left[\frac{\sigma_s^2}{\sigma_s^2 + (v_0 c_s)^2} \right] + 4c_e \frac{f_0}{v_0^2} \Delta(hG') \quad (1)$$

where f_0 is the initial resonant frequency of the SAW sensor; v_0 is the initial propagation velocity of the surface acoustic wave; c_m denotes the mass sensitivity coefficient; $\Delta(m/A)$ represents the mass change per unit area; K^2 is the electromechanical coupling coefficient of the piezoelectric substrate; σ_s refers to the electrical conductivity of the sensitive film; c_s is the surface

capacitance per unit length; c_e is the elasticity sensitivity coefficient of the substrate; h is the thickness of the sensitive film; and G' is the real part of the storage modulus of the thin film. Therefore, the overall response of a SAW sensor is determined by the combined effects of mass loading, electrical conductivity, and viscoelasticity in the sensitive film. By quantitatively decoupling and analyzing these three factors, the SAW sensor can simultaneously detect multiple parameters, which is one of the foremost strengths of RF sensors in e-nose/e-tongue applications.

2.1.2. BAW Resonators. BAW resonators represent another key class of piezoelectric resonators employed in high-sensitivity mass-sensing applications, which operate based on the propagation of acoustic waves through the bulk of a piezoelectric material.²¹ Unlike SAW devices, which confine energy to the surface, BAW resonators operate in a different manner. This bulk propagation mode enables BAW resonators to achieve improved stability, as their energy distribution is less affected by surface defects or environmental fluctuations.⁵⁷

Specifically, when an alternating voltage is applied to the electrodes on both sides of the piezoelectric crystal, the inverse piezoelectric effect drives mechanical vibrations along the crystal's thickness direction. These exciting bulk acoustic waves propagate through the crystal. When the applied voltage frequency matches the crystal's inherent mechanical resonant frequency, the system enters resonance, producing a stable BAW signal. Sensing materials are often coated onto the BAW electrodes to detect target analytes through adsorption and desorption processes. Upon adsorption of target analytes onto the sensing film, the crystal's effective mass increases, reducing its inherent resonant frequency, desorption of analytes reverses this frequency shift. The selective interaction between the sensing film and target analytes is critical to enhancing the sensor's specificity, while the magnitude of the frequency shift directly correlates with the amount of adsorbed analytes. This frequency shift serves as the direct quantitative signal for detection, described by the Sauerbrey equation,⁵⁸ which relates the added mass to the frequency change under conditions of a rigid, thin, and uniformly distributed film,

$$\Delta f = -\frac{2f_0^2}{A\sqrt{\rho_q\mu_q}}\Delta m \quad (2)$$

$$f_0 = \frac{1}{2d}\sqrt{\frac{\mu_q}{\rho_q}} \quad (3)$$

In this equation, Δf is the resonant frequency shift of the BAW (Hz), f_0 is the fundamental resonant frequency (Hz), Δm is the mass change (g) of the sensitive film, A is the electrode surface area (cm²), d is the thickness of the crystal, μ_q and ρ_q are its shear modulus (g/cm²s) and density (g/cm³). It should be noted that the above relationship is only applicable to describe the correlation between Δf of BAW resonators and Δm the sensitive film. This relationship holds only under the following conditions: (i) the adsorbed layer is thin and rigid, (ii) the added mass is rigid, and (iii) the mass is uniformly distributed over the sensing area. However, in practical sensing scenarios, the viscoelastic effect of the sensitive film exerts a non-negligible impact on the sensor response, which can be quantitatively characterized by the change in resonant frequency shift or the variation of Q-factor⁵⁹—this is the

core origin of the multidimensional response capability of BAW resonators.

The two mainstream implementations of BAW resonators are the QCM and the FBAR. The QCM is one of the most prevalent electroacoustic sensors operating on the bulk acoustic shear mode. Its core component is a circular single-crystal quartz disc with electrodes centrally and symmetrically positioned on both sides, generating an applied electric field across the disc's thickness. The thickness of the quartz disc typically ranges from hundreds of micrometers to several millimeters, resulting in a fundamental resonant frequency generally between 5 and 20 MHz,^{34,60,61} with 10 MHz being the typical value for most devices. The Q-factor of QCM sensors typically lies within the order of thousands to tens of thousands,^{63–65} which is critical for achieving low noise and high frequency resolution, resulting in a lower limit of detection (LOD).

FBARs share the same fundamental operational principle as QCM, while operating at significantly higher resonant frequencies, typically in the range of sub-GHz to approximately 10 GHz.^{66,67} This is achieved by employing a piezoelectric film with a substantially reduced thickness, ranging from submicrometer to few micrometers (e.g., 0.5–3 μm).⁶⁸ The thinner piezoelectric substrate is the direct cause of these higher resonant frequencies, which, according to the Sauerbrey equation, implies a theoretically superior mass sensitivity compared to QCM. However, the Q-factor of FBARs is generally lower, approximately ranging from 200 to 2500,^{30,69,70} primarily attributable to differences in piezoelectric materials and potentially higher acoustic losses at elevated frequencies.

2.1.3. The Impedance Model of the Piezoelectric Resonator. The Butterworth–Van Dyke (BVD) model, illustrated in Figure 3a, is the most widely employed equivalent

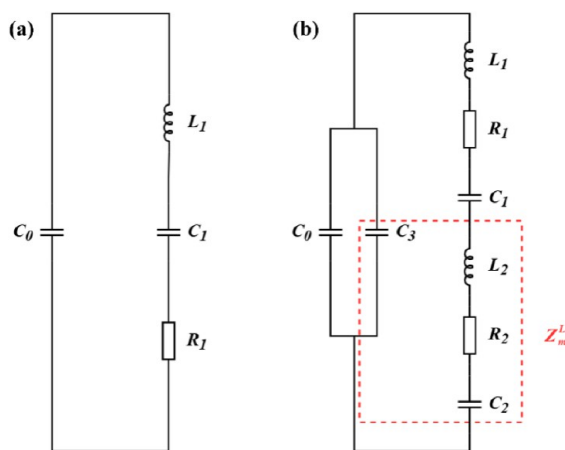


Figure 3. Impedance models of piezoelectric resonators. (a) Standard BVD model of a pristine piezoelectric resonator; (b) mBVD model of a resonator functionalized with a sensitive film, which accounts for the mechanical and electrical contributions of the sensing layer.

circuit for analyzing the electrical impedance of piezoelectric resonators.⁷¹ As a classical theoretical framework for describing electro-acoustic coupling behavior, the BVD model simplifies the complex multiphysics coupling process of piezoelectric resonators into a parallel configuration: a static electrical branch and a series resonant branch corresponding to the mechanical vibration of the device. The original BVD model

consists of C_0 , R_1 , L_1 , and C_1 , where C_0 arises from the static capacitance of the piezoelectric material and electrodes, and R_1 , L_1 , and C_1 are motional components: R_1 accounts for energy dissipation during resonance, L_1 reflects the substrate's equivalent inertial mass, and C_1 represents the substrate's elastic stiffness. When the piezoelectric resonator is functionalized with a sensitive film for analyte detection, its impedance characteristics can be further interpreted using the modified BVD (mBVD) model,^{72,73} shown in Figure 3b. The introduced Z_m^L denotes the load surface impedance, which captures the additional mechanical and electrical contributions induced by the sensitive film upon analyte interaction: an inertial component L_2 arising from mass loading, an energy dissipation term R_2 resulting from viscous loss, an elastic stiffness term C_2 corresponding to the mechanical elasticity of the film, and a static capacitance component C_3 derived from the underlying variation in the electrical conductivity of the film.

During the resonance process, the resonant frequency (f), characterizing the inherent rate of energy exchange between inductive and capacitive components, and the Q-factor (Q) defined as the ratio of energy stored to energy dissipated per oscillation cycle, can be precisely quantified utilizing lumped element parameters.⁷³ For piezoelectric resonators, analysis based on the mBVD model reveals that the characteristic impedance response exhibits two distinct frequency extrema: the series resonant frequency (f_s), which is dominated exclusively by the motional branch components (L_m and C_m), and the parallel resonant frequency (f_p), which is jointly determined by the motional branch and the static capacitance. The mechanical and viscoelastic dissipation, arising from both the resonator itself and the porous film upon analyte adsorption, maps directly onto the motional resistance (R_m), which in turn governs the Q of the system. The above relationships are as follows:⁷⁴

$$f_s = \frac{1}{2\pi\sqrt{L_m C_m}} = \frac{1}{2\pi\sqrt{(L_1 + L_2)(C_1 + C_2)}} \quad (4)$$

$$f_p = f_s \sqrt{1 + \frac{C_m}{C_0 + C_3}} \quad (5)$$

$$Q = \frac{\omega L_m}{R_m} = \frac{2\pi f_s (L_1 + L_2)}{R_1 + R_2} \quad (6)$$

Furthermore, beyond the aforementioned fundamental parameters, other feature signals of RF sensors—such as the peak magnitudes of scattering parameters (e.g., S_{11} or S_{12}) and phase—can also be rigorously derived and parametrized based on the impedance model.

2.2. Microwave Resonators

In contrast to piezoelectric resonators, which operate at lower frequencies and rely on mechanical resonance, microwave resonators operate at frequencies ranging from 300 MHz to tens of GHz⁷⁵ and sense changes via dielectric perturbation, where electromagnetic energy oscillates between electric and magnetic fields. Although the Q-factors of microwave resonators are lower than those of piezoelectric counterparts, their dielectric perturbation-based sensing mechanism endows them with unique technical merits: most notably the inherent noninvasive and nondestructive detection capability enabled by the deep penetrability of electromagnetic waves,⁷⁶ along with

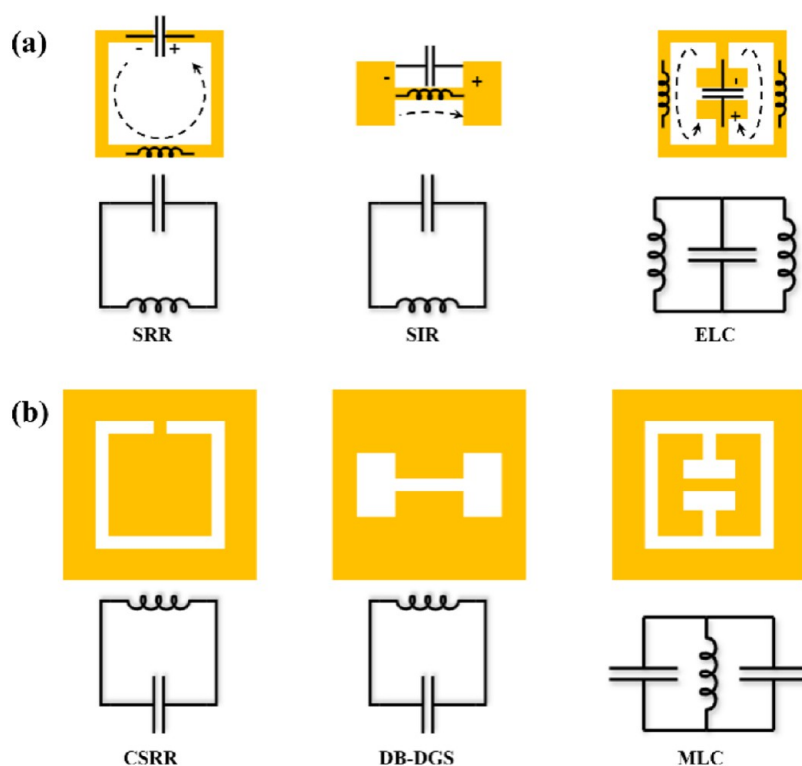


Figure 4. Typical topologies of some planar microwave resonators and their equivalent circuit models. (a) Metallic resonators. (b) Complementary resonators.

the high compatibility of their GHz operating band with modern wireless communication systems.⁷⁷

Among the various resonator types, planar microwave resonators offer advantages of compact size, ease of fabrication, cost-effectiveness, and straightforward system integration⁷⁸ over bulk resonators such as waveguide cavities and dielectric resonators, making them widely applicable in various sensing applications. Through carefully engineered resonant structures, planar microwave resonators can generate standing waves at target frequencies, effectively localizing the electromagnetic field and forming regions with strong near-field coupling that are highly sensitive for detection purposes. When the target analytes are adsorbed by a porous film close to or directly covering the resonant region, it perturbs the electromagnetic boundary conditions, leading to measurable variations in resonant frequency, amplitude, phase, or Q-factor, thereby enabling sensing.

Planar microwave resonators are broadly classified into metallic resonators and complementary resonators based on their structural configuration relative to the ground plane.⁷⁹ Metallic resonators consist of patterned conductive traces on the substrate surface and are sensitive to magnetic field perturbations. Complementary resonators are formed by etching slots into the ground plane, rendering them highly responsive to electric field variations. This structural duality enables the design of tailored sensors for specific analyte interactions and application environments.

2.2.1. Metallic Resonators. Metallic resonators are composed of conductive metallic patterns situated on one side of a dielectric substrate, with the metallic ground plane residing on the opposite side. Representative topologies include split-ring resonators (SRRs), stepped impedance resonators (SIRs), and electric-LC (ELC) resonators,⁷⁹

among others. In RF sensing applications, these metallic traces generate localized surface currents near the resonant frequency, forming a magnetic field distribution perpendicular to the resonator plane. This makes them sensitive to magnetic perturbations, suitable for detecting magnetic materials or low-loss dielectrics. Their lumped circuit model typically manifests as an inductor-capacitor (LC) resonant tank, as illustrated in Figure 4a.

2.2.2. Complementary Resonators. Complementary resonators, also referred to as slot resonators or defected ground structure (DGS) resonators, are formed by etching slot patterns that are geometrically complementary to those of their metallic counterparts directly into the ground plane. Representative complementary resonators include complementary split-ring resonators (CSRRs), double dumbbell-shaped DGSs (DB-DGS), and magnetic-LC (MLC) resonators.⁷⁹ According to the Babinet–Booker principle,⁸⁰ equivalent magnetic current loops are induced within the slots of complementary resonant structures that generate an axial electric field, yielding electromagnetic field distributions that are dual to those of the original metallic structures. As a result, complementary resonators exhibit pronounced sensitivity to electric-field perturbations, with the resonant frequency shifting significantly in response to variations in the permittivity of the overlaying material. They are therefore particularly effective for interacting with polar molecules or high-permittivity samples. The equivalent circuit model of a complementary resonator is likewise represented by a parallel LC tank, as depicted in Figure 4b.

2.2.3. The Impedance Model of the Microwave Resonator. As previously discussed, a planar microwave resonator can be modeled as a lumped-parameter parallel LC resonant circuit. In practical sensing applications, the modified

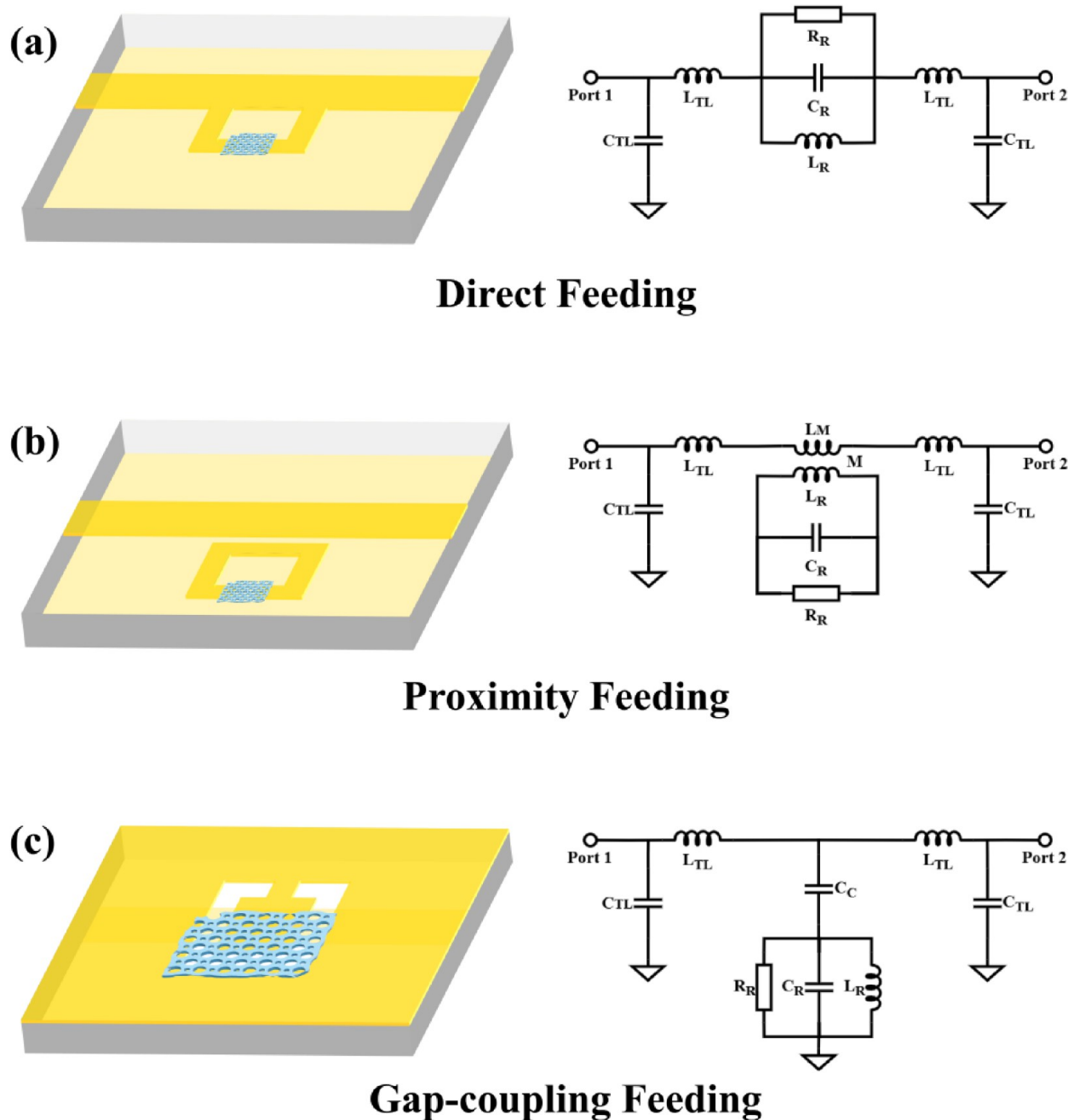


Figure 5. Schematic diagrams and impedance models of microwave resonators. (a) Direct feeding. (b) Proximity feeding. (c) Gap-coupling feeding.

equivalent capacitance (C_R) must be considered, as the sensitive film is positioned within the resonator's strong electric field region.⁸¹ Additionally, energy dissipation during resonance, caused by intrinsic dielectric losses from both the resonator substrate and the functional sensitive film, must be accounted for. This dissipation is represented by introducing an equivalent parallel resistance (R_R) into the circuit.⁸¹ The independent variations of C_R and R_R provide multiple sensing dimensions for the microwave sensor, overcoming the limitations of single-parameter detection. Furthermore, the dielectric properties of materials are frequency-dependent due to intrinsic dielectric dispersion,⁸² with different target analytes displaying distinct frequency-dependent dispersion patterns.⁸³ By analyzing these property variations across multiple

frequency points, the sensing dimensionality of the microwave sensor can be further enhanced.

Apart from the resonant unit, the feeding structure is also an essential component of the planar microwave sensor. The resonant unit relies on a dedicated feeding structure to excite electromagnetic resonance and transmit the resonant signal; therefore, the feeding structure and its coupling effect with the resonant unit must be fully considered in the establishment and analysis of the impedance model for microwave resonator sensors. For planar microwave resonators, various excitation schemes have been reported, including single-port microstrip lines,⁸⁴ dual-port microstrip lines,⁸⁵ coplanar waveguides,⁸⁶ substrate-integrated waveguides,⁸⁷ and power dividers/combiners,⁸⁸ among others. Despite the diverse feeding

Table 1. Comparative Characteristics of Common RF Sensors: Frequency, Sensitivity, Advantages, and Compatibility

RF sensors	Common frequency range	Typical sensitivity scale	Principal advantages	Gas/liquid compatibility
SAW	100 MHz–1.5 GHz ⁵³	$\sim 10^\circ$ Hz/ng ⁹⁹	Planar configuration Facile array integration	liquid detection depends on the vibration mode
QCM	5–20 MHz ^{34,60–62}	$\sim 10^{-1}$ – 10° Hz/ng ⁹⁹	Low cost Excellent resistance to liquid-phase viscosity	good
FBAR	500 MHz–10 GHz ^{66,67}	$\sim 10^5$ – 10^7 Hz/ng ⁹⁹	Ultrahigh mass sensitivity Outstanding miniaturization capability	liquid detection depends on the vibration mode
SRR	300 MHz–19 GHz ⁷⁵	$\sim 1\%$ – 4% $\Delta f_0/\Delta \epsilon_r$ ⁷⁵	Highly concentrated electric field distribution Relatively high Q-factor	good
CSRR	300 MHz–19 GHz ⁷⁵	$\sim 1\%$ – 5% $\Delta f_0/\Delta \epsilon_r$ ⁷⁵	Larger sensing area Facilitates functionalization of sensitive materials	good

structures, the feeding methods can be broadly classified into three categories based on the relationship between the feeding structure and the resonator: direct feeding, gap-coupling feeding, and proximity feeding. In direct feeding, the resonator is integrally connected to the transmission line, forming a unified structure that enables simple fabrication without introducing extraneous capacitance and inductance. This implies that the resonance is primarily related to the capacitances and inductances within the resonator.⁸⁹ In contrast, both gap-coupling feeding and proximity feeding belong to noncontact feeding, which respectively supply energy to the resonator through magnetic coupling and electrical coupling. This noncontact method not only achieves DC isolation, which facilitates the integration of RF circuits, but also features a wider operating bandwidth that is conducive to feeding multiple resonators.⁹⁰ Taking the SRR and CSRR as examples, Figure 5 presents the schematic diagrams and lumped circuit models of these feeding methods.

Based on these established equivalent circuits, the resonant performance indicators of microwave sensors can also be quantitatively correlated with the lumped impedance parameters. Taking the proximity-fed resonator as an illustrative example, where the noncontact energy transfer is typically represented by a coupling capacitance (C_C) in the equivalent circuit, the mathematical relationships can be formulated as follows:⁹¹

$$f = \frac{1}{2\pi\sqrt{L_R(C_R + C_C)}} \quad (7)$$

$$Q = \omega R_R(C_R + C_C) \quad (8)$$

2.3. RF Sensor Comparison

Due to differences in working principles and structures, various RF sensors have their own performance boundaries and applicable scenarios. Piezoelectric resonators typically operate at frequencies ranging from several MHz to a few GHz,⁵³ offering high Q-factors and exceptional mass sensitivity, making them suitable for detecting trace gases and biomolecules. In contrast, microwave resonators operate at higher frequencies (ranging from 300 MHz to several tens of GHz⁷⁵), exhibiting strong dielectric sensitivity and high penetrability, which are advantageous for analyzing polar molecules and complex media. Specifically, among piezoelectric sensors, SAW resonators, benefit from well-established fabrication techniques, enabling easy multichannel integration for applications in e-noses and e-tongues.⁹² QCM is widely used for its low cost and high stability due to the mature

processes for preparing quartz crystals.⁹³ FBAR achieve significant advances in mass sensitivity, operating at GHz frequencies with extremely small resonant masses, while also featuring a highly compact device structure.⁹⁴ Within the category of planar microwave resonators, metallic structures represented by SRRs concentrate a localized electric field within the minute volume of the split gap, offering a high Q-factor and making them ideal for detecting variations in the dielectric properties of small-volume materials.⁹⁵ Complementary resonators provide a more uniform distribution of electric fields across the resonator area, offering a larger effective sensing area and superior structural compatibility for integration with coplanar microwave circuits.⁷⁹

Furthermore, the adaptability of RF sensors differs in gas-phase and liquid-phase detection. For piezoelectric resonators, their sensing compatibility fundamentally depends on the vibrational mode and the acoustic coupling loss characteristics of the liquid medium.⁹⁶ When operating in longitudinal vibration modes (e.g., the thickness extensional mode), particle displacement is perpendicular to the solid–liquid interface, causing significant acoustic energy loss into the liquid and leading to viscous damping. This results in sharp Q-factor attenuation, limiting sensitivity and stability in liquid-phase detection. Consequently, with the exception of QCM devices operating in the thickness shear mode, SAW and FBAR resonators intended for liquid-phase detection require orientation control of the piezoelectric crystal to excite shear vibration modes (such as SH-SAW⁹⁷ or thickness shear mode FBAR⁹⁸), thereby suppressing longitudinal acoustic energy loss. In contrast, microwave resonators rely on the interaction between electromagnetic fields and the dielectric properties of the substance, bypassing the issues of viscous damping or acoustic radiation loss inherent to mechanically vibrating piezoelectric systems. This enables them to offer broader applicability across diverse detection environments.

Table 1 Systematically compares the characteristics of common RF sensors in terms of common operating frequencies, typical sensitivity levels, principal advantages, and gas/liquid compatibility, providing a reference basis for the rational selection of sensor types in different application scenarios.

3. POROUS SENSITIVE MATERIALS

An appropriate sensing material can significantly enhance the performance of RF sensors in terms of key metrics, such as sensitivity, selectivity, and response speed. Porous materials are intrinsically well-suited for integration with RF sensors, as their unique structural attributes align precisely with the multiparameter modulation

mechanism that governs RF sensing.¹⁰⁰ This synergy yields richer sensing information and comprehensively enhances core performance metrics. The functionalization of RF sensors with porous materials typically simultaneously alters their multiple inherent parameters, such as the dielectric constant and dielectric loss,¹⁰¹ thereby modifying the equivalent capacitance and inductance of the resonators, and affecting key performance indicators, including the resonant frequency and Q-factor, etc.

To adapt to various application scenarios, porous sensing materials are often required to be equipped with selectivity, high adsorption capacity, rapid adsorption/desorption kinetics, chemical/mechanical/thermal stability, compatibility with the sensors, and the ability to produce significant physical parameter changes. This section provides a detailed overview of porous materials classified by spatial dimensionality (Figure 6) and discusses their pivotal roles in RF

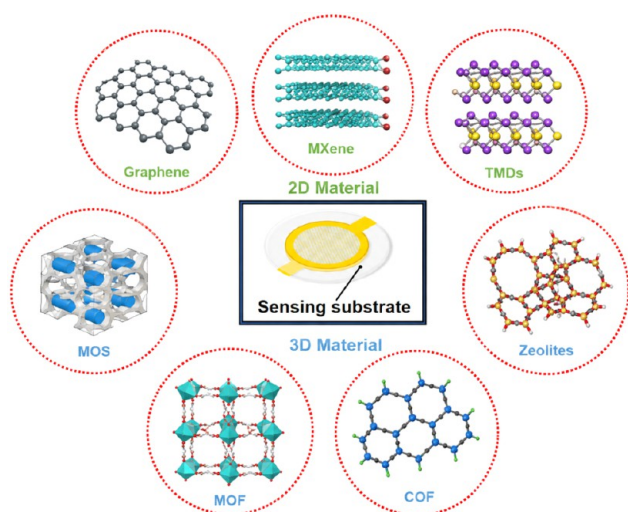


Figure 6. Schematic classification of porous materials according to spatial dimensionality.

sensors. Our discussion is specifically focused on representative two-dimensional (2D)¹⁰² and three-dimensional (3D)¹⁰³ porous materials, and other structural categories such as one-dimensional (1D) porous materials are not discussed in detail herein owing to their limited applicability in RF sensor platforms.¹⁰⁴ Structurally, this section first reviews the specific merits, sensing mechanisms, and applications of typical 2D and 3D candidates. Subsequently, a comparative analysis is presented to highlight their respective advantages, accompanied by a discussion on engineering strategies to further amplify their impedance modulation capabilities.

3.1. 2D Porous Materials

2D materials, characterized by atomic-scale thickness, ultralarge specific surface area, outstanding mechanical strength, and surface tunability, demonstrate significant potential for gas sensing and integrating with RF sensors.¹⁰⁵ 2D porous materials exhibit superior sensing performance when loaded with RF sensors, primarily including graphene and its derivatives, transition metal dichalcogenides (TMDs), and transition metal carbides/nitrides (MXenes).

3.1.1. Graphene and Its Derivatives. Graphene was the first 2D atomic crystal material discovered by Geim and Novoselov, characterized by a six-sided honeycomb lattice structure composed of single-layer carbon atoms with sp^2 hybridization.¹⁰⁶ Due to its π -electron cloud formed by a long-range π -conjugation structure, graphene exhibits remarkable electrical conductivity, which is conducive to a rapid gas sensing response and recovery speed. Coupled with its extensive specific surface area, graphene exhibits high sensitivity in gas adsorption, making it an exceptional material for gas detection.¹⁰⁵ Going beyond physical adsorption, reduced graphene oxide (rGO) with multiple oxygen-containing functional groups enables chemical adsorption, enhancing both gas sensing efficiency

and specificity. Although rGO exhibits high stability in thermal, acidic, and oxidative environments, it often requires doping with other materials (such as Au nanoparticles, carbon-containing compounds, and metal oxides) to form composite systems for enhanced detection performance, due to its lack of specific adsorption sites. Kuang et al. developed a QCM sensor based on SiC/rGO nanosheet composites for ammonia detection by introducing mesoporous structures on the nanosheets through annealing, which provided abundant active sites and effectively enhanced its NH_3 sensing performance.¹⁰⁷ In terms of the lumped model of an RF CSRR sensor, the adsorption of gas molecules by the ZnO-doped graphene composite as sensing material alters its capacitance and conductance, thereby enabling high sensitivity and rapid detection.⁴⁴

3.1.2. TMDs. TMDs are a class of remarkable 2D layered materials for chemical sensing (especially gas sensing) due to their unique properties like good conductivity, large surface area, fast electron transfer kinetics, high signal/noise ratio, and feasibility for forming composites.¹⁰⁸ Due to their sandwich structure of X-M-X held by van der Waals force, where M stands for the transition metal element (W, Mo, Ti, etc.) and X represents the chalcogens (S, Se, or Te), TMDs exhibit a trigonal prismatic (2H) and an octahedral phase (1T), representing the semiconductive and the metallic properties, respectively.¹⁰⁵

Inspired by approaches used with graphene, hybridizing TMDs with functional nanostructures such as metal nanoparticles and organic compounds, which introduces key functional groups, represents a highly promising strategy for enhancing their analytical sensing performance. Gaggiotti et al. developed an e-nose using QCM sensors functionalized with 16 composite materials containing organic compounds (e.g., ellagic acid, tannic acid) and TMDs (e.g., MoS_2 , WSe_2) for VOC determination.¹⁰⁹ The strong affinity between TMDs and the gold surface of QCMs enhances the overall mechanical strength of the sensor, resulting in low energy loss and providing stable, repeatable, and exact sensing performance.

3.1.3. MXenes. As a novel class of 2D sensing porous material, 2D MXenes have demonstrated a remarkable ability for gas sensing in recent research. The formal formulas of MXenes are $M_{n+1}X_nT_x$, where M represents transition metal (such as Ti, Mo, and V), X demonstrates carbon or nitrogen, n ($= 1, 2$, or 3) represents the stacking sequence of the M layer and the X layer, and T stands for various surface terminations (like O, F, and OH) attached to the 2D surface during the etching process.^{105,110} MXenes are synthesized by selectively etching the A-layer element (e.g., Al, Si, S) from the MAX phase precursor, resulting in atomically thin 2D sheets that exhibit outstanding properties such as large specific surface area, hydrophilicity, narrow bandgap, high electrical conductivity, abundant functional groups, porous structure, and facile functionalization, showing low limits of detection for gas and humidity sensing application.^{110,111} Specifically, $Ti_3C_2T_x$ has been demonstrated to exhibit excellent ammonia detection capabilities at room temperature, achieving a ppb-level LOD. Employing an MXene/ZnO hybrid as the sensitive material on a QCM sensor resulted in an approximately 8-fold increase in sensitivity to ammonia compared to a pure ZnO-based counterpart.^{112,113}

3.2. 3D Porous Materials

Compared to 2D materials, 3D porous materials benefit from their structure, exhibiting higher adsorption capacity, ultramassive specific surface area, superior mechanical stability, and abundant adsorption sites enabled by numerous functional groups.¹¹⁴ Notably, the size-sieving effect of their porous framework allows for the selective entry of molecules based on pore dimensions, thereby enhancing the materials' selectivity and specificity.^{39,40} Additionally, the diversity of materials and building blocks enables the creation of countless structurally and functionally distinct 3D materials, offering the potential for tailor-made designs targeting specific analytes.¹¹⁵

3.2.1. Metal Oxide Semiconductor (MOS) Nanomaterials. MOS nanomaterials are among the most extensively studied 3D porous materials for analyte sensing, including ZnO, TiO_2 , WO_3 , and SnO_2 . They are commonly employed in electrochemical sensors by

utilizing the resistive changes that occur during redox reactions with specific target gases. Notably, their diverse 3D nanostructures promote the adsorption of target analytes, thereby inducing pronounced changes in electrical and mechanical properties, modulating impedance characteristics, and ultimately enabling their widespread application in RF sensors. For example, through particular fabrication approaches, 3D highly porous ZnO nanostructures can be synthesized with various morphologies (such as nanotetrapods, nanosheets, nanowires, and nanorods), resulting in an ultrahigh specific surface area and a high number of adsorption sites, which enhances performance in gas sensing.^{116,117} Sibel et al. developed a QCM sensor functionalized with CuO/ZnO composite nanofibers for formaldehyde detection, resulting in a specific surface area exceeding 90 m²/g and achieving a LOD as low as 41 ppb.¹¹⁸ Cha et al. in situ grew superhydrophilic ZnO nanoneedle arrays on QCM electrodes via a facile chemical deposition method, achieving an ultrahigh sensitivity of 21.4 Hz/%RH and a fast response within seconds due to physical adsorption effects between water molecules and the nanomaterials.¹¹⁹

3.2.2. Metal-Organic Frameworks (MOFs). MOFs are porous crystalline materials formed by the coordinative self-assembly of metal ions or clusters as core nodes with organic ligands. They combine the rigidity of inorganic materials with the design flexibility of organic components, resulting in highly ordered pore structures. Endowed with unique 3D architectures, ultrahigh porosity, exceptional specific surface area, abundant functional groups, and numerous coordinatively unsaturated sites, MOFs become fascinating sensing materials.^{120,121} Gao et al. deposited MIL-101(Cr) as a sensing material onto an FBAR sensor for VOC gas detection, resulting in exceptional sensitivity due to its ultrahigh specific surface area (3067.8 m²/g) and extensive adsorption sites.³⁰ Based on the mBVD equivalent circuit model, the adsorption of target gases by a MOF sensitive layer alters the sensor's impedance and scattering parameters, which serve as the fundamental prerequisite for constructing a VSA utilizing FBAR. Further, the highly tunable feature of MOFs allows them to be engineered through precise structural modifications for deployment in virtually all detection scenarios. For instance, coating MOF surfaces with PDMS achieves hydrophobic functionalization, effectively suppressing moisture interference during gas detection and yielding highly humidity-resistant sensing materials.^{45,122} Mohammad et al. developed a MOF-functionalized microwave resonator as a CO₂ sensor, where the MOF translates the specific adsorption of CO₂ into a significant change in its own effective dielectric constant, thereby modulating the capacitive parameters in its equivalent circuit and ultimately leading to a shift in the resonant frequency.¹²³ MOF porous materials, with their abundant specific adsorption sites, can actively and efficiently tune the core electromagnetic parameters of the RF sensor, demonstrating sensing performance far superior to that of traditional materials.

3.2.3. Covalent-Organic Frameworks (COFs). COFs are fully organic crystalline porous materials formed by the self-assembly of organic building blocks (boronic acids, aldehyde-amines, etc.) via robust covalent bonds, possessing ordered periodic network architectures and tunable pore structures. Due to the high bond energy of covalent linkages, COFs exhibit superior thermal stability, maintaining functionality at temperatures of 400 ~ 500 °C.¹²⁴ Their ultralow density (typically <0.2 g/cm³) minimizes mass-loading on resonator surfaces while enhancing analyte adsorption per unit area, critical for high-resolution mass-sensitive devices, such as QCM. Similarly, functionalizing COF at specific sites is also a common strategy to enhance its selectivity and sensitivity for detecting targeted analytes. Li et al. designed a bipyridine-containing hydrazone-linked COF termed Bpda-Bth as a sensing material to functionalize a QCM sensor for Hg(II) ion detection with LOD of 58 nM.¹²⁵ However, due to poor electrical conductivity, COFs can be combined with conductive materials to form composites, thereby enhancing their electrical nature for device integration.^{124,126}

3.2.4. Zeolites. Zeolites, a class of naturally occurring or synthetically produced crystalline microporous aluminosilicate minerals, are renowned for their unique channel structures, high specific

surface area, ion-exchange capacity, adsorptive selectivity, and catalytic activity. Zeolites are composed of TO₄ tetrahedral units (where T typically represents Si or Al atoms) interconnected through oxygen atoms to form cage-like structures of varying sizes and morphologies. Utilizing their molecular sieve effect and selective adsorption capabilities, zeolite nanomaterials are widely applied in mass-sensitive sensors. Furthermore, through modification with metal ions, the adsorption capacity of zeolites for specific substances can be significantly enhanced, thereby improving the sensing performance of these devices.^{120,127}

3.3. Porous Materials Comparison

It should be noted that the foregoing classification of porous materials based on spatial dimensionality (i.e., 2D and 3D) is not exclusive. Pore-size classification (micropores (pore size < 2 nm), mesopores (2 ~ 50 nm), and macropores (>50 nm)) is likewise a widely adopted standard in the field.¹²⁸ Moreover, emerging porous materials such as 3D MXenes¹²⁹ and 2D MOFs¹³⁰ have further blurred the boundaries between conventional dimensional categories in recent years. The overview provided above is therefore intended as a general framework; practical applications of porous materials in RF sensing require tailored analysis based on the specific structural attributes of the material in question. To facilitate systematic comparison and performance assessment of porous materials for RF sensor applications, a comparative summary is presented below (Table 2).

Beyond pristine porous materials, the performance of RF sensors can be further refined through deliberate material engineering strategies. Common modification routes include the following approaches.

Morphology and pore engineering, which are distinctive to porous architectures, enable precise control over pore size distribution, interlayer spacing, and the construction of 3D nanostructures. By tuning the ratio of micropores, mesopores, and macropores and optimizing mass transport pathways, this approach maximizes specific surface area and the density of active sites.¹ These enhancements amplify adsorption-induced variations in mass loading, viscoelasticity, or dielectric permittivity, thereby simultaneously improving both the magnitude and the response rate of multidimensional impedance signals in RF sensing.

Composite engineering combines materials with distinct advantages to construct hybrid porous frameworks, endowing the composite with high specific surface area, abundant active sites, and favorable carrier transport properties.¹⁵⁸ Moreover, p-n, n-n p-p, or Schottky junctions form at the interfaces between heterogeneous components.¹⁵⁹ These built-in potentials and tunable energy barriers modulate carrier transport and interfacial polarization upon target adsorption, enabling amplified electrical responses, richer sensing dimensions, and significantly enhanced sensitivity for RF sensor platforms.

Surface functionalization¹ and defect/doping engineering^{160,161} represent two additional and widely adopted strategies for optimizing porous materials, offering effective means to regulate surface chemistry and electronic structure. Specific functional groups and molecular recognition sites enhance selective adsorption while suppressing environmental interference, whereas lattice defects and heteroatom doping introduce supplementary active sites and charge-transfer pathways. Both approaches further amplify the analyte's influence on the impedance response of RF sensors.

4. DESIGN GUIDE AND APPLICATION OF HIGH-PERFORMANCE POROUS MATERIAL-BASED RF SENSORS

As previously analyzed, RF devices possess inherent advantages in extracting multidimensional feature information. Porous materials, serving as the preferential sensitive films, further amplify these advantages through their unique structural and interfacial effects. Consequently, porous material-based RF sensors have demonstrated immense application potential across diverse fields, including environmental monitoring,^{162,163} biomedical diagnostics,^{164,165} and

Table 2. Comparative Summary of Porous Materials for RF Sensor Applications

Material type	Specific surface area (m ² /g)	Typical pore size range	Stability	Adsorption/desorption kinetics
Graphene and derivatives	high (250 ~ 1500, theoretical value: 2630) ¹³¹	micropores, mesopores, macropores ¹³²	pristine graphene exhibits excellent thermal stability. Graphene oxide (GO) with abundant oxygen-containing groups shows reduced stability, while reduced graphene oxide (rGO) possesses superior environmental stability ¹³³	fast adsorption at room temperature, slow desorption ¹³⁴
TMDs	moderate (20 ~ 240) ^{135,136}	micropores, mesopores, macropores ^{137,138}	good environmental stability, but may undergo oxidation in high-temperature or oxidizing environments ¹³⁹	adsorption relies on surface interactions, desorption is generally slow ^{134,140}
MXenes	moderate (tens to hundreds) ¹⁴¹	micropores, mesopores, macropores ¹⁴²	poor environmental stability, prone to oxidation in aqueous and oxygen-containing environments ¹⁴³	ultrafast adsorption response at room temperature, slow desorption for single-component analytes ¹⁴⁴
MOS Nanomaterials	moderate (10 ~ 500) ¹⁴⁵	mesopores, macropores ¹⁴⁶	excellent thermal stability and good chemical stability ¹⁴⁷	fast kinetics at high temperatures, kinetics are hindered at room temperature ¹³⁴
MOFs	extremely high (1000 ~ 10000) ¹⁴⁸	micropores, mesopores, macropores ¹⁴⁹	stability is correlated with metal–ligand bond strength; some MOFs are sensitive to humidity, which can be significantly improved via structural design ¹⁵⁰	highly reversible adsorption/desorption kinetics, pore channels facilitate rapid molecular diffusion ¹⁵¹
COFs	extremely high (100 ~ 4000) ¹⁵²	micropores, mesopores ¹⁵³	excellent thermal and chemical stability ¹⁵⁴	similar to MOFs, covalent frameworks enable fast reversible mass transfer and desorption ¹⁵⁴
Zeolites	high (200 ~ 1000) ¹⁵⁵	micropores ¹⁵³	exceptional thermal, chemical, and humidity stability ¹⁵⁶	dependent on molecular size: small molecules diffuse rapidly in micropores, while large molecules are diffusion-limited with slow desorption ¹⁵⁷

process control and quality assessment.^{166,167} Successfully deploying these sensors in such complex scenarios requires precisely decoding the intricate interactions at the material–device interface. It is here that the equivalent impedance model emerges as an indispensable theoretical bridge. It not only links the physical perturbations of the porous materials with the RF device platform to guide rational sensor design, but also serves as a powerful tool to fully exploit the inherent multidimensional sensing capabilities. When further integrated with sensor arrays and advanced algorithms, this impedance-driven approach paves the way for the broader implementation of intelligent e-nose and e-tongue systems.

Accordingly, this section first introduces a comprehensive design framework and performance optimization methodology for RF sensors grounded in impedance models. Subsequently, it systematically reviews recent advances in porous material–based RF sensors, organized according to the three aforementioned application domains.

4.1. Design and Optimization of RF Sensors Based on Impedance Model

The impedance model can be harnessed as a core instrument for guiding the full-process design and optimization of high-performance RF sensors. Adopting this model-driven methodology offers distinct analytical and practical advantages. Integrating impedance analysis with circuit simulations enables the precise prediction of sensor responses prior to physical fabrication, thereby substantially streamlining development cycles and minimizing costs. Moreover, the inherent structural simplicity and clear physical interpretability of the model provide a robust theoretical foundation for identifying key optimization pathways to maximize device performance. Specifically, the sensor design and optimization workflow guided by impedance analysis is illustrated in Figure 7, systematically encompassing three progressive stages: performance specification mapping, resonator-material matching, and comprehensive optimization strategies.

4.1.1. Sensor Performance Specifications. To effectively guide the structural design of RF devices and the selection of porous materials, it is a prerequisite to quantitatively map macroscopic sensor performance specifications—such as sensitivity, LOD, and resolution—onto the microscopic lumped parameters of the equivalent circuit model.

For RF gas or liquid sensors, the sensitivity (*S*) is conventionally defined as the magnitude of change in the resonant characteristic, typically the resonant frequency, per unit change in analyte concentration (Δc).¹⁶⁸ From the perspective of the impedance model, the *S* can be further expressed as

$$S = \frac{\Delta f}{\Delta c} = \frac{\Delta f}{\Delta Z} \cdot \frac{\Delta Z}{\Delta c} \quad (9)$$

where ΔZ denotes the core lumped-parameter change associated with the sensing mechanism—specifically, an increase in motional inductance *L* arising from mass loading in piezoelectric sensors,⁴⁷ and an increase in equivalent capacitance *C* resulting from permittivity changes in microwave sensors.⁴⁸ The two terms in this equation define the two core directions for sensor optimization: the latter determines the analyte-induced intrinsic impedance variation, fundamentally governed by specific physicochemical interactions between the porous sensitive film and target analytes; the former quantifies the device's transduction efficiency, characterizing the sensor's inherent resonant frequency response to perturbations in a specific impedance parameter. Furthermore, the sensor's LOD is closely correlated with above-defined sensitivity, which is conventionally calculated via the 3-Sigma rule.¹⁶⁹ Regarding sensor resolution, it is typically dictated by the minimum detectable frequency

$$\Delta f_m = \frac{\alpha 3\sqrt{3}f_0}{4A_0Q} \cdot \Delta A \quad (10)$$

where α is correction constant, f_0 and A_0 are the Lorentzian resonant peak's resonant frequency and amplitude, *Q* is the Q-factor, and ΔA is

Design Flowchart of Porous Material-Empowered RF Sensor Based on Impedance Model

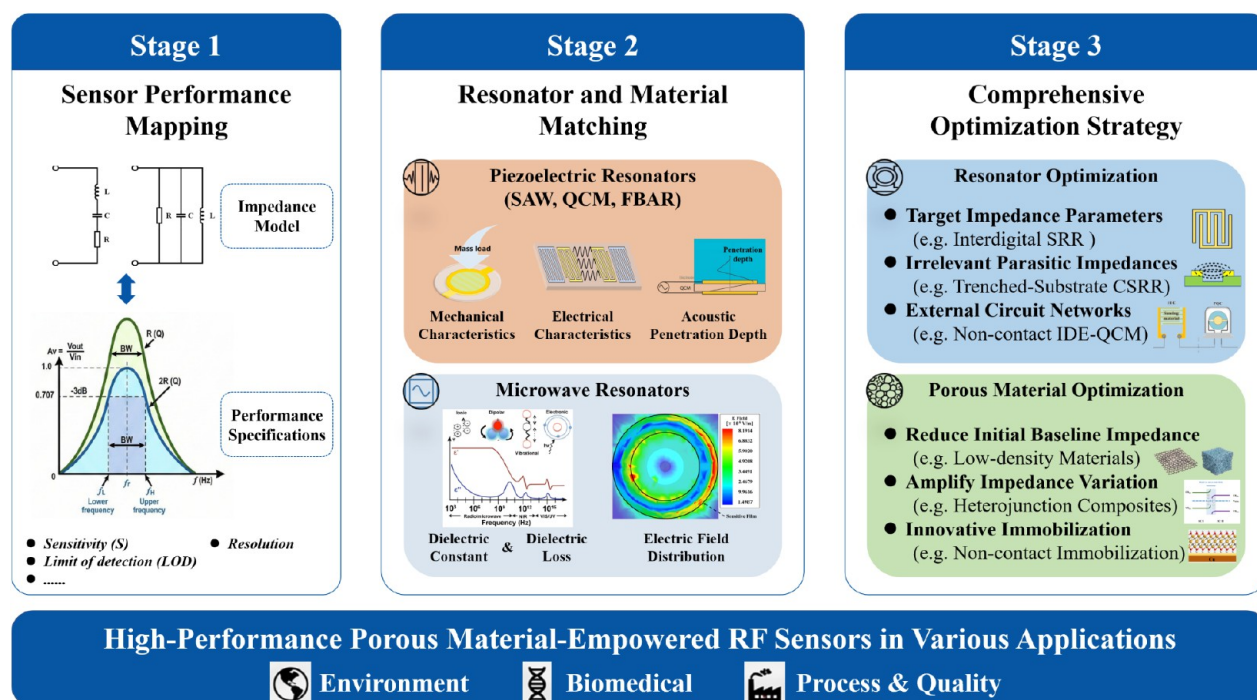


Figure 7. General design and optimization framework for RF sensors based on impedance model theory.

the minimum variation in resonance profile amplitude, which is considered three times the amplitude of the electric white-noise.¹⁷⁰

Beyond the aforementioned quantifiable core performance metrics, the multicomponent recognition capability highly prioritized in e-noses/e-tongues—can also be profoundly analyzed and evaluated utilizing the impedance model. By fitting the actual resonant responses to extract equivalent circuit parameters that independently correspond to the analytes' distinct physicochemical properties (e.g., mass loading, viscoelastic damping, permittivity), a Virtual Sensor Array (VSA) can be constructed with a single RF sensor, which has become a popular frontier strategy for RF e-noses/e-tongues.¹⁷¹ While the VSA's feature dimensionality can be further expanded via multiharmonic to greatly enhance pattern recognition performance.¹⁷² However, it should be noted that when the VSA is constructed from impedance parameters, rigorous parasitic de-embedding and precise modeling procedures are imperative, rather than the mere application of a classical ideal impedance model.¹⁷³

4.1.2. Porous Materials and RF Resonators Matching. In the design of high-performance RF sensors, the physicochemical evolution of the porous material must be precisely matched with the core sensing impedance of the device, thereby fulfilling the overarching objective of maximizing the effective impedance perturbation induced by target analytes. Given the fundamental differences in the underlying transduction mechanisms among various RF sensor types, their corresponding material selection guidelines and tailored design strategies naturally diverge.

For piezoelectric resonators, the sensing mechanism primarily relies on the mass loading effect described by the Sauerbrey equation, where the resonant frequency shift is proportional to the mass variation on the resonator surface. Additionally, fluctuations in the conductivity of the sensitive film can perturb the resonance, such as the acoustoelectric effect in SAW resonators¹⁷⁴ or altering the effective overlapping area of the electrodes in BAW resonators.⁴⁵ Therefore, theoretically, as long as the porous material can adsorb target analytes and introduce a resolvable mass loading, a measurable frequency shift can be generated; meanwhile, the concurrent changes in the electrical properties of the porous film can further enhance the multidimen-

sional response capability of the sensor. Benefiting from this, piezoelectric resonators demonstrate remarkable adaptability to a diverse range of porous material systems.

Specifically, in SAW sensors, the acoustic wave penetrates to a depth on the order of the wavelength, rendering the device intrinsically responsive to both mass loading and acoustoelectric interactions occurring at or near the surface.⁵¹ In terms of equivalent impedance, these interactions synergistically modulate the motional components and electrical component. Accordingly, porous materials that deliver ultrahigh adsorption capacity while undergoing a pronounced conductivity change upon analyte adsorption represent the optimal sensitive candidates for such systems, include MXene nanosheets¹⁷⁵ and conductive MOFs.¹⁶⁶ For QCMs, which typically operate in the thickness-shear mode, the acoustic wave exhibits a penetration depth ranging from hundreds of nanometers to the micrometer scale.¹⁷⁶ This substantial decay length ensures high sensitivity to both mass and viscoelastic variations of the sensing layer—which are directly mapped to the L_m and R_m parameters in the mBVD model—while offering high tolerance for diverse, cost-effective film-forming processes. Consequently, porous materials featuring high specific surface areas, uniform pore structures, and excellent solution-processability—such as 3D MOFs and COFs¹⁷⁷—are highly preferred. Conversely, FBARs boast a mass sensitivity several orders of magnitude higher than conventional QCMs. However, their acoustic penetration depth is extremely shallow—typically on the order of tens of nanometers¹⁷⁶—rendering their high-frequency resonance hypersensitive to the mechanical loss and viscoelastic damping of the sensing film. Such viscoelastic damping translates to a drastic surge in motional resistance, which fatally degrades the Q-factor. Thus, ultrathin and highly uniform porous materials are strictly required to prevent severe quality factor degradation, with typical examples including atomically thin 2D porous nanosheets (e.g., graphene oxide¹⁷⁸ and Mxenes¹⁷⁹) and ultrathin MOF nanofilms.³⁰

In contrast to piezoelectric devices, microwave resonator sensors rely predominantly on the perturbation of the near-field electromagnetic waves distributed around the resonant structure.⁷⁹ Guided

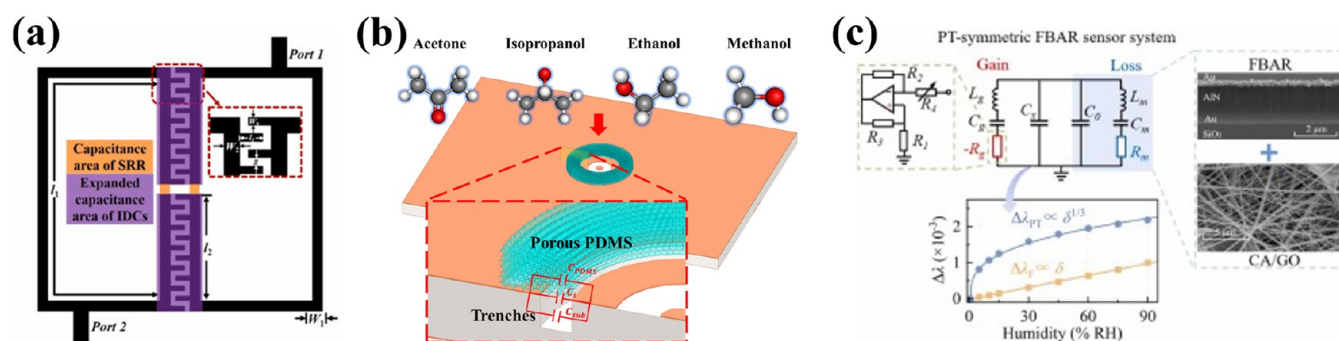


Figure 8. Optimization strategies for RF sensors based on resonators and impedance models. (a) Parameter tuning (interdigital SRR¹⁸⁶); (b) Parasitic impedance elimination (trenched-substrate CSRR¹⁸⁸); (c) External circuit construction (parity-time symmetric system¹⁹⁰). Adapted with permission from ref 186. Copyright 2022 Elsevier. Adapted with permission from ref 188. Copyright 2024 Elsevier. Adapted with permission from ref 190. Copyright 2025 Elsevier.

by impedance analysis, the rational selection of sensing films dictates matching the material's properties to maximize the perturbation of specific impedance components (C_R and L_R). Consequently, discernible sensing signals are obtained only when molecular adsorption induces sufficiently pronounced variations in the effective permittivity, dielectric loss, or conductivity of the sensing film. Based on this transduction mechanism, 2D materials such as graphene¹⁸⁰ and MXenes,¹⁸¹ owing to their intrinsic high conductivity and robust electrical responses (e.g., charge transfer) to molecular adsorption, are universally regarded as ideal candidates for microwave sensing. Furthermore, owing to their intrinsically low baseline dielectric constant and substantial pore adsorption capacity,¹⁸² 3D framework materials such as MOFs can undergo pronounced macroscopic dielectric changes upon the enrichment of target molecules. Within the equivalent circuit model, these variations translate directly into significant relative impedance changes, rendering them well-suited for microwave sensing applications.¹²³

Given the specific spatial distribution of microwave electromagnetic fields, the porous sensitive film must be highly spatially matched with the device's field distribution. In microwave sensing theory, the efficiency of this spatial overlap is quantitatively characterized by the filling factor, defined as the volume fraction of the sensing material occupying the region of intensified electric field energy.¹⁸³ From the impedance model perspective, maximizing the filling factor most efficiently translates the localized dielectric variations induced by analyte adsorption into profound perturbations of the equivalent impedance. For instance, Kumar et al.¹⁸⁴ precisely etched specific microwell structures within the region of maximum electric field intensity of the microwave resonator for the targeted deposition of sensing materials, which markedly increased the filling factor and successfully lowered the sensor's detection limit for methanol vapor to 61 ppb.

4.1.3. Sensor Performance Optimization Strategies. As established, the impedance model provides the theoretical bridge linking sensor performance specifications with the physical sensor configuration. Accordingly, model-driven sensor optimization can be systematically pursued from two complementary perspectives: the intrinsic resonator itself, and the porous sensitive material.

(1) From the resonator perspective, fundamentally, the core impedance parameters that play a crucial role in determining the macroscopic performance indicators of the sensor can be identified. These impedance parameters correspond to the actual physical design parameters of the sensor, enabling precise modulation. For instance, in FBARs, the Q -factor is critically constrained by the parasitic dissipation of series contact resistance, which can be mitigated by systematically optimizing electrode materials, physical thickness, geometric layout, and electrical contact.¹⁸⁵ For planar microwave resonators, a stronger intrinsic equivalent capacitance in the circuit model directly translates to amplified sensitivity toward dielectric perturbations. This principle fundamentally drives the widespread

adoption of interdigital SRR topologies (Figure 8a) in high-performance microwave sensing architectures.^{186,187}

Furthermore, beyond tuning existing parameters, researchers can structurally reconfigure the sensor to actively eliminate irrelevant parasitic impedances from the model, thereby maximizing the response weight of the target transduction impedance. As shown in Figure 8b, Ma et al.¹⁸⁸ etched a physical trench within the region of maximum electric field intensity of a microwave resonator, thereby effectively removing the nonresponsive parasitic substrate capacitance from the equivalent circuit. By integrating porous polydimethylsiloxane as the sensing medium, this strategy achieved a remarkable 175.5% enhancement in sensitivity toward volatile organic compounds (VOCs).

Recently, a more advanced optimization paradigm has emerged. By constructing external circuit networks derived directly from the sensor's equivalent circuit model, the underlying response mechanism can be fundamentally altered, enabling a substantial leap in performance. Liu et al.¹⁸⁹ proposed a novel sensing architecture integrating an IDE with a QCM, fundamentally circumventing the severe acoustic attenuation and viscoelastic damping caused by direct contact between the analyte-adsorbed film and the crystal. By leveraging the drastic equivalent impedance variations of chitin nanofibers upon moisture adsorption for the purely electrical modulation of the QCM's resonant frequency, the device achieved an exceptional sensitivity of 171.7 Hz/% RH. Similarly, Wei et al.¹⁹⁰ engineered a second-order nonlinear parity-time symmetric system by bridging an FBAR sensor and an external nonlinear negative-resistance LC circuit via a coupling capacitor, as illustrated in Figure 8c. This architecture achieved a 17-fold sensitivity enhancement over a single FBAR, alongside a remarkable 4 orders of magnitude increase in the Q -factor. Furthermore, this highly adaptable network paradigm can be seamlessly extended to other RF sensing platforms.

(2) From the porous material perspective, the core optimization objective is to maximize the impact of impedance variations induced by target analyte adsorption on the resonant performance of the sensor. On the one hand, sensor sensitivity can be enhanced by reducing the initial baseline impedance of the sensitive layer itself, thereby amplifying the relative impedance change induced by adsorption. This strategy effectively mitigates the parasitic damping on the initial resonant characteristics, ensuring the resonator maintains a high Q -factor. For instance, Zhang et al.¹⁹¹ synthesized a poly(sodium 4-styrenesulfonate) (PSS)-doped ZIF-8-derived porous carbon/polyaniline (PANI) composite by MOFs template-carbonization and in situ polymerization. Upon coating this composite material onto a QCM sensor, its low initial density and optimized electrical properties yield a Q -factor approximately twice that of a sensor coated with the pure material.

On the other hand, beyond merely suppressing the background impedance, innovative material engineering strategies can directly amplify the absolute impedance variation by the adsorption of target molecules from a physicochemical mechanistic level. For example,

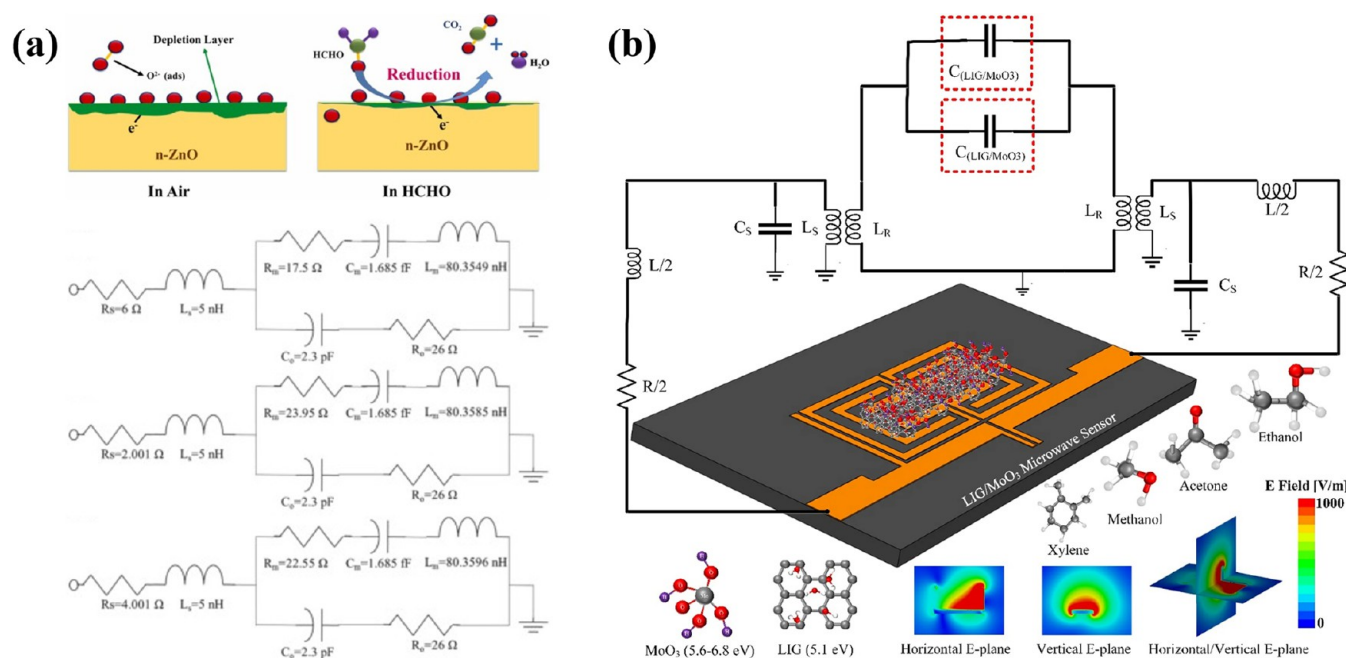


Figure 9. Representative impedance analysis cases of porous material-empowered RF sensors for VOC detection. (a) mBVD model elucidation of the dual impedance response (R_m and L_m variations) in a ZnO nanostructure-functionalized SAW formaldehyde sensor;¹⁹⁶ (b) Equivalent circuit analysis of the synergistic impedance enhancement effect in a LiG/MoO₃ composite-based SRR ethanol sensor.¹⁹⁷ Adapted with permission from ref 196. Copyright 2024 Elsevier. Adapted with permission from ref 197. Copyright 2025 Springer Nature.

within a microwave resonant sensing architecture, Zhou et al.¹⁹² engineered SnO₂/Co₃O₄ composite nanofibers enriched with p-n heterojunction interfaces. By leveraging the robust charge transfer and energy-band modulation effects occurring at these interfaces, a drastic increase in the equivalent resistance of the porous film upon acetone adsorption is realized, yielding a remarkable 4.26-fold enhancement in sensing sensitivity relative to its pristine counterpart. Notably, Bahoumina et al.¹⁹³ perfectly amalgamated both optimization strategies by developing a conductive PEDOT:PSS-MWCNTs composite, wherein robust π - π interactions and hydrogen bonding facilitate efficient charge transfer to boost the intrinsic conductivity from 4.64 to 40.17 S/cm, thereby minimizing the equivalent baseline resistance to ensure a high Q-factor and distinct resonant clarity. Concurrently, the incorporation of MWCNTs not only provides an expansive specific surface area but also elevates the electrochemical energy storage capacity via enhanced conductivity, fundamentally multiplying the absolute impedance perturbations triggered by target analyte adsorption.

Furthermore, innovating the immobilization methods of porous materials can effectively isolate the negative impacts of sensitive materials on the resonator's intrinsic impedance in physical space, thereby maximizing sensing performance without sacrificing the device's baseline Q-factor. When developing a planar microwave sensor for acetone detection, Ali et al.¹⁹⁴ proposed a novel noncontact immobilization method for Ti₃C₂T_x MXene with a polyimide film as the key dielectric spacer. This approach not only decreased the initial capacitance of the device, but also eliminated the significant parasitic dielectric loss and damping effects caused by the sensitive material, thus increasing the sensor's acetone sensitivity from 8.23 kHz/ppm to 17.85 kHz/ppm.

In summary, the systematic optimization strategy based on the impedance model has laid a solid physical foundation for breaking through the performance limit of RF sensing. Benefiting from these theoretical guidelines and innovations in device engineering, porous material-enabled RF sensors have demonstrated groundbreaking application potential in core fields including environmental monitoring, biomedical analysis, as well as industrial and food quality control. The subsequent chapters will systematically review on these cutting-edge application scenarios.

4.2. RF Sensors in Environmental Monitoring Applications

The escalating global crisis of environmental degradation necessitates continuous and highly sensitive monitoring of ubiquitous pollutants in both gaseous and aqueous ecosystems. Traditional analytical techniques are often ill-suited to the demands of real-time, in situ detection of trace-level contaminants.¹⁹⁵ Consequently, porous material-enabled RF sensors present an excellent solution for meeting these requirements. By capitalizing on the immense specific surface area and customizable pore chemistry of porous frameworks, these devices transduce minute molecular adsorption events involving VOCs, toxic inorganic gases, and aquatic pollutants into pronounced impedance perturbations. Harnessing the exceptional multidimensional impedance transduction capability inherent to RF sensors further empowers high-performance sensing. This synergy directly addresses the environmental sector's critical demand for ultralow limits of detection and broad geographic deployability.

4.2.1. Monitoring of VOCs. VOCs as a globally significant class of air pollutants, not only contribute substantially to environmental degradation through their participation in atmospheric photochemical reactions but also pose potential risks to human health. Porous material-empowered RF sensors have emerged as ideal platforms for high-precision VOC monitoring owing to their exceptional sensing performance. For instance, Singh et al.¹⁹⁶ proposed a SAW formaldehyde sensor functionalized with an n-type ZnO nanostructured film. Based on the mBVD model, a dual impedance response mechanism triggered by the sensitive layer was profoundly elucidated (Figure 9a): redox reactions occurring between formaldehyde and oxygen vacancies on the ZnO surface release electrons, leading to an abrupt increase in conductivity (characterized by a pronounced variation in motional resistance, R_m); meanwhile, the exceptionally high specific surface area of the nanostructure significantly amplifies the acoustic mass loading effect induced by gas adsorption (characterized by an increase in motional inductance, L_m). Owing to this optimal impedance-level compatibility, an ultralow LOD of 80 ppb was achieved. As shown in Figure 9b, Ali et al.¹⁹⁷ developed a SRR sensor dedicated to ethanol detection. For the design of sensitive materials, the team composited laser-induced graphene (LiG), featuring high conductivity and large specific surface area, with MoO₃ nanosheets rich in active adsorption sites—a design

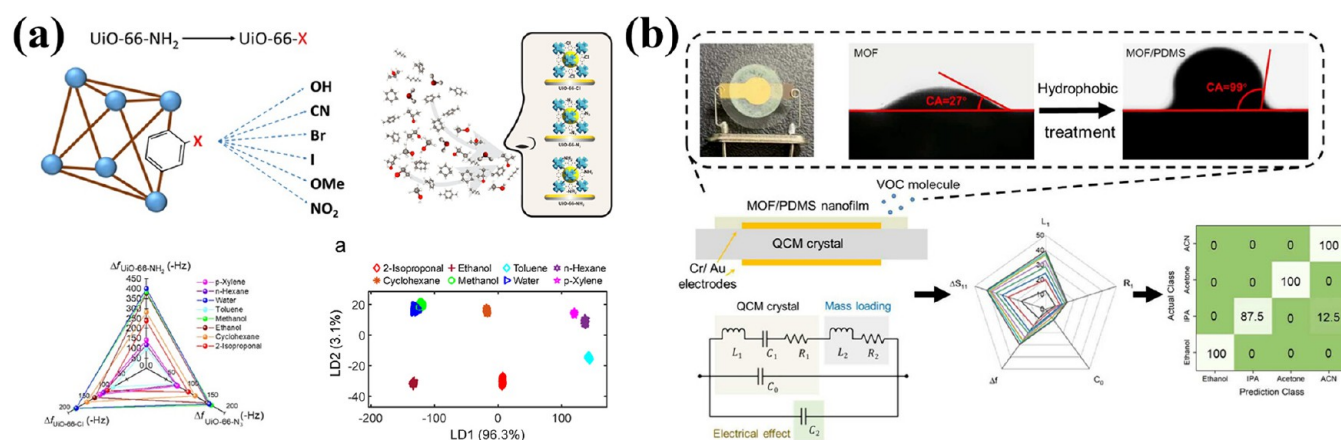


Figure 10. RF sensor-based e-nose technologies for VOC classification and multicomponent recognition. (a) MSA constructed by QCM resonators with differentially functionalized MOF sensitive layers;²⁰⁰ (b) Single-device VSA enabled by advanced impedance parameter extraction and modeling.⁴⁵ Adapted from ref²⁰⁰. Copyright 2024 American Chemical Society. Adapted from ref⁴⁵. Copyright 2024 American Chemical Society.

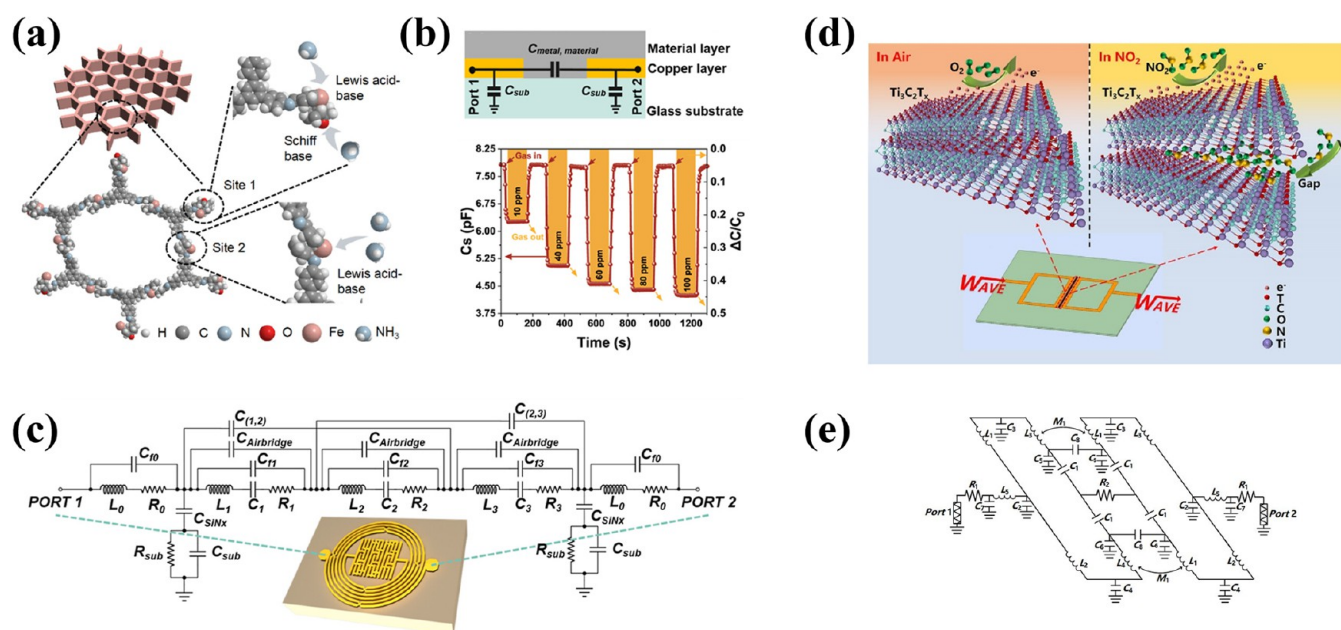


Figure 11. Porous material-based RF sensors for toxic inorganic gas detection and their impedance analysis. (a) Ferrocene-functionalized COF-based QCM sensor with dual-interaction sites for enhanced NH₃ sensing;²⁰³ (b) Dielectric alteration mechanism of porous PANI/SnO₂ composite heterojunction microwave sensor upon NH₃ adsorption;²⁰⁵ (c) IPD-based microwave sensor with optimized interdigital capacitor array and air-bridge structure;²⁰⁵ (d) MXene-integrated coupled SRR sensor for ppb-level NO₂ detection;²⁰⁶ (e) Equivalent circuit model illustrating the dielectric loss-sensitive coupling mechanism, where R_2 corresponds to the dielectric loss of MXene.²⁰⁶ Adapted with permission from ref²⁰³. Copyright 2025 IEEE. Adapted from ref²⁰⁵. Copyright 2023 American Chemical Society. Adapted from ref²⁰⁶. Copyright 2025 American Chemical Society.

that can be equivalently modeled as introducing a pair of parallel capacitances (C_{LIG} and C_{MO_3}) into the sensor's equivalent circuit model—to significantly enhance the amplitude of adsorption-induced impedance variation. Furthermore, at the resonator level, meandering coupled arms and microstrip stubs were incorporated into the SRR structure to enlarge the above intrinsic equivalent capacitance. Through this synergistic material-structure impedance optimization, the sensor achieved a remarkable frequency shift sensitivity of 193 kHz/ppm, providing a highly reliable paradigm for the sensitive discrimination of environmental VOCs.

Given the immense variety and physicochemical similarity of VOCs, practical applications urgently demand robust classification and multicomponent recognition capabilities, and e-nose technology based on RF sensors has thus been extensively developed. Typically, constructing a multisensor array (MSA) with resonant units coated

with distinct sensitive materials generates a unique fingerprint upon analyte adsorption. Coupled with machine learning algorithms, this approach enables highly accurate discrimination of individual and mixed gases.^{198–200} For example, Okur et al.²⁰⁰ developed a three-channel QCM array functionalized with UiO-66-X MOFs, as shown in Figure 10a, where the linker functional groups were systematically tailored via postsynthetic modification to diversify guest adsorption affinities. By leveraging differential frequency shifts to construct specific fingerprints via linear discriminant analysis, they achieved an exceptional recognition accuracy of 95% for eight VOCs (e.g., ethanol, isopropanol) at low concentrations (50–100 ppm). Beyond physical arrays, the inherent multiparameter nature of RF sensors (e.g., frequency shift, Q-factor, amplitude, and impedance) naturally facilitates the construction of single-device virtual sensor arrays (VSAs).²⁰¹ Furthermore, strategies such as temperature modulation (which exploits the temperature-dependent affinities of materials)²⁰²

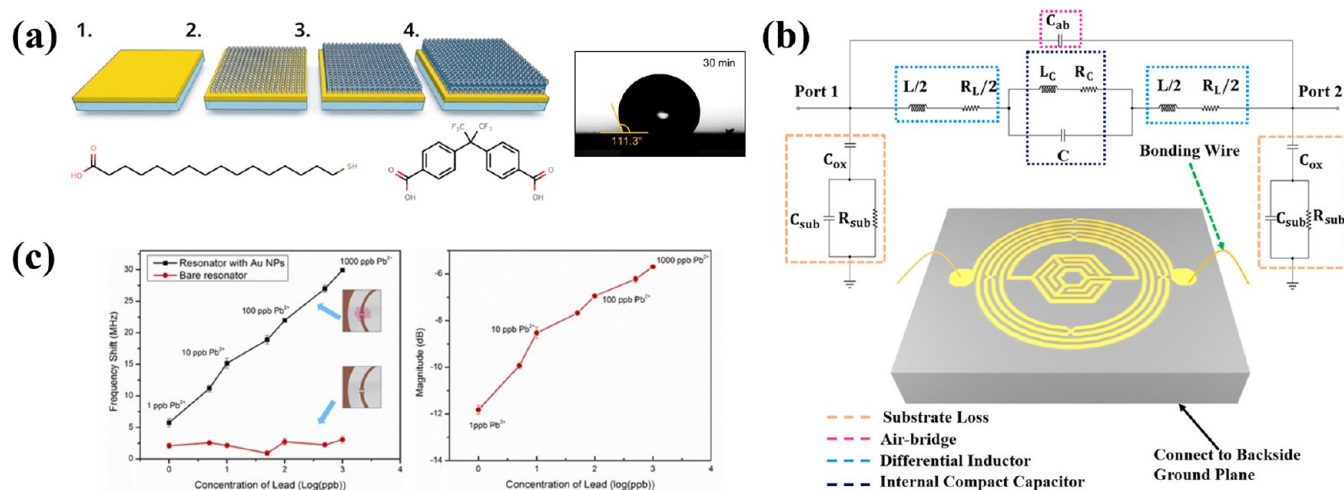


Figure 12. RF sensor technologies for liquid-phase water quality monitoring. (a) Layer-by-layer grown hydrophobic MOF sensor shielding against water-induced acoustic dissipation;¹⁶³ (b) Microwave ELC sensor exploiting dielectric perturbations for nanoparticle quantification;²⁰⁹ (c) SRR sensor with Au nanoparticle porous film for heavy metal detection, achieving ppb-level detection through simultaneous resonant frequency and amplitude modulation.²¹⁰ Adapted from ref¹⁶³ Copyright 2025 American Chemical Society. Adapted from ref²⁰⁹. Copyright 2023 American Chemical Society. Adapted with permission from ref²¹⁰. Copyright 2023 Elsevier.

and power modulation (which leverages the adsorption/desorption dynamics of materials under varying oscillation amplitudes)¹⁶² have been actively pursued to substantially expand the feature dimensionality of RF-based electronic noses. Notably, when utilizing a porous PDMS-integrated CSRR sensor to classify homologous VOCs (methanol, ethanol, isopropanol), Ma et al.¹⁸⁸ demonstrated that classification models based on equivalent circuit parameters outperform those based on raw RF resonant parameters. This superiority stems from the fact that extracted impedance parameters are not only less susceptible to measurement equipment precision but also correspond more directly to the intrinsic permittivity and dielectric loss of the analytes. Subsequently, anchored by this advanced impedance modeling paradigm, the same team successfully achieved stable VOC discrimination in complex high-humidity environments using a QCM⁴⁵ (Figure 10b) and constructed a high-performance single-chip VSA based on an FBAR³⁰ both functionalized with a highly porous MOF film, achieving an exceptional classification accuracy of 94.6% across mixed VOCs.

4.2.2. Monitoring of Toxic Inorganic Gases. Toxic inorganic gases, characterized by their minuscule molecular sizes and high hazard levels, mandate that porous material-based RF sensors possess profoundly exceptional surface affinity and acute impedance resolution. Guo et al.²⁰³ synthesized a novel ferrocene-containing COF for QCM sensing. As shown in Figure 11a, the sensitive film operated through a dual-interaction mechanism: a reversible Schiff base condensation between the aldehyde groups of the framework and NH_3 , coupled with strong Lewis acid–base coordination provided by the iron centers of the ferrocene moieties. This synergy maximized the rate of motional inductance variation per unit volume without appreciably increasing the baseline physical thickness of the coating, ultimately yielding a notable frequency shift of 65 Hz toward 10 ppm of NH_3 .

RF impedance analysis offers a powerful tool for elucidating the sensing interactions between sensitive materials and inorganic gas molecules and guiding the structural optimization of RF sensors. Utilizing the mBVD model and conductance spectrum analysis, Wang et al.²⁰⁴ convincingly demonstrated that MOF-74 porous films adsorbing CO_2 exhibit significant viscoelastic characteristics. Similarly, when designing a microwave sensor, Tang et al.²⁰⁵ profoundly revealed via experimental capacitance measurements that NH_3 adsorption effectively modulates the depletion region within the porous PANI/ SnO_2 composite heterojunction, thereby triggering inducing dielectric alterations (Figure 11b). Moreover, during the optimization of the sensor's configuration based on integrated passive

device (IPD) technology, the team consistently utilized the impedance model. As shown in Figure 11c, by incorporating three sets of interdigital capacitors (C_{f1} , C_{f2} , C_{f3}) their respective coupling capacitors (C_{12} , C_{23}) and an air-bridge structure ($C_{\text{Airbridge}}$), they maximally expanded the effective sensitive capacitance within the network, achieving an exceptional NH_3 detection sensitivity of 0.925 MHz/ppm. In another microwave sensing paradigm, the $\text{Ti}_3\text{C}_2\text{T}_x$ MXene was precisely positioned within the coupling region of two SRR (Figure 11d).²⁰⁶ Owing to the extreme sensitivity of the coupling circuit to dielectric loss (equivalent to resistance R_2 , as shown in Figure 11e), they achieved an ultralow NO_2 trace detection limit of 2 ppb.

4.2.3. Monitoring of Water Quality. A diverse array of contaminants, including heavy metal ions, hazardous organic compounds, and suspended particulate matter, is ubiquitously present in aquatic environments. Consequently, high-performance water quality monitoring has emerged as a critical research priority. As noted earlier, significant mechanical damping restricts the operating mode of piezoelectric sensors in liquid environments. Even for QCM sensors, which are compatible with the liquid phase, resolving weak signals under conditions of strong acoustic damping largely depends critically maximizing the impedance perturbations induced by target analyte adsorption within the porous sensitive layer. For instance, Li et al.¹²⁵ utilized the abundant $\text{N}_3\text{N}'$ -chelating sites provided by bipyridine units on the COF pore walls to undergo specific coordination with Hg^{2+} in water. This formed stable five-membered ring complexes, which excited a significant increment in motional inductance successfully achieving trace mercury ion detection at 58 nM. To overcome liquid-phase interference, as shown in Figure 12a, Malhotra et al.¹⁶³ deposited a highly hydrophobic crystalline MOF film via a layer-by-layer growth process, successfully preventing bulk water from penetrating the pores and effectively shielding the porous film from water-induced viscoelastic mutations and extra acoustic dissipation. Concurrently, owing to the similarity in size between the pore aperture of the porous material (4.79 Å) and the kinetic diameter of methane molecules (~ 4 Å), dissolved methane can be rapidly transferred from the aqueous phase into the water-free pores, thereby enabling ultrahigh sensitivity at the ppb level in liquid environments.

Beyond QCM, SAW and FBAR devices based on specific cuts and wave modes are accelerating their deployment in liquid-phase detection. Choudhari et al.²⁰⁷ utilized a shear-horizontal wave SAW architecture paired with a molecularly imprinted polymer (MIP)—a custom porous sensitive layer—to achieve an ultrawide detection

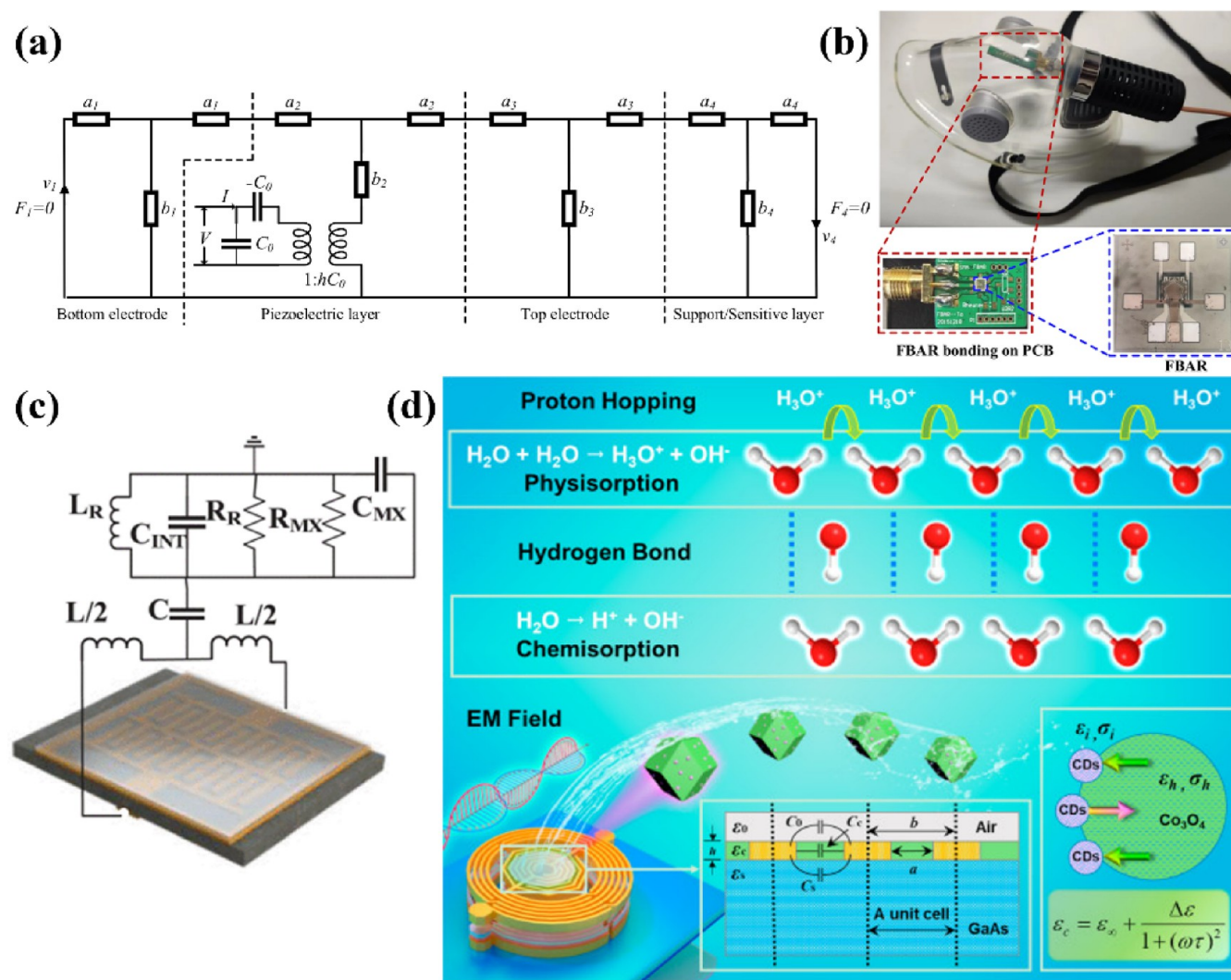


Figure 13. Advances in RF humidity sensors for respiratory monitoring applications. (a) and (b) Wearable FBAR sensor for mask-based respiratory dynamics monitoring;²¹² (c) MXene/ELC sensor with parallel impedance components for dual-parameter humidity detection;²¹³ (d) High-performance IPD ELC humidity sensor with porous composite functionalization.²¹⁵ Adapted from ref²¹². Available under a CC-BY 4.0 license. Copyright 2022 Yusi Zhu et al. Adapted with permission from ref²¹³. Copyright 2023 IEEE. Adapted with permission from ref²¹⁵. Copyright 2021 Elsevier.

range from 10^{-7} ppm to 100 ppm for the organic pollutant 4-aminomethylpyridine in complex turbid waters. Takuli et al.²⁰⁸ employed (002)-oriented high-quality ZnO thin-film planar electrodes to excite a coplanar shear wave mode, and integrated an air cavity structure into the device design to significantly reduce the motional resistance (R_m) in the impedance model, enabling the bare device to maintain a quality factor of over 1900 even under highly damped liquid-phase operating conditions.

In contrast, microwave RF sensors interact directly with substance via electromagnetic fields, inherently evading severe mechanical acoustic damping and thus exhibiting broader applicability in liquid-phase monitoring. For example, Qiang et al.²⁰⁹ developed a microwave ELC sensor composed of a spiral inductor and a hexagonal capacitor, which relied exclusively on abrupt changes in the effective dielectric constant and loss of aqueous solutions—driven by variations in the concentration, size, and surface charge of high-resistance, low-dielectric nanoparticles (NPs)—to selectively modulate the capacitance (C) and resistance (R_C), as shown in Figure 12b. Through this sensing mechanism, highly sensitive and repeatable quantitative detection of NPs was successfully realized. To further push the microwave liquid-phase detection limits to the extreme, Cui et al.²¹⁰ functionalized a SRR using a drop-cast, self-assembled gold

nanoparticle (Au-NPs) porous film, realizing ultrasensitive trace heavy metal detection down to 1 ppb within a complex tap water background. Leveraging the asymmetric surface atomic distribution and the prominent electron confinement effect, this functional film not only markedly enhanced and extended the localized electric field in the sensing region, but also amplified the dielectric and loss perturbations induced by Pb^{2+} adsorption, which directly modulates both the resonant frequency and amplitude response of the sensor (Figure 12c).

4.3. RF Sensors in Biomedical and Bioanalysis Applications

In the realm of clinical diagnostics and personalized healthcare, the diagnostic paradigm is rapidly shifting toward point-of-care testing (POCT).²¹¹ This shift urgently demands sensing platforms capable of seamlessly bridging biological microenvironments with electronic signal readouts. Porous material-based RF sensors occupy this critical niche effectively by combining ultrafast molecular binding kinetics³³ with their inherent wireless and passive attributes,²⁰ thereby rendering them highly suitable for wearable electronic devices and point-of-care diagnostic applications. Whether used for monitoring dynamic respiratory patterns, detecting trace physiological biomarkers in exhaled breath, or performing label-free bioanalysis in complex biological fluids, rationally designed porous interfaces ensure

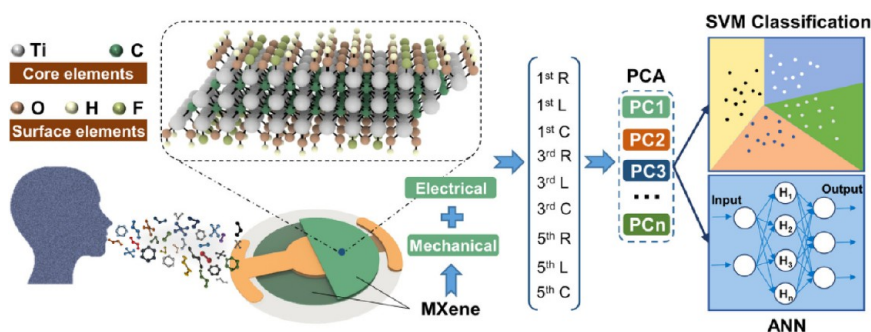


Figure 14. Multidimensional impedance sensing technology for exhaled breath analysis.¹⁷² Reproduced from ref 172. Copyright 2021 American Chemical Society.

exceptional molecular specificity, while RF resonators provide real-time, multidimensional signal transduction.

4.3.1. Respiration Monitoring via Humidity Sensing. Monitoring respiratory dynamics is clinically essential for evaluating core human vital signs, and porous material-enabled RF sensors demonstrate exceptional suitability in this domain owing to their ultrafast molecular adsorption/desorption kinetics as well as their inherent potential for wearable integration. Zhu et al.²¹² utilized a porous PI film simultaneously as the structural support and humidity-sensitive layer for a FBAR sensor. By analyzing mass loading and viscoelastic variations via an impedance model, this sensor, which was embedded in the mask, enabled real-time dual-parameter respiration monitoring through frequency shift and return loss amplitude, as shown in Figure 13a and b. Similarly, by integrating a MXene material onto an ELC resonator and introducing parallel R_{MX} and C_{MX} components into the equivalent circuit model (Figure 13c), dual-parameter humidity detection with high sensitivities of 1.04 MHz/% RH and 30 mdB/%RH was successfully achieved.²¹³

Continuous optimization of porous materials has further pushed sensing performance to its limits. Wang et al.²¹⁴ employed an Au/TiO₂/PEDOT porous ternary composite film to concurrently trigger mass loading and viscoelastic response mechanisms, enabling a single SAW device to distinctly differentiate humidity from CO₂ in exhaled breath through bidirectional frequency shifts, while also delivering remarkably rapid response kinetics—specifically, response times of 9 s for CO₂ and 15 s for RF. Furthermore, by introducing a composite—comprising physisorption-dominant porous Co₃O₄ (a MOF derivative) and superhydrophilic carbon dots (CDs) inspired by diapers—into an IPD ELC sensor, highly sensitive dual-parameter detection (3.40 MHz/%RH and 0.15 dB/%RH) was remarkably realized,²¹⁵ as shown in Figure 13d. Luo et al.²¹⁶ employed a comprehensive SRR-IDC impedance model that accounted for parasitic capacitance. By optimizing the ratio of α -Al₂O₃ nanoflakes to chitosan to balance porosity with the density of hydrophilic functional groups, and by precisely controlling the film thickness to confine the active layer within the region of high dielectric sensitivity, the sensor achieved an outstanding sensitivity of 4.463 MHz/%RH across a broad relative humidity range of 5% to 95%.

In addition to elucidating sensing mechanisms and informing performance optimization, the abundant multidimensional impedance signatures of RF sensors can be integrated with machine learning algorithms within electronic nose systems to enable in-depth analysis of respiratory patterns. For instance, based on a QCM sensor coated with a UiO-66-NH₂/V₂CT_x (MOF/MXene) composite, six distinct respiration patterns (e.g., normal, fast, slow, and Biot breathing) were successfully discriminated using a decision tree algorithm, yielding an exceptional classification accuracy of 97.7%.²¹⁷

4.3.2. Tracking of Physiological Biomarkers. Beyond environmental monitoring, specific gaseous constituents present in human breath frequently reflect underlying metabolic status, rendering their accurate detection as physiological biomarkers profoundly significant in clinical medicine.²¹⁸ For instance, acetone concentrations in the breath of diabetic patients—typically ranging from several hundred ppb to several ppm—are marginally elevated relative to those of

healthy individuals, thereby serving as an ideal biomarker for noninvasive diagnosis.²¹⁹ Given that human exhaled breath is fundamentally a complex matrix comprising hundreds of background VOCs, reliable noninvasive diagnosis dictates that the sensor must precisely discriminate the target biomarker from this highly interfering VOC mixture. Addressing this challenge, as illustrated in Figure 14, Li et al.¹⁷² employed a MXene-functionalized QCM sensor and extracted multidimensional impedance signatures reflecting both electrical and mechanical characteristics from the higher-harmonic responses on the basis of the mBVD model. Through integration with machine learning algorithms, the e-nose system achieved a recognition rate of 90.6% across various interfering VOCs as well as their mixtures, ultimately culminating in a clinical diagnostic accuracy of 95.0% for actual exhaled breath samples.

In contrast, while Li et al. emphasized the discrimination of complex VOC mixtures, Katsuno et al.¹⁶⁵ focused on improving the LOD for acetone to ensure the precise quantification of trace-level biomarker fluctuations within the breath. Building upon a UiO-66 MOF characterized by a high specific surface area of 1120 m²/g, they designed a two-stage sensing system that integrates a gas concentrator with a QCM detector. Following the concentrator's adsorption of low-concentration gas in substantial quantities and subsequent instantaneous thermal desorption, the variations in motional inductance and the resonant frequency shifts of the QCM are dramatically enhanced, enabling the detection of acetone within a concentration range of 200 ppb to 8 ppm. Other hardware-assisted detection accessories have also been employed in RF gas sensing. For instance, gas mouthpieces have been utilized in the measurement of breath alcohol concentrations to ensure flow stability through mechanical design, thereby enhancing measurement repeatability.²²⁰

Furthermore, various functionalized materials have been synthesized to enhance the adsorption of gaseous biomarkers and amplify the impedance response of RF sensors. Tasaltin et al.²²¹ first functionalized thiol-terminated calix[4]arenes onto high-surface-area gold nanorods and silver nanocubes, respectively. SAW sensors combined with these materials exhibited a 6- to 8-fold enhancement in sensitivity toward various exhaled VOCs, with differences in selectivity corresponding essentially to distinct impedance variation pathways arising from the interaction between the VOCs and the porous film. Lu et al.²²² developed a differential CSRR sensor integrated with Pd-doped CdSnO₃ nanoparticles and demonstrated that Pd doping not only enhanced the chemical selectivity of the sensing layer but also furnished catalytically active Pd centers that promote the dissociation of target gas molecules followed by their migration to the surface of the metal oxide support—a phenomenon known as the spillover effect. The above effect amplified the variation in the imaginary part of the resonator's complex impedance, ultimately enabling a sensitivity of 0.49 dB/ppm toward H₂S, a biomarker associated with both asthma and halitosis, at ultralow concentrations ranging from 1 ppb to 0.2 ppm.

4.3.3. Label-Free Bioanalysis. By directly detecting impedance perturbations induced by the interaction between analytes and sensitive films, label-free biochemical analysis is enabled with RF sensors, thereby obviating the need for fluorescence labeling or

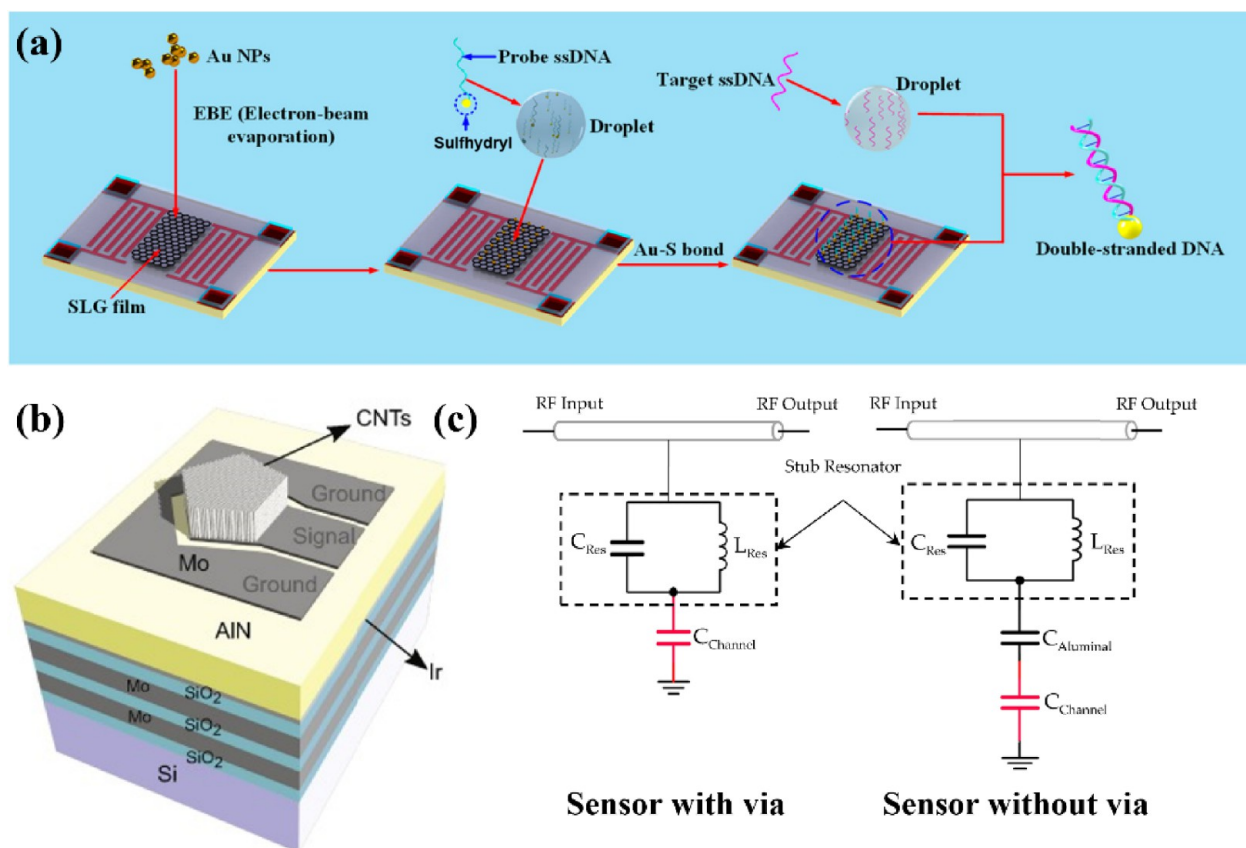


Figure 15. Multifunctional porous materials enabling high-performance RF biosensors. (a) SLG as an ideal biorecognition element immobilization platform for SH-SAW DNA sensors;²²⁵ (b) VA-CNTs suppressing liquid-phase energy dissipation in FBAR sensors;²²⁶ (c) Comparison of impedance models with and without porous alumina template-based conductive vias.²²⁷ Adapted from ref²²⁵. Copyright 2020 American Chemical Society. Adapted with permission from ref 226. Copyright 2018 Elsevier. Adapted from ref 227. Available under a CC-BY 4.0 license. Copyright 2022 Atena Amanati Shahri et al.

complex pretreatment typically required in conventional biochemical assays. Since biological analyses are predominantly conducted in liquid environments, the energy loss arising from the liquid phase must also be addressed. For instance, Ooi et al.²²³ developed a QCM sensor based on MIPs, in which the MIP layer was weakly coupled to the resonator and exhibited characteristic viscoelastic behavior. By employing a coupled oscillator model and performing modified impedance analysis, this study successfully elucidated the positive resonant frequency shift observed experimentally. SH-SAW biosensors integrated with CVD-grown monolayer graphene (SLG) films have been developed for the targeted detection of endotoxins²²⁴ and specific DNA sequences.²²⁵ Characterized by its exceptionally high specific surface area, ultrathin SLG provides an ideal platform for the uniform and efficient immobilization of biorecognition elements while simultaneously minimizing device loss and equivalent resistance. Specifically, through the functionalization of SLG with chitosan for improved hydrophilicity and glutaraldehyde cross-linking for aptamer immobilization, a linear detection range of 0–100 ng/mL with an LOD of 0.15 ng/mL was achieved for endotoxins. Concurrently, by decorating SLG with gold nanoparticles and thiol-modified ssDNA probes, as shown in Figure 15a, the sensor demonstrated a highly sensitive response to target gene sequences with an LOD of 0.45 pM (12.4 pg/mL).

Beyond serving as immobilization platforms, porous materials play critical roles in modulating acoustic properties and eliminating parasitic parameters in RF bioanalysis. Flewitt et al.²²⁶ exploited the exceptionally low mass density of vertically aligned CNTs (VA-CNTs) together with the pronounced acoustic impedance contrast among the nanotubes, the piezoelectric substrate, and the liquid medium to suppress energy dissipation. Through the introduction of the VA-CNTs (Figure 15b), the Q-factor of an FBAR operating in

water in the thickness longitudinal mode was raised from approximately 3 to over 160. Shahri et al.²²⁷ utilized porous alumina as a template to grow metallic nanowires, forming conductive vias that “physically extend” the resonator tip to the top of the microfluidic channel. This design completely eliminated the parasitic capacitance of the substrate ($C_{Alumina}$), as illustrated in Figure 15c, enabling high-sensitivity detection of single cancer cell.

Furthermore, the unique wireless sensing capabilities of microwave sensors combined with flexible porous materials have expanded the boundaries of wearable biochemical monitoring. Li et al.¹⁶⁴ immobilized carcinoembryonic antigen (CEA) antibodies using a gold nanoparticle-decorated vertical graphene film on a dual-symmetric bending line antenna (Figure 16a). Based on lumped equivalent circuit analysis, the CEA recognition events were directly mapped to variable capacitances, achieving wireless detection of CEA from 0.01 ng/mL to 100 ng/mL. Zhuang et al.²²⁸ designed a square spiral antenna on a flexible PDMS substrate featuring an electropolymerization dopamine (ePDA)/SiO₂ nanowall film that reversibly adsorbs lactate through hydrogen bonding (N–H), altering the antenna’s equivalent capacitance and resonant frequency with a sensitivity of 150 kHz/mM (Figure 16b). In these physiological monitoring applications, interference from nontarget constituents present in human serum or sweat is effectively mitigated by the physical filtration and specific recognition capabilities of the porous membranes, thereby ensuring reliable detection.

4.4. RF Sensors in Process and Quality Control Applications

Modern industrial and agricultural workflows routinely operate under highly dynamic and frequently extreme conditions.²²⁹ In these scenarios, ensuring operational safety and product quality hinges

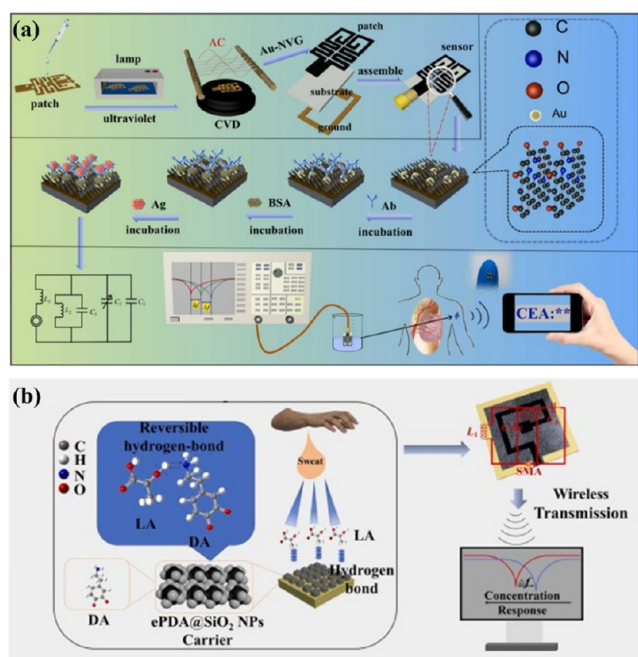


Figure 16. Wireless wearable biochemical sensors enabled by flexible porous materials and microwave antenna technology. (a) Gold nanoparticle/vertical graphene-modified antenna for label-free CEA detection;¹⁶⁴ (b) ePDA/SiO₂ nanowall-functionalized flexible antenna for reversible lactate sensing through hydrogen bonding interactions.²²⁸ Adapted with permission from ref 164. Copyright 2024 Elsevier. Adapted with permission from ref 228. Copyright 2024 Elsevier.

critically on the real-time, stable capture of incipient molecular emissions. In contrast to ambient environmental or controlled clinical settings, process control imposes more stringent requirements for sensing robustness against severe cross-interferences, including drastic temperature fluctuations and elevated humidity levels.²³⁰ Through advanced material engineering, porous material-based RF sensors have proven capable of meeting these exacting industrial and agricultural demands. Via structural optimization of physical stability and the integration of e-nose technology, these sensing systems maintain reliable operation in harsh process environments and effectively decouple complex odor mixtures, thereby facilitating precise leak detection and food quality assessment in real-world industrial and agricultural applications.

4.4.1. Monitoring of Hazardous Substance Leakage and Emissions.

In modern process control operations across industrial

and agricultural sectors, real-time monitoring of the release and emission of toxic and hazardous substances is of paramount safety importance. Relative to environmental and biomedical detection scenarios, the harsh operating conditions characteristic of such settings impose stringent requirements on the physical stability and anti-interference robustness of porous-material-based RF sensors. For instance, a porous WO₃ film with outstanding thermal stability was deposited on a QCM, enabling the robust monitoring of industrial VOC leaks (benzene, toluene, xylene).²³¹ The film effectively prevented early stage mass loss of the acoustic resonator at 100–200 °C, and significantly mitigated its potential degradation and adverse chemical reactions at temperatures up to 500–600 °C, as shown in Figure 17a. Targeting formic acid—a highly corrosive industrial VOC pervasive in chemical manufacturing that poses severe occupational health hazards and machinery corrosion risks—Ahmed et al.²³² exploited the specific binding between this acidic gas and the high-density basic nitrogen sites within the pores of a phenylenediamine-based COF to maximize the resulting motional impedance changes. This sensor not only achieved an exceptional LOD of 1.18 ppm—well below the Occupational Safety and Health Administration workplace threshold of 5.0 ppm—but its remarkable stability also ensured that the COF maintained its structural integrity after 2 months of continuous operation, with the sensing response to 53 ppm formic acid degrading by merely 6.59%. Similarly, targeting H₂S gas widely present in oil extraction and rubber manufacturing, Cui et al.¹⁶⁶ utilized a piezoelectric substrate with a high electromechanical coupling coefficient to fabricate an acoustoelectric-dominant SAW device, and synthesized a conductive Cu₃(HHTP)₂ porous MOF material. Benefiting from the perfect impedance matching between this material and the device, a theoretical LOD of 6 ppb was achieved. Furthermore, the team constructed a phase discrimination circuit to directly detect the influence of minute impedance changes induced by molecular adsorption on the transmission phase, thereby thoroughly circumventing the susceptibility of amplitude readout approaches to environmental fluctuations and parasitic film impedance.

Notably, while porous materials offer exceptional adsorption capacities, their inherent high specific surface area makes them highly prone to adsorbing ambient moisture, leading to severe performance drift and unpredictable impedance fluctuations in RF sensors under different humidity levels.^{233,234} To address this in the context of monitoring VOCs emitted in petrochemical and manufacturing plants, Yan et al.¹²² constructed an ultrathin (~10 nm) gas-permeable hydrophobic PDMS layer atop the HKUST-1 MOF serving as the VOC-sensitive film of an FBAR. Benefiting from a high Q-factor of 3000, this hybrid surface engineering strategy enabled the sensitive film to retain its morphology and sensing response following 4 h of water immersion without loss of its VOC adsorption capacity. From a more fundamental chemical design perspective, Ji et al.¹⁶⁷ employed a SIFSIX-1-Cu MOF, which chemically and physically repels water molecules through the complete enclosure of metal sites (non-

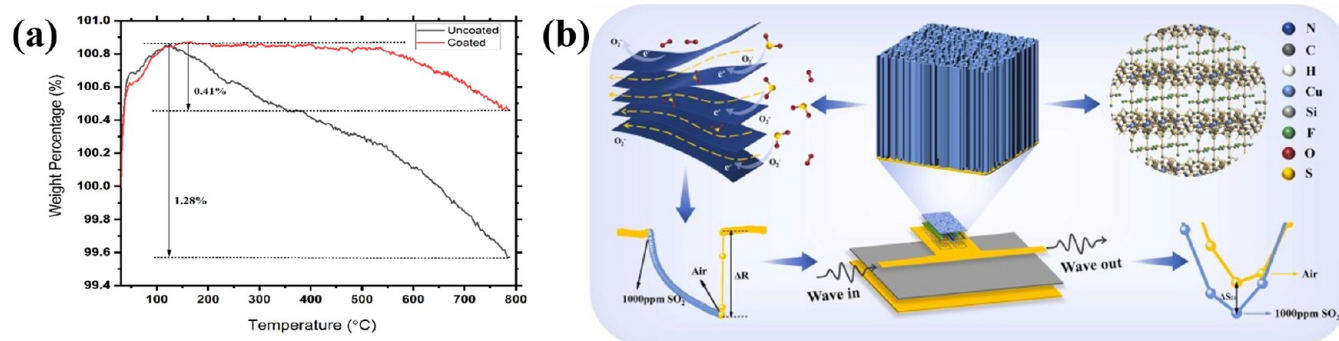


Figure 17. Porous material-based RF sensors for hazardous substance monitoring in harsh industrial environments. (a) High-temperature stable WO₃-coated QCM sensor preventing resonator mass loss and degradation up to 600 °C;²³¹ (b) Hydrophobic SIFSIX-1-Cu MOF sensor with pure conductance response (ΔR), eliminating parasitic humidity interference in high-humidity industrial exhaust.¹⁶⁷ Adapted with permission from ref 231. Copyright 2025 Elsevier. Adapted with permission from ref 167. Copyright 2025 Elsevier.

accessible Cu^{2+}) and the hydrophobic character of SiF_6^{2-} groups and aromatic rings, thereby eliminating parasitic interference with the system's equivalent capacitance and dielectric loss. This allows the MOF film to be approximated as a pure variable conductance within the equivalent circuit during the detection of SO_2 —a highly toxic and corrosive byproduct ubiquitous in industrial flue gas and fossil fuel combustion (Figure 17b). This moisture-resilient impedance fidelity enhanced the reliability of the RF system in complex, high-humidity environments such as industrial exhaust emission stacks, pushing the LOD down to an impressive 10 ppb. Recently, an e-nose system combining a 6-channel QCM array (including a reference channel to offset environmental temperature and humidity fluctuations) loaded with five different sensitive materials and backpropagation neural network algorithms has also been successfully deployed for the intelligent emission monitoring of mixed hazardous gases such as CO_2 , H_2S , and NH_3 .²³⁵

4.4.2. Evaluation of Food Freshness and Ripeness. The processes of food ripening and spoilage are typically accompanied by the release of a broad spectrum of characteristic gases whose concentrations evolve dynamically over time.²³⁶ Indirect assessment of food status through the detection of these characteristic gases has become a highly promising nondestructive strategy. Timely identification of these gaseous markers in the early stages enables proactive interventions to avert severe spoilage and mitigate substantial economic losses, rendering the online, precise capture of trace early emissions especially critical. The integration of porous materials with RF sensing technology—capitalizing on their vast adsorptive specific surface area, ultrafast passive response kinetics, and high degree of integrability—has emerged as an ideal solution for meeting the complex demands of food quality assessment.

In meat monitoring, total volatile basic nitrogen (TVB-N, which consists primarily of biogenic amines such as ammonia, trimethylamine (TMA), and dimethylamine (DMA) serves as the “gold standard” for assessing spoilage,²³⁷ and numerous QCM sensors functionalized with porous materials have been widely deployed for the trace monitoring of meat freshness.^{238–240} For example, Cheng et al.²³⁸ synthesized a nanoporous 0.5Co-Mn-BTC MOF composite and coated it onto a QCM surface. In addition to providing an abundance of active adsorption sites, this material efficiently translated weak chemical adsorption into pronounced impedance perturbations through coordination between target gases and central metal ions, ultimately enabling rapid detection (within 30 s) and subppm discrimination of spoilage markers such as TMA and DMA. To further address the cross-interference of complex gas mixtures in real-world food environments, Chen et al.²⁴¹ constructed a four-channel QCM array comprising four distinct porous nanomaterials (graphene oxide, MXene, hydroxylated multiwalled carbon nanotubes, and graphene oxide-alkane), each of which exhibited pronounced selectivity toward TMA, DMA, formaldehyde, and ammonia, respectively. Upon continuous monitoring of salmon samples stored at 4 °C, this array—when integrated with principal component analysis (PCA) for dimensionality reduction and a quadratic support vector machine (QSVM) model—achieved a classification accuracy of 100% across six distinct storage stages spanning 0 to 5 days.

Similarly, in the field of fruit and vegetable storage and transportation, e-nose systems based on porous RF sensors have demonstrated immense potential in ripeness monitoring. Yen et al.²⁴² constructed a four-channel SAW-based array system that integrated four sensitive films with distinct selectivities toward esters, alcohols, and acids, thereby successfully extracting characteristic VOC fingerprints. Beyond mere VOC screening, Tao et al.²⁴³ utilized a microwave sensing array to extend real-time monitoring to encompass multiple environmental variables (ethanol, CO_2 , and temperature) encountered during cold chain transportation. They coupled three SRRs of different sizes to a single microstrip line to produce three discrete resonant frequencies, and coated the resonators with three distinct rGO-supported porous nanocomposites. The exceptionally high specific surface area and excellent conductivity of rGO ensured high overall impedance sensitivity, while the unique impedance modulation mechanisms of the three specific metal oxide combina-

tions toward their respective target variables reliably preserved the independent selectivity of each sensing channel. To advance portable on-site detection, Wu et al.²⁴⁴ developed a smartphone-sized e-nose system specifically for assessing banana ripeness in cold chains, based on an FBAR functionalized with a porous layer by layer film and a field-programmable gate array (FPGA) digital circuit. This system achieved sensitive capture of ethylene down to 5 ppm and isoamyl esters down to 0.5 ppm. Notably, to overcome the impact of temperature fluctuations during cold chain transportation on the system, the e-nose not only incorporated a Si_xO_y compensation layer with an extremely low temperature coefficient (2.2213 ppm/°C) within the FBAR structure, but also innovatively introduced a differential reference FBAR, thereby profoundly stabilizing the system's baseline impedance from both physical and circuit dimensions.

Looking forward, given the exceedingly complex mixtures of odor components encountered in real-world food detection scenarios, the further deep mining and decoupling of the rich multidimensional impedance features inherent in the aforementioned RF e-nose systems, through the application of advanced algorithms, will constitute the pivotal breakthrough toward commercial deployment in this field.

5. FUTURE DIRECTIONS AND CHALLENGES

While porous-material-based RF sensors have progressed considerably, their translation from well-controlled laboratory settings into real-world e-nose and e-tongue systems continues to face critical bottlenecks. Decoupling intricate cross-sensitivities in multicomponent mixtures and compensating for dynamic environmental fluctuations in wearable scenarios remain particularly formidable. Addressing these challenges calls for a fundamental shift in how the equivalent impedance model is deployed: rather than serving merely as an analytical descriptor, it must act as a design compass. By directly mapping specific physical and chemical perturbations onto distinct electrical parameters, impedance-centric strategies can chart a deliberate path through both hardware constraints and algorithmic complexities. Future development of high-performance RF e-noses and e-tongues should therefore be steered toward the following pivotal paradigms.

5.1. Synergy of Cascaded Resonator Systems and VSAs for Enhanced Selectivity

The advancement of RF-based e-nose/e-tongue technology is increasingly transcending the conventional paradigm of single-parameter readout from individually functionalized resonators, shifting toward multifeature sensing architectures that integrate physical MSA with VSA.^{245–247} This hybrid design beautifully leverages two complementary strengths: MSA utilizes the chemical diversity of various porous materials at the interface to capture rich chemical interactions with target molecules; concurrently, VSA delves into the physical essence by extracting high-dimensional intrinsic impedance features from a single physical sensor via equivalent circuit models. Merging these dimensions exponentially expands the feature space, profoundly enhancing multianalyte discrimination capabilities in complex mixtures.

However, traditional physical arrays typically rely on parallel polling/multiplexing readout methods.^{245–247} This not only creates an efficiency bottleneck for multichannel real-time detection but also faces severe signal integrity challenges at high RF frequencies. Furthermore, high-density integrated RF components suffer from severe electromagnetic/acoustic crosstalk, leading to resonant peak distortion and erratic impedance baseline drift. Therefore, future efforts must focus

on circuit-level cascading strategies grounded in impedance equivalent models. Through the introduction of interdevice coupling impedance parameters, a global-impedance-equivalent-model can be established for the cascaded array, thereby quantitatively relating the distribution of resonant peaks to the strength of electromagnetic coupling and enabling predictable and repeatable multipeak configurations. Architecturally, such a cascaded network permits the simultaneous acquisition of multiple resonant peaks through single-port readout, thereby fully addressing concerns regarding polling efficiency and signal synchronization.

5.2. Structural-Functional Convergence in Flexible and Wearable E-Nose/E-Tongue Systems

Concurrent with the rise of the Internet of Things (IoT) and embodied AI, fully leveraging the structural engineering advantages offered by porous materials for the development of miniaturized, wireless, and flexible detection platforms has become an urgent necessity.^{248,249} Through the combination of flexible substrates and the intrinsic porosity of the porous films, the RF sensor is endowed with excellent mechanical stress buffering capability and long-term cycling stability, rendering it perfectly suited for wearable scenarios. More importantly, emerging conductive porous materials such as MXenes and laser-induced graphene (LIG) can be employed directly as core RF structural components including microwave resonant electrodes.^{197,250,251} This monolithic structural-functional convergence not only substantially reduces the device footprint but also fundamentally minimizes the intrinsic parasitic impedances present within the circuit model. Consequently, it enables the target analyte to perturb the core electromagnetic or acoustic fields directly and maximally, and driving theoretical sensitivity toward its ultimate limit.

However, the introduction of flexible deformations such as bending and stretching inevitably induces dynamic impedance fluctuations. Specifically, alterations in electrode spacing and the deformation of the porous framework can induce significant drift in the static capacitance, distributed inductance, and equivalent resistance represented within the equivalent circuit, thereby becoming confounded with the impedance signatures triggered by molecular adsorption. Therefore, future sensor designs can incorporate differential RF resonant structures based on impedance symmetry. Through such a common-mode rejection mechanism, static baseline drift caused by mechanical deformation and environmental fluctuations can be physically suppressed. From the model perspective, a dual-field equivalent circuit model encompassing both stress perturbation and molecular adsorption can be established. By leveraging the high-dimensional sensing capabilities intrinsic to RF sensors to distinguish the distinct parameters governed predominantly by mechanical strain versus chemical adsorption.

6. CONCLUSION

This review comprehensively outlines the significant advancements in leveraging porous materials to enhance the performance of RF-based e-nose and e-tongue systems. The combination of the distinct advantages of RF transduction mechanisms and the unique physicochemical properties of porous sensitive layers has been thoroughly elucidated. From the fundamental principles of piezoelectric and microwave resonators, along with their corresponding lumped impedance models, to the detailed classification of 2D and 3D porous

materials, this work provides a comprehensive framework for understanding the impedance-driven design and systematic optimization of these sophisticated sensors. The practical applicability of these sensors has been demonstrated across a broad spectrum of fields, including environmental monitoring, biomedical analysis, and process and quality control. Despite considerable progress, challenges remain in achieving high selectivity in complex mixtures, and realizing robust flexible material-device integration for portable or wearable use. Future efforts should fundamentally focus on two key trajectories: the development of hybrid sensing architectures that synergize cascaded physical resonators with VSAs derived from impedance models, and the pursuit of flexible material-device monolithic integration. The continued evolution of porous material-based RF sensors holds great promise for driving innovations toward intelligent, wireless, and deployable e-nose and e-tongue systems, ultimately contributing to smarter and sustainable solutions in environmental protection, personalized healthcare, industrial efficiency, and food safety.

AUTHOR INFORMATION

Corresponding Author

Yunqi Cao – College of Control Science and Engineering, Zhejiang University, Hangzhou 310027, China;
 orcid.org/0000-0001-8063-0931; Email: caoyunqi@zju.edu.cn

Authors

Zhiqiang Ma – College of Control Science and Engineering, Zhejiang University, Hangzhou 310027, China;
 orcid.org/0009-0009-6841-1117
Yancheng Bi – College of Control Science and Engineering, Zhejiang University, Hangzhou 310027, China
Chenyang Gao – College of Control Science and Engineering, Zhejiang University, Hangzhou 310027, China
Mengyao Fu – College of Control Science and Engineering, Zhejiang University, Hangzhou 310027, China
Qingqing Zeng – Hangzhou Ecological and Environmental Monitoring Centre of Zhejiang Province, Hangzhou 310027, China
Lili Shi – Hangzhou Ecological and Environmental Monitoring Centre of Zhejiang Province, Hangzhou 310027, China
Gang Wu – Zhejiang Province of Environmental Protection Group Co., Ltd, Hangzhou 310027, China
Wei Li – College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing 210046, China
Dibo Hou – College of Control Science and Engineering, Zhejiang University, Hangzhou 310027, China;
 orcid.org/0000-0002-3948-9072
Hongjian Zhang – College of Control Science and Engineering, Zhejiang University, Hangzhou 310027, China

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acsami.6c08905>

Author Contributions

Z.M.: Conceptualization, Methodology, Formal Analysis, Investigation, Validation, Writing—Original Draft; Y.B.: Investigation, Validation, Writing—Original Draft; C.G.: Conceptualization, Investigation, Validation, Writing—Original Draft; M.F.: Investigation, Writing—Original Draft; Q.Z.: Project Administration; L.S.: Project Administration; G.W.:

Project Administration; W.L.: Investigation, Project Administration; D.H.: Project Administration, Funding Acquisition; H.Z.: Supervision, Project Administration; Y.C.: Conceptualization, Supervision, Writing—Review and Editing, Project Administration, Funding Acquisition.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported in part by the Pioneer and Leading Goose R & D Program of Zhejiang (2026C02A1210), the National Natural Science Foundation of China under Grants 62273304 and U21A20519, and the Zhejiang Province Basic Public Welfare Research Program under Grant TGY23F030004. The authors would also like to thank Hangzhou Chengxi Sci-tech Innovation Corridor Management Committee for facilitating investigation and project administration in this study.

REFERENCES

- (1) He, S.; Wen, J.; Cao, B.; Shi, G.; Zhang, M. Porous Material-Based Electronic Noses for the Sensing of Volatile Organic Compounds. *ACS Appl. Mater. Interfaces* **2025**, *17* (44), 60055–60103.
- (2) Kim, H. J.; Hyun, J. U.; Jang, H. W. Electronic Tongue: Active Channels, Molecular Sieves, Receptors and Arrays. *Sustainable Food Technol.* **2025**, *3* (6), 1681–1704.
- (3) Day, B. A.; Ahualli, N. I.; Wilmer, C. E. Multipressure Sampling for Improving the Performance of MOF-Based Electronic Noses. *ACS Sens.* **2024**, *9* (7), 3531–3539.
- (4) Yuan, H.; Li, N.; Fan, W.; Cai, H.; Zhao, D. Metal-Organic Framework Based Gas Sensors. *Adv. Sci.* **2022**, *9* (6), 2104374.
- (5) Garay-Rairan, F. S.; Baharfar, M.; Wang, Q.; Qian, J.; Tricoli, A. Emerging Electronic Nose Design for Breath-Based Cancer Diagnostics: Advances in Machine Learning Approaches and Sensor Architecture Design. *ACS Sens.* **2026**, *11*, 3566.
- (6) Zhang, L.; Qi, Z.; Yang, Y.; Lu, N.; Tang, Z. Enhanced “Electronic Tongue” for Dental Bacterial Discrimination and Elimination Based on a DNA-Encoded Nanozyme Sensor Array. *ACS Appl. Mater. Interfaces* **2024**, *16* (9), 11228–11238.
- (7) Wang, D.; Zhang, D.; Li, Y.; Zhang, H.; Lin, L.; Zhang, X.; Wang, H.; Yu, Q. V2CTx MXene/Ni-Metal Organic Framework-Derived V2O5/NiO Nanohybrid Methane Sensors for Fault Detection of Lithium-Ion Battery. *Sens. Actuators, B* **2025**, *428*, 137261.
- (8) Mahanti, N. K.; Shivashankar, S.; Chhetri, K. B.; Kumar, A.; Rao, B. B.; Aravind, J.; Swami, D. Enhancing Food Authentication through E-nose and E-tongue Technologies: Current Trends and Future Directions. *Trends Food Sci. Technol.* **2024**, *150*, 104574.
- (9) Yin, P.; Wu, X.; Xiao, Y.; Feng, C.; Wang, X.; Klyuyev, D.; Wu, Z.; Zhao, Y.; Hu, B. SERS-Based E-Tongue Data Analysis Methods: From Spectrum Preparation to Qualitative and Quantitative Modeling. *ACS Sens.* **2026**, *11* (3), 1831–1859.
- (10) Chiu, S.-W.; Tang, K.-T. Towards a Chemiresistive Sensor-Integrated Electronic Nose: A Review. *Sensors* **2013**, *13* (10), 14214–14247.
- (11) Ye, Z.; Liu, Y.; Li, Q. Recent Progress in Smart Electronic Nose Technologies Enabled with Machine Learning Methods. *Sensors* **2021**, *21* (22), 7620.
- (12) Ji, H.; Mi, C.; Yuan, Z.; Liu, Y.; Zhu, H.; Meng, F. Multicomponent Gas Detection Method Via Dynamic Temperature Modulation Measurements Based on Semiconductor Gas Sensor. *IEEE Trans. Ind. Electron.* **2023**, *70* (6), 6395–6404.
- (13) Rivera, P.; Della Pelle, F.; Stonyte, J.; Martins Antunes de Melo, W. D. C.; Abouhagger, A.; Pauliukaite, R. Overcoming Challenges in Electrochemical Sensing: Toward Continuous Monitoring. *ACS Sens.* **2025**, *10* (12), 9183–9202.
- (14) Meng, K.; Zhu, J.; Zhang, X.; Zhang, Y.; Chen, X.; Li, F.; Tong, Y.; Zhang, S.; Qiu, D.; et al. Nanostructure-Gated Organic Electrochemical Transistors for Accurate Glucose Monitoring in Dynamic Biological pH Conditions. *Biosens. Bioelectron.* **2025**, *287*, 117677.
- (15) Sharma, A.; Eadi, S. B.; Noothalapati, H.; Otyepka, M.; Lee, H.-D.; Jayaramulu, K. Porous Materials as Effective Chemiresistive Gas Sensors. *Chem. Soc. Rev.* **2024**, *53* (5), 2530–2577.
- (16) Niu, G.; Zhuang, Y.; Hu, Y.; Liu, Z.; Wu, B.; Wang, F. Selective Discrimination of VOCs Gases at ppb-Level Using MOS Gas Sensor in Temperature-Pulsed Operation Mode with Modified Hill Equation. *Surf. Interfaces* **2024**, *44*, 103761.
- (17) Chen, Y.; Su, X.; Zhang, Y.; Wei, X.; Zhang, J.; Zhang, Z.; Deng, X.; Sun, S. Synergistic Integration of Self-Cleaning Interface and Molecular Imprinting in Ratiometric Electrochemical Biosensor: Toward Ultrasensitive BSA Monitoring. *Biosens. Bioelectron.* **2025**, *287*, 117764.
- (18) Radha Shanmugam, N.; Rainbow, J.; Lee, J.-C.; Jolly, P.; Ingber, E. D. Label-Free Electrochemical Impedance Spectroscopy for Biosensing: Evolving Interfaces and Mechanistic Insights. *Small Sci.* **2025**, *5* (12), No. e202500380.
- (19) Mehdizadeh, M. *Microwave/RF Applicators and Probes: for Material Heating, Sensing, and Plasma Generation*; William Andrew, 2015.
- (20) Alam, T.; Cheffena, M. Integrated Microwave Antenna/Sensor for Sensing and Communication Applications. *IEEE Trans. Microwave Theory Technol.* **2022**, *70* (11), 5289–5300.
- (21) Lange, K. Bulk and Surface Acoustic Wave Sensor Arrays for Multi-Analyte Detection: A Review. *Sensors* **2019**, *19* (24), 5382.
- (22) Juan, C. G.; Grenier, K.; Zarifi, M.; Ebrahimi, A.; Martin, F. Planar Microwave Sensors: State of the Art and Applications. *IEEE Access* **2025**, *13*, 143985–144038.
- (23) Rasouli, R.; Villegas, K. M.; Tabrizian, M. Acoustofluidics—Changing Paradigm in Tissue Engineering, Therapeutics Development, and Biosensing. *Lab Chip* **2023**, *23* (5), 1300–1338.
- (24) Han, X.; Liu, K.; Peng, P.; Zhang, S. Novel High-Sensitivity Differential Microwave Sensor for Permittivity Determination of Ethanol-Water Solutions. *IEEE Trans. Instrum. Meas.* **2025**, *74*, 1–11.
- (25) Speller, N. C.; Siraj, N.; Regmi, B. P.; Marzoughi, H.; Neal, C.; Warner, I. M. Rational Design of QCM-D Virtual Sensor Arrays Based on Film Thickness, Viscoelasticity, and Harmonics for Vapor Discrimination. *Anal. Chem.* **2015**, *87* (10), 5156–5166.
- (26) Juan, C. G.; Potelon, B.; Aquino, A.; Garca-Martnez, H.; Quendo, C. Multi-Parameter Simultaneous Extraction with a Novel Microwave Sensor Based on Coupled Resonators. *Sci. Rep.* **2024**, *14* (1), 23076.
- (27) Rafi, M. A.; Niknahad, F.; Wiltshire, B. D.; Salim, A.; Zarifi, M. H. Dynamically Tunable Planar Microwave Resonator Sensor Using Liquid Metal Alloy. *IEEE Trans. Microwave Theory Technol.* **2024**, *72* (7), 4103–4113.
- (28) Chen, P.; Mitkova, M.; Tenne, D. A.; Wolf, K.; Georgieva, V.; Vergov, L. Study of the Sorption Properties of Ge20Se80 Thin Films for NO₂ Gas Sensing. *Thin Solid Films* **2012**, *525*, 141–147.
- (29) Sakti, S. P.; Masruroh; Istiroyah; Kamasi, D. D.; Zafirah, T. N. Ultra Thick Polystyrene Coating on Quartz Crystal Microbalance Sensor. *Mater. Today Proc.* **2021**, *44*, 3217–3220.
- (30) Gao, C.; Fu, M.; Fan, S.; Ma, Z.; Tang, Y.; Hou, D.; Cao, Y. High-Performance Virtual Sensors Array Based on a Single-Chip FBAR for Volatile Organic Compounds (VOCs) Detection and Classification. *Sens. Actuators, B* **2025**, *422*, 136687.
- (31) Lei, X.; Pan, F.; Hua, C.; Wang, S.; Xiong, B.; Liu, Y.; Fu, Z.; Xiang, B.; Lu, Y. Oxide-Doped Hierarchically Porous Carbon for High-Performance Supercapacitor. *J. Alloys Compd.* **2022**, *901*, 163624.
- (32) Bian, H.; Li, D.; Jing, Y.; Li, P.; Zhang, N. Amino Functionalization of Porous Aromatic Frameworks for Efficient and Stable Iodine Adsorption. *Inorg. Chem. Commun.* **2024**, *164*, 112388.

- (33) Wang, W.; Virkar, A. V. A Conductimetric Humidity Sensor Based on Proton Conducting Perovskite Oxides. *Sens. Actuators, B* **2004**, *98* (2–3), 282–290.
- (34) Rodahl, M.; Höök, F.; Fredriksson, C.; Keller, C. A.; Krozer, A.; Brzezinski, P.; Voinova, M.; Kasemo, B. Simultaneous Frequency and Dissipation Factor QCM Measurements of Biomolecular Adsorption and Cell Adhesion. *Faraday Discuss* **1997**, *107*, 229–246.
- (35) Douillard, J.; Maurin, G.; Henn, F.; Devautour-Vinot, S.; Giuntini, J.-C. Use of Dielectric Relaxation for Measurements of Surface Energy Variations During Adsorption of Water on Mordenite. *J. Colloid Interface Sci.* **2007**, *306* (2), 440–448.
- (36) Underwood, T. R.; Bourg, I. C. Dielectric Properties of Water in Charged Nanopores. *J. Phys. Chem. B* **2022**, *126* (14), 2688–2698.
- (37) Chappanda, K. N.; Shekhah, O.; Yassine, O.; Patole, S. P.; Eddaoudi, M.; Salama, K. N. The Quest for Highly Sensitive QCM Humidity Sensors: The Coating of CNT/MOF Composite Sensing Films as Case Study. *Sens. Actuators, B* **2018**, *257*, 609–619.
- (38) Wei, J.; Wu, E.-K.; Zhang, J.-H.; Zhang, Z.; Tang, X.-C.; Qiang, T.; Jang, Y.; Gu, X.-F.; Zhao, M.; Von Gratoski, S.; et al. Microwave Ultra-Narrow Band Antenna Sensor Based on TiO₂/MXene for Room-Temperature Acetone Detection. *J. Environ. Chem. Eng.* **2025**, *13* (5), 118803.
- (39) Zhou, T.; Sang, Y.; Wang, X.; Wu, C.; Zeng, D.; Xie, C. Pore Size Dependent Gas-Sensing Selectivity Based on ZnO@ ZIF Nanorod Arrays. *Sens. Actuators, B* **2018**, *258*, 1099–1106.
- (40) Han, Z.; Wang, K.; Min, H.; Xu, J.; Shi, W.; Cheng, P. Bifunctionalized Metal–Organic Frameworks for Pore-Size-Dependent Enantioselective Sensing. *Angew. Chem.* **2022**, *134* (25), No. e202204066.
- (41) Wang, M.; Duan, M.; Yu, F.; Fu, X.; Gu, M.; Chi, K.; Li, M.; Xia, X.; Hu, R.; Yang, Y.; et al. Development of Aflatoxin B1 Aptamer Sensor Based on Iron Porphyrin Organic Porous Material. *Food Anal. Methods* **2021**, *14* (3), 537–544.
- (42) Yi, K.; Wang, Y.; Shi, K.; Chi, J.; Lyu, J.; Zhao, Y. Aptamer-Decorated Porous Microneedles Arrays for Extraction and Detection of Skin Interstitial Fluid Biomarkers. *Biosens. Bioelectron.* **2021**, *190*, 113404.
- (43) Bujran, M. A.; Dar, F.; Tahir, A.; Want, B. Terbium-Doped Metal–Organic Frameworks: Dielectric and Ferroelectric Properties with Enhanced Electrochemical Sensing for Heavy Metal Ion Detection. *J. Solid State Chem.* **2025**, *341*, 125100.
- (44) Singh, S. K.; Tiwari, N. K.; Yadav, A. K.; Akhtar, M. J.; Kar, K. K. Design of ZnO/N-Doped Graphene Nanohybrid Incorporated RF Complementary Split Ring Resonator Sensor for Ammonia Gas Detection. *IEEE Sens. J.* **2019**, *19* (18), 7968–7975.
- (45) Cao, Y.; Fu, M.; Fan, S.; Gao, C.; Ma, Z.; Hou, D. Hydrophobic MOF/PDMS-Based QCM Sensors for VOCs Identification and Quantitative Detection in High-Humidity Environments. *ACS Appl. Mater. Interfaces* **2024**, *16* (6), 7721–7731.
- (46) Wu, J.; Wu, Z.; Tao, K.; Liu, C.; Yang, B.-R.; Xie, X.; Lu, X. Rapid-Response Reversible and Flexible Humidity Sensing Platform Using a Hydrophobic and Porous Substrate. *J. Mater. Chem. B* **2019**, *7* (12), 2063–2073.
- (47) Martin, S. J.; Granstaff, V. E.; Frye, G. C. Characterization of a Quartz Crystal Microbalance with Simultaneous Mass and Liquid Loading. *Anal. Chem.* **1991**, *63* (20), 2272–2281.
- (48) Yang, C.-L.; Lee, C.-S.; Chen, K.-W.; Chen, K.-Z. Noncontact Measurement of Complex Permittivity and Thickness by Using Planar Resonators. *IEEE Trans. Microwave Theory Technol.* **2016**, *64* (1), 247–257.
- (49) Tu, C.; Lee, J. E.-Y.; Zhang, X.-S. Dissipation Analysis Methods and Q-Enhancement Strategies in Piezoelectric MEMS Laterally Vibrating Resonators: A Review. *Sensors* **2020**, *20* (17), 4978.
- (50) Kumar, A.; Prajesh, R. The Potential of Acoustic Wave Devices for Gas Sensing Applications. *Sens. Actuators, A* **2022**, *339*, 113498.
- (51) Devkota, J.; Ohodnicki, P. R.; Greve, D. W. SAW Sensors for Chemical Vapors and Gases. *Sensors* **2017**, *17* (4), 801.
- (52) Tao, X.; Jin, H.; Mintken, M.; Wolff, N.; Wang, Y.; Tao, R.; Li, Y.; Torun, H.; Xie, J.; Luo, J.; et al. Three-Dimensional Tetrapodal ZnO Microstructured Network Based Flexible Surface Acoustic Wave Device for Ultraviolet and Respiration Monitoring Applications. *ACS Appl. Nano Mater.* **2020**, *3* (2), 1468–1478.
- (53) Ezre, S.; Reyna, M. A.; Anguiano, C.; Avitia, R. L.; Márquez, H. Lab-on-a-Chip Platforms for Airborne Particulate Matter Applications: A Review of Current Perspectives. *Biosensors* **2022**, *12* (4), 191.
- (54) Chen, Z.; Zhou, J.; Tang, H.; Liu, Y.; Shen, Y.; Yin, X.; Zheng, J.; Zhang, H.; Wu, J.; Shi, X.; et al. Ultrahigh-Frequency Surface Acoustic Wave Sensors with Giant Mass-Loading Effects on Electrodes. *ACS Sens.* **2020**, *5* (6), 1657–1664.
- (55) Grate, J. W.; Martin, S. J.; White, R. M. Acoustic Wave Microsensors. *Anal. Chem.* **1993**, *65* (21), 940A–948A.
- (56) Cheeke, J. D. N.; Tashtoush, N.; Eddy, N. Surface Acoustic Wave Humidity Sensor Based on the Changes in the Viscoelastic Properties of a Polymer Film. *1996 IEEE Ultrasonics Symposium. Proceedings IEEE* **1996**, 1449–452.
- (57) Drafts, B. Acoustic Wave Technology Sensors. *IEEE Trans. Microwave Theory Technol.* **2001**, *49* (4), 795–802.
- (58) Sauerbrey, G. Verwendung Von Schwingquarzen Zur Wägung Dünner Schichten Und Zur Mikrowägung. *Z. Phys.* **1959**, *155* (2), 206–222.
- (59) Rodahl, M.; Höök, F.; Krozer, A.; Brzezinski, P.; Kasemo, B. Quartz Crystal Microbalance Setup for Frequency and Q-Factor Measurements in Gaseous and Liquid Environments. *Rev. Sci. Instrum.* **1995**, *66* (7), 3924–3930.
- (60) He, H.; Zhou, L.; Wang, Y.; Li, C.; Yao, J.; Zhang, W.; Zhang, Q.; Li, M.; Li, H.; Dong, W.-F. Detection of Trace Microcystin-Lr on a 20 MHz QCM Sensor Coated with in Situ Self-Assembled MIPs. *Talanta* **2015**, *131*, 8–13.
- (61) Percival, C.; Stanley, S.; Braithwaite, A.; Newton, M.; McHale, G. Molecular Imprinted Polymer Coated QCM for the Detection of Nandrolone. *Analyst* **2002**, *127* (8), 1024–1026.
- (62) Michalzik, M.; Wendler, J.; Rabe, J.; Büttgenbach, S.; Bilitewski, U. Development and Application of a Miniaturised Quartz Crystal Microbalance (QCM) as Immunosensor for Bone Morphogenetic Protein-2. *Sens. Actuators, B* **2005**, *105* (2), 508–515.
- (63) Zhou, D.; Kang, Z.; Liu, X.; Yan, W.; Cai, H.; Xu, J.; Zhang, D. High Sensitivity Ammonia QCM Sensor Based on ZnO Nanoflower Assisted Cellulose Acetate-Polyaniline Composite Nanofibers. *Sens. Actuators, B* **2023**, *392*, 134072.
- (64) Yao, Y.; Chen, Q.; Li, Y.-Q.; Huang, X.-H.; Ling, W.-W.; Xie, Z.-M.; Wang, J.-Q.; Chen, C.-M. Nanodiamond/Ti₃C₂MXene-Coated Quartz Crystal Microbalance Humidity Sensor with High Sensitivity and High Quality Factor. *Rare Met* **2024**, *43* (6), 2719–2729.
- (65) Yu, X.; Chen, X.; Yu, X.; Chen, X.; Ding, X. A Quartz Crystal Microbalance (QCM) Humidity Sensor Based on a Pencil-Drawn Method with High Quality Factor. *IEEE Trans. Electron. Devices* **2021**, *68* (10), 5149–5154.
- (66) Nor, N. I. M.; Khalid, N.; Ramli, M. M.; Isa, M. M.; Ahmad, N.; Isa, S. S. M.; Kasjoo, S. Film Bulk Acoustic Wave Resonator in 10–20 GHz Frequency Range. In *2016 3rd International Conference on Electronic Design (ICED)*; IEEE, 2016, pp. 481–485.
- (67) Mirea, T.; Olivares, J.; Clement, M.; Iborra, E. Impact of FBAR Design on Its Sensitivity as in-Liquid Gravimetric Sensor. *Sens. Actuators, A* **2019**, *289*, 87–93.
- (68) Larson, J. D.; Oshrynsky, Y. Measurement of Effective K/Sub T//Sup 2/, Q, R/Sub P/, R/Sub S/Vs. Temperature for Mo/AlN FBAR Resonators. In *2002 IEEE Ultrasonics Symposium, 2002. Proceedings; IEEE, 2002; Vol.: 1*, pp. 939–943.
- (69) Zhao, W.; Asadi, M. J.; Li, L.; Chaudhuri, R.; Nomoto, K.; Xing, H. G.; Hwang, J. C.; Jena, D. 15-GHz Epitaxial AlN FBARs on SiC Substrates. *IEEE Electron Device Lett.* **2023**, *44* (6), 903–906.
- (70) Liu, J.; Zhao, Z.; Fang, Z.; Liu, Z.; Zhu, Y.; Du, L. High-Performance FBAR Humidity Sensor Based on the PI Film as the Multifunctional Layer. *Sens. Actuators, B* **2020**, *308*, 127694.
- (71) Van Dyke, K. S. The Piezo-Electric Resonator and Its Equivalent Network. *Proc. IRE* **1928**, *16* (6), 742–764.

- (72) Arnau, A.; Jiménez, Y.; Sogorb, T. An Extended Butterworth Van Dyke Model for Quartz Crystal Microbalance Applications in Viscoelastic Fluid Media. *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* **2001**, *48* (5), 1367–1382.
- (73) Arnau, A. A Review of Interface Electronic Systems for at-Cut Quartz Crystal Microbalance Applications in Liquids. *Sensors* **2008**, *8* (1), 370–411.
- (74) Zhang, H.; Kim, E. S. Micromachined Acoustic Resonant Mass Sensor. *J. Microelectromech. Syst.* **2005**, *14* (4), 699–706.
- (75) Almajali, E.; Ansari, S.; Lakshmanaperumal, A.; Mahmoud, S.; Jarndal, A.; Majzoub, S.; Bonny, T. Performance Analysis and Design Parameter Optimization of Resonance-Based Microwave Sensors for Material Characterization: A Review. *IEEE Sens. J.* **2025**, *25* (22), 41006–41026.
- (76) Origlia, C.; Rodriguez-Duarte, D. O.; Tobon Vasquez, J. A.; Bolomey, J.-C.; Vipiana, F. Review of Microwave near-Field Sensing and Imaging Devices in Medical Applications. *Sensors* **2024**, *24* (14), 4515.
- (77) Zhang, Z.; Zhang, G.; Tam, K.-W.; Chio, C.-H.; Zhou, X.; Zhang, W.; Teng, C.; Hong, J. Microwave Linear Displacement Sensors Integrated with Filtering Antenna for Simultaneous Sensing and Wireless Communication. *IEEE Trans. Microwave Theory Technol.* **2025**, *73* (8), 5406–5418.
- (78) Alahnomi, R. A.; Zakaria, Z.; Yusoff, Z. M.; Althuwayb, A. A.; Alhegazi, A.; Alsariera, H.; Rahman, N. A. Review of Recent Microwave Planar Resonator-Based Sensors: Techniques of Complex Permittivity Extraction, Applications, Open Challenges and Future Research Directions. *Sensors* **2021**, *21* (7), 2267.
- (79) Muñoz-Enano, J.; Vélez, P.; Gil, M.; Martín, F. Planar Microwave Resonant Sensors: A Review and Recent Developments. *Appl. Sci.* **2020**, *10* (7), 2615.
- (80) Falcone, F.; Lopetegi, T.; Laso, M.; Baena, J.; Bonache, J.; Beruete, M.; Marqués, R.; Martín, F.; Sorolla, M. Babinet Principle Applied to the Design of Metasurfaces and Metamaterials. *Phys. Rev. Lett.* **2004**, *93* (19), 197401.
- (81) Ebrahimi, A.; Withayachumnankul, W.; Al-Sarawi, S.; Abbott, D. High-Sensitivity Metamaterial-Inspired Sensor for Microfluidic Dielectric Characterization. *IEEE Sens. J.* **2014**, *14* (5), 1345–1351.
- (82) Kremer, F.; Schönhals, A. *Broadband Dielectric Spectroscopy*; Springer Science & Business Media, 2002.
- (83) Gregory, A. P.; Clarke, R. N. A Review of RF and Microwave Techniques for Dielectric Measurements on Polar Liquids. *IEEE Trans. Dielectr. Electr. Insul.* **2006**, *13* (4), 727–743.
- (84) Amer, R. A. B.; Shairi, N. A.; Said, M. A. M.; Zakaria, Z.; Misran, M. H.; Othman, A.; Alam, S.; Rahim, S. K. A. Single-Port Microwave Sensor Using Defected Ground Structure Complementary Split Ring Resonator for Solid Material Characterization. *Prog. Electromagn. Res. C* **2025**, *158*, 139–149.
- (85) Mirzaei, H.; McClelland, J.; Sharma, D.; Arjmand, M.; Zarifi, M. H. A Microwave Voyage into Swelling Phenomenon: Investigation of Polydimethylsiloxane and VOCs Interaction. *ACS Appl. Mater. Interfaces* **2023**, *15* (31), 38008–38017.
- (86) Elsheikh, M. A.; Safwat, A. M. Wideband Modeling of SRR-Loaded Coplanar Waveguide. *IEEE Trans. Microwave Theory Technol.* **2019**, *67* (3), 851–860.
- (87) Farzami, F.; Noroozariab, M. Experimental Realization of Tunable Transmission Lines Based on Single-Layer Siws Loaded by Embedded SRRs. *IEEE Trans. Microwave Theory Technol.* **2013**, *61* (8), 2848–2857.
- (88) Vélez, P.; Su, L.; Grenier, K.; Mata-Contreras, J.; Dubuc, D.; Martín, F. Microwave Microfluidic Sensor Based on a Microstrip Splitter/Combiner Configuration and Split Ring Resonators (SRRs) for Dielectric Characterization of Liquids. *IEEE Sens. J.* **2017**, *17* (20), 6589–6598.
- (89) *Smart Sensors Measurement and Instrumentation*; Mukhopadhyay, S. C.; Mason, A., Eds.; School of Engineering and Advanced Technology: Massey, 2013.
- (90) Joshi, K.; Yadav, D.; Bhardwaj, D. Design of Coplanar Proximity Coupled Feed Hexagonal Shaped Circular Polarized Microstrip Patch Antenna. *Trends Sci.* **2022**, *19* (10), 4170.
- (91) Pozar, D. M. *Microwave Engineering*; John Wiley & Sons, 2011.
- (92) Hao, H.; Tang, K.; Ku, P.; Chao, J.; Li, C.; Yang, C.; Yao, D. Development of a Portable Electronic Nose Based on Chemical Surface Acoustic Wave Array with Multiplexed Oscillator and Readout Electronics. *Sens. Actuators, B* **2010**, *146* (2), 545–553.
- (93) Rianjanu, A.; Fauzi, F.; Triyana, K.; Wasisto, H. S. Electrospun Nanofibers for Quartz Crystal Microbalance Gas Sensors: A Review. *ACS Appl. Nano Mater.* **2021**, *4* (10), 9957–9975.
- (94) Zhang, Y.; Luo, J.; Flewitt, A. J.; Cai, Z.; Zhao, X. Film Bulk Acoustic Resonators (FBARs) as Biosensors: A Review. *Biosens. Bioelectron.* **2018**, *116*, 1–15.
- (95) Withayachumnankul, W.; Jaruwongrunsee, K.; Tuantranont, A.; Fumeaux, C.; Abbott, D. Metamaterial-Based Microfluidic Sensor for Dielectric Characterization. *Sens. Actuators, A* **2013**, *189*, 233–237.
- (96) Pang, W.; Zhao, H.; Kim, E. S.; Zhang, H.; Yu, H.; Hu, X. Piezoelectric Microelectromechanical Resonant Sensors for Chemical and Biological Detection. *Lab Chip* **2012**, *12* (1), 29–44.
- (97) Länge, K.; Rapp, B. E.; Rapp, M. Surface Acoustic Wave Biosensors: A Review. *Anal. Bioanal. Chem.* **2008**, *391* (5), 1509–1519.
- (98) Wingqvist, G.; Bjurström, J.; Liljeholm, L.; Yantchev, V.; Katardjiev, I. Shear Mode AlN Thin Film Electro-Acoustic Resonant Sensor Operation in Viscous Media. *Sens. Actuators, B* **2007**, *123* (1), 466–473.
- (99) Mujahid, A.; Afzal, A.; Dickert, F. L. An Overview of High Frequency Acoustic Sensors—QCMs, Saws and FBARs—Chemical and Biochemical Applications. *Sensors* **2019**, *19* (20), 4395.
- (100) Liang, J.; Yue, W.; Kim, N.-Y.; Wang, C.; Wang, L. Sensing Performance Enhancement Based on High-Frequency Polarization of Materials. *Acc. Mater. Res.* **2024**, *5* (5), 544–559.
- (101) Li, F.; Zheng, Y.; Hua, C.; Jian, J. Gas Sensing by Microwave Transduction: Review of Progress and Challenges. *Front. Mater.* **2019**, *6*, 101.
- (102) Zhao, B.; Fu, J.; Zhou, C.; Yu, L.; Qiu, M. Emerging Porous Two-Dimensional Materials: Current Status, Existing Challenges, and Applications. *Small* **2023**, *19* (39), 2301917.
- (103) Shaukat, R. A.; Tamim, A. M.; Hwang, G.-T.; Jeong, C. K. Humidity Sensors Using 2D and 3D Nanomaterials: From Materials Selection to Technological Aspects. *Trans. Electr. Electron. Mater.* **2024**, *25* (2), 123–140.
- (104) Chen, Z.; Wang, K.; Tang, Y.; Li, L.; Hu, X.; Han, M.; Guo, Z.; Zhan, H.; Chen, B. Reticular Synthesis of One-Dimensional Covalent Organic Frameworks with 4-C S_q Topology for Enhanced Fluorescence Emission. *Angew. Chem.* **2023**, *135* (1), No. e202213268.
- (105) Wu, P.; Li, Y.; Yang, A.; Tan, X.; Chu, J.; Zhang, Y.; Yan, Y.; Tang, J.; Yuan, H.; Zhang, X.; et al. Advances in 2D Materials Based Gas Sensors for Industrial Machine Olfactory Applications. *ACS Sens.* **2024**, *9* (6), 2728–2776.
- (106) Liu, Y.; Ping, J.; Ying, Y. Recent Progress in 2D-Nanomaterial-Based Triboelectric Nanogenerators. *Adv. Funct. Mater.* **2021**, *31* (17), 2009994.
- (107) Kuang, Y.; Shen, Y.; Yuan, H.; Zhang, W.; Gu, F.; Wang, L. Improved Room Temperature Gas Sensing Performance of SiC/rGO Nanosheets Based on QCM. *J. Mater. Sci.: Mater. Electron.* **2024**, *35* (36), 2281.
- (108) Ping, J.; Fan, Z.; Sindoro, M.; Ying, Y.; Zhang, H. Recent Advances in Sensing Applications of Two-Dimensional Transition Metal Dichalcogenide Nanosheets and Their Composites. *Adv. Funct. Mater.* **2017**, *27* (19), 1605817.
- (109) Gaggiotti, S.; Scroccarello, A.; Della Pelle, F.; Ferraro, G.; Del Carlo, M.; Mascini, M.; Cichelli, A.; Compagnone, D. An Electronic Nose Based on 2D Group VI Transition Metal Dichalcogenides/Organic Compounds Sensor Array. *Biosens. Bioelectron.* **2022**, *218*, 114749.

- (110) Bhardwaj, R.; Hazra, A. MXene-Based Gas Sensors. *J. Mater. Chem. C* **2021**, 9 (44), 15735–15754.
- (111) Li, D.; Shao, Y.; Zhang, Q.; Qu, M.; Ping, J.; Fu, Y.; Xie, J. A Flexible Virtual Sensor Array Based on Laser-Induced Graphene and MXene for Detecting Volatile Organic Compounds in Human Breath. *Analyst* **2021**, 146 (18), 5704–5713.
- (112) Zhang, B.; Li, Z.; Li, C.; Li, M.; Fu, C.; Tao, R.; Zha, X.-H.; Li, H.; Luo, J. High-Sensitive ppb-Level Ammonia QCM Sensor Based on Sulfur Doped Ti3C2Tx MXene. *Sens. Actuators, A* **2023**, 350, 114138.
- (113) Lu, J.; Yin, L.; Liu, P.; Feng, L.; Yin, H. Miniaturized High-Frequency QCM Ammonia Sensor Based on MXene/ZnO Heterojunction at Room Temperature. *IEEE Sens. J.* **2024**, 24 (16), 25501–25509.
- (114) Furukawa, H.; Cordova, K. E.; O'Keeffe, M.; Yaghi, O. M. The Chemistry and Applications of Metal-Organic Frameworks. *Science* **2013**, 341 (6149), 1230444.
- (115) Kim, M.; Pander, M.; Moon, H. R. Advancing Metal–Organic Framework Designs for Room-Temperature Chemiresistive Gas Sensors. *ACS Appl. Electron. Mater.* **2024**, 6 (5), 3024–3038.
- (116) Paulowicz, I.; Postica, V.; Lupan, O.; Wolff, N.; Shree, S.; Cojocaru, A.; Deng, M.; Mishra, Y. K.; Tiginyanu, I.; Kienle, L.; et al. Zinc Oxide Nanotetrapods with Four Different Arm Morphologies for Versatile Nanosensors. *Sens. Actuators, B* **2018**, 262, 425–435.
- (117) Minh, V. A.; Tuan, L. A.; Huy, T. Q.; Hung, V. N.; Van Quy, N. Enhanced NH₃ Gas Sensing Properties of a QCM Sensor by Increasing the Length of Vertically Orientated ZnO Nanorods. *Appl. Surf. Sci.* **2013**, 265, 458–464.
- (118) Dilemiz, S. E.; Ecevit, K. High-Performance Formaldehyde Adsorption on CuO/ZnO Composite Nanofiber Coated QCM Sensors. *J. Alloys Compd.* **2019**, 783, 608–616.
- (119) Cha, X.; Yu, F.; Fan, Y.; Chen, J.; Wang, L.; Xiang, Q.; Duan, Z.; Xu, J. Superhydrophilic ZnO Nanoneedle Array: Controllable in Situ Growth on QCM Transducer and Enhanced Humidity Sensing Properties and Mechanism. *Sens. Actuators, B* **2018**, 263, 436–444.
- (120) Wales, D. J.; Grand, J.; Ting, V. P.; Burke, R. D.; Edler, K. J.; Bowen, C. R.; Mintova, S.; Burrows, A. D. Gas Sensing Using Porous Materials for Automotive Applications. *Chem. Soc. Rev.* **2015**, 44 (13), 4290–4321.
- (121) Kökçam-Demir, Ü.; Goldman, A.; Esrafil, L.; Gharib, M.; Morsali, A.; Weingart, O.; Janiak, C. Coordinatively Unsaturated Metal Sites (Open Metal Sites) in Metal–Organic Frameworks: Design and Applications. *Chem. Soc. Rev.* **2020**, 49 (9), 2751–2798.
- (122) Yan, X.; Qu, H.; Chang, Y.; Pang, W.; Wang, Y.; Duan, X. Surface Engineering of Metal–Organic Framework Prepared on Film Bulk Acoustic Resonator for Vapor Detection. *ACS Appl. Mater. Interfaces* **2020**, 12 (8), 10009–10017.
- (123) Zarifi, M. H.; Gholidoust, A.; Abdolrazzaghi, M.; Shariaty, P.; Hashisho, Z.; Daneshmand, M. Sensitivity Enhancement in Planar Microwave Active-Resonator Using Metal Organic Framework for CO₂ Detection. *Sens. Actuators, B* **2018**, 255, 1561–1568.
- (124) Liu, X.; Huang, D.; Lai, C.; Zeng, G.; Qin, L.; Wang, H.; Yi, H.; Li, B.; Liu, S.; Zhang, M.; et al. Recent Advances in Covalent Organic Frameworks (COFs) as a Smart Sensing Material. *Chem. Soc. Rev.* **2019**, 48 (20), 5266–5302.
- (125) Li, D.-M.; Li, S.-Q.; Huang, J.-Y.; Yan, Y.-L.; Zhang, S.-Y.; Tang, X.-H.; Fan, J.; Zheng, S.-R.; Zhang, W.-G.; Cai, S.-L. A Recyclable Bipyridine-Containing Covalent Organic Framework-Based QCM Sensor for Detection of Hg (II) Ion in Aqueous Solution. *J. Solid State Chem.* **2021**, 302, 122421.
- (126) Peng, Y.; Zhao, M.; Chen, B.; Zhang, Z.; Huang, Y.; Dai, F.; Lai, Z.; Cui, X.; Tan, C.; Zhang, H. Hybridization of MOFs and COFs: A New Strategy for Construction of MOF@COF Core–Shell Hybrid Materials. *Adv. Mater.* **2018**, 30 (3), 1705454.
- (127) Ji, X.; Yao, W.; Peng, J.; Ren, N.; Zhou, J.; Huang, Y. Evaluation of Cu-ZSM-5 Zeolites as QCM Sensor Coatings for DMMP Detection. *Sens. Actuators, B* **2012**, 166, 50–55.
- (128) Thommes, M.; Kaneko, K.; Neimark, A. V.; Olivier, J. P.; Rodriguez-Reinoso, F.; Rouquerol, J.; Sing, K. S. Physisorption of Gases, with Special Reference to the Evaluation of Surface Area and Pore Size Distribution (IUPAC Technical Report). *Pure Appl. Chem.* **2015**, 87 (9–10), 1051–1069.
- (129) Li, K.; Liang, M.; Wang, H.; Wang, X.; Huang, Y.; Coelho, J.; Pinilla, S.; Zhang, Y.; Qi, F.; Nicolosi, V.; et al. 3D MXene Architectures for Efficient Energy Storage and Conversion. *Adv. Funct. Mater.* **2020**, 30 (47), 2000842.
- (130) Yücer, S.; Sarac, B.; Ciftci, F. 2D MOF Structure, Electrochemical Biosensing in Bioelectronic Applications. *Npj 2D Mater. Appl.* **2025**, 9 (1), 81.
- (131) Lee, S.-Y.; Park, S.-J. Comprehensive Review on Synthesis and Adsorption Behaviors of Graphene-Based Materials. *Carbon Lett.* **2012**, 13 (2), 73–87.
- (132) Ramos Ferrer, P.; Mace, A.; Thomas, S. N.; Jeon, J.-W. Nanostructured Porous Graphene and Its Composites for Energy Storage Applications. *Nano Converg.* **2017**, 4 (1), 29.
- (133) Singh, E.; Nalwa, H. S. Stability of Graphene-Based Heterojunction Solar Cells. *RSC Adv.* **2015**, 5 (90), 73575–73600.
- (134) Ko, J.-K.; Park, I.-H.; Hong, K.; Kwon, K. C. Recent Advances in Chemoresistive Gas Sensors Using Two-Dimensional Materials. *Nanomaterials* **2024**, 14 (17), 1397.
- (135) Zhang, X.; Hou, L.; Ciesielski, A.; Samori, P. 2D Materials Beyond Graphene for High-Performance Energy Storage Applications. *Adv. Energy Mater.* **2016**, 6 (23), 1600671.
- (136) Wang, K.; Lian, G.; Qiao, Y. Adsorption Behavior and Mechanism of Heavy Metal Ions from Acid Mine Drainage Using Two-Dimensional MoS₂ Nanosheets. *Chalcogenide Lett.* **2025**, 22 (10), 889–904.
- (137) Opetubo, O. R.; Kitalu, R.; Oviroh, P. O.; Oyinbo, S. T.; Imoisili, P. E.; Jen, T.-C. A Mini-Review on MoS₂ Membrane for Water Desalination: Recent Development and Challenges. *Nano-technol. Rev.* **2023**, 12 (1), 20220563.
- (138) Lee, Y. Y.; Park, G. O.; Choi, Y. S.; Shon, J. K.; Yoon, J.; Kim, K. H.; Yoon, W.-S.; Kim, H.; Kim, J. M. Mesoporous Transition Metal Dichalcogenide Me₂ (Me = Mo, W; E = S, Se) with 2-D Layered Crystallinity as Anode Materials for Lithium Ion Batteries. *RSC Adv.* **2016**, 6 (17), 14253–14260.
- (139) Yang, J.; Liu, X.; Dong, Q.; Shen, Y.; Pan, Y.; Wang, Z.; Tang, K.; Dai, X.; Wu, R.; Jin, Y.; et al. Oxidations of Two-Dimensional Semiconductors: Fundamentals and Applications. *Chin. Chem. Lett.* **2022**, 33 (1), 177–185.
- (140) Huang, B.; Li, Y.; Zeng, W. Application of Metal-Organic Framework-Based Composites for Gas Sensing and Effects of Synthesis Strategies on Gas-Sensitive Performance. *Chemosensors* **2021**, 9 (8), 226.
- (141) Akhter, R.; Maktedar, S. S. MXenes: A Comprehensive Review of Synthesis, Properties, and Progress in Supercapacitor Applications. *J. Materiomics* **2023**, 9 (6), 1196–1241.
- (142) Chen, C.; Tang, X.; Wang, M.; Ma, Y.; Wang, S.; Yue, Y. Multiscale Porous Structured MXene: Synthesis, Design and Applications in Batteries and Supercapacitors. *Mater. Today* **2025**, 88, 526–551.
- (143) Cao, F.; Zhang, Y.; Wang, H.; Khan, K.; Tareen, A. K.; Qian, W.; Zhang, H.; Ågren, H. Recent Advances in Oxidation Stable Chemistry of 2D MXenes. *Adv. Mater.* **2022**, 34 (13), 2107554.
- (144) Cai, Z.; Kim, H. Recent Advances in MXene Gas Sensors: Synthesis, Composites, and Mechanisms. *Npj 2D Mater. Appl.* **2025**, 9 (1), 66.
- (145) Bushell, M.; Beauchemin, S.; Kunc, F.; Gardner, D.; Ovens, J.; Toll, F.; Kennedy, D.; Nguyen, K.; Vladislavjevic, D.; Rasmussen, P. E.; et al. Characterization of Commercial Metal Oxide Nanomaterials: Crystalline Phase, Particle Size and Specific Surface Area. *Nanomaterials* **2020**, 10 (9), 1812.
- (146) Chen, K.; Zhu, L.; Wang, J.; Xie, W.; Deng, Y.; Xue, L.; Long, H.; Wan, H.; Ren, J.; Yuan, K.; et al. Active Polymer-Templated Porous Metal Oxide Nanospheres with Tailored Single-Atom Modification for Olfactory Intelligence. *J. Am. Chem. Soc.* **2025**, 147 (51), 47616–47626.

- (147) Riente, P.; Noël, T. Application of Metal Oxide Semiconductors in Light-Driven Organic Transformations. *Catal. Sci. Technol.* **2019**, *9* (19), 5186–5232.
- (148) Jiao, L.; Seow, J. Y. R.; Skinner, W. S.; Wang, Z. U.; Jiang, H.-L. Metal–Organic Frameworks: Structures and Functional Applications. *Mater. Today* **2019**, *27*, 43–68.
- (149) Goel, N.; Kunal, K.; Kushwaha, A.; Kumar, M. Metal Oxide Semiconductors for Gas Sensing. *Eng. Rep.* **2023**, *5* (6), No. e12604.
- (150) Trejo, S.; Antonio, J. J.; Kraka, E.; Chen, H. Assessing the Stability of Metal–Organic Frameworks with Local Vibrational Mode Theory. *J. Phys. Chem. C* **2026**, *130* (7), 2847–2854.
- (151) Nandi, S.; Adimule, V.; Kerur, S. S.; Dhanalakshmi, S.; Barmavatu, P. Recent Advances in Metal–Organic Framework Derived Nanocomposites in Gas and Chemical Sensors: A Review. *Sens. Technol.* **2024**, *2* (1), 2317385.
- (152) Ting, Z.; Yi-Fan, G.; Jia, G. Covalent Organic Frameworks: Design, Synthesis and Applications. *J. Funct. Polym.* **2018**, *31* (3), 189–215.
- (153) Sacchetti, A. *Synthesis of Mesoporous Zeolites for the Selective Conversion of Sugars into Methyl Lactate and Other Hydroxyesters*; M.S. Thesis, University of Bologna: Bologna, Italy, 2018.
- (154) Vardhan, H.; Rummer, G.; Deng, A.; Ma, S. Large-Scale Synthesis of Covalent Organic Frameworks: Challenges and Opportunities. *Membranes* **2023**, *13* (8), 696.
- (155) Zhang, S.; Xu, L.; Xu, J.; Shen, B. A Mini-Review of Recent Progress in Zeolite-Based Catalysts for Photocatalytic or Photo-thermal Environmental Pollutant Treatment. *Catalysts* **2025**, *15* (2), 158.
- (156) Rosseinsky, M. Recent Developments in Metal–Organic Framework Chemistry: Design, Discovery, Permanent Porosity and Flexibility. *Microporous Mesoporous Mater.* **2004**, *73* (1–2), 15–30.
- (157) Su, X. *Diffusion Mechanism of Aromatic Hydrocarbons within Zeolites in Liquid Phase and Sub-and Super-Critical Fluids*; Ph.D. Thesis, Hokkaido University: Sapporo, Japan, 2022.
- (158) Bag, A.; Lee, N.-E. Gas Sensing with Heterostructures Based on Two-Dimensional Nanostructured Materials: A Review. *J. Mater. Chem. C* **2019**, *7* (43), 13367–13383.
- (159) Miller, D. R.; Akbar, S. A.; Morris, P. A. Nanoscale Metal Oxide-Based Heterojunctions for Gas Sensing: A Review. *Sens. Actuators, B* **2014**, *204*, 250–272.
- (160) Bhardwaj, R.; Pumera, M. Surface-Engineered 2D Nanomaterials in Gas Sensors: Advancement and Challenges. *Small* **2025**, *21* (34), 2410360.
- (161) An, D.; Chen, L.; Liang, Y.; Hou, J.; Chen, J. Defect-Containing Metal–Organic Framework Materials for Sensor Applications. *J. Mater. Chem. A* **2023**, *12* (1), 38–58.
- (162) Wang, Z.; Zhao, Z.; Jin, S.; Bian, F.; Chang, Y.; Duan, X.; Men, X.; You, R. Investigation of Sorptive Interactions between Volatile Organic Compounds and Supramolecules at Dynamic Oscillation Using Bulk Acoustic Wave Resonator Virtual Sensor Arrays. *Microsyst. Nanoeng.* **2024**, *10* (1), 99.
- (163) Malhotra, J. S.; Dávila Duarte, C.; Reichert, P.; Krishnan, D.; Sundberg, J. Quantification of Methane in Water at Parts Per Billion Sensitivity Using a Metal–Organic Framework-Functionalized Quartz Crystal Resonator. *ACS Appl. Nano Mater.* **2025**, *8* (9), 4542–4552.
- (164) Chen, Y.; Xuan, X.; Li, M.; Jiang, D.; Li, H. Implantable Antenna Immunosensor Based on Au-Decorated Graphene Film for Wireless CEA Detection. *Sens. Actuators, B* **2024**, *416*, 136037.
- (165) Ito, H.; Hashimoto, S.; Katsuno, T. Quartz Crystal Microbalance Sensing System with a UiO-66 Metal–Organic Framework Gas Concentrator for Detecting Low-Concentration Acetone Biomarker Gases. *Emergent Mater.* **2025**, *8*, 8073–8088.
- (166) Cui, B.; Lang, X.; Ren, Z.; Cheng, L.; Yang, D.; Wang, W. Conductive Metal–Organic Framework for ppb-Concentration and Highly Selective SAW Hydrogen Sulfide Sensing at Room Temperature. *Sens. Actuators, B* **2025**, *423*, 136817.
- (167) Ji, Y.; Ma, T.; Yang, X.; Wang, R.; Wang, K.; Liu, X.; Wang, X.; Jin, Q. Innovative SIFSIX-1-Cu-Based Microwave Sensor with Directionally Ordered Interconnected Nanochannels: Revolutionizing High-Performance SO₂ Detection. *J. Hazard. Mater.* **2025**, *496*, 139517.
- (168) Ebrahimi, A.; Scott, J.; Ghorbani, K. Ultrahigh-Sensitivity Microwave Sensor for Microfluidic Complex Permittivity Measurement. *IEEE Trans. Microwave Theory Technol.* **2019**, *67* (10), 4269–4277.
- (169) Ott, C. E. Strategies for Assessing the Limit of Detection in Voltammetric Methods: Comparison and Evaluation of Approaches. *Analyst* **2024**, *149* (16), 4295–4309.
- (170) Zarifi, M. H.; Daneshmand, M. Liquid Sensing in Aquatic Environment Using High Quality Planar Microwave Resonator. *Sens. Actuators, B* **2016**, *225*, 517–521.
- (171) Potyailo, R. A. Multivariable Sensors for Ubiquitous Monitoring of Gases in the Era of Internet of Things and Industrial Internet. *Chem. Rev.* **2016**, *116* (19), 11877–11923.
- (172) Li, D.; Xie, Z.; Qu, M.; Zhang, Q.; Fu, Y.; Xie, J. Virtual Sensor Array Based on Butterworth–Van Dyke Equivalent Model of QCM for Selective Detection of Volatile Organic Compounds. *ACS Appl. Mater. Interfaces* **2021**, *13* (39), 47043–47051.
- (173) Lv, S.; Sun, X.; Zhang, Y.; Zhang, K.; Feng, C.; Cao, Y.; Sun, H.; Zhao, J. Laterally Excited Bulk Wave Resonators on ZY-Plane Linbo3: Optimization, Fabrication, and De-Embedding Structure for Accurate Parameter Extraction. *Microelectron. J.* **2025**, *156*, 106554.
- (174) Yuan, Y.; Yang, T.; Chen, X.; Yu, L.; Hou, X.; Zhang, G.; Dong, W.; Lu, Z.; Li, H.; Reindl, L.; et al. Analysis of the Acoustoelectric Response of SAW Gas Sensors Using a COM Model. *Microsyst. Nanoeng.* **2024**, *10* (1), 69.
- (175) Li, X.; Feng, Y.; Long, J.; Lv, H.; Guo, Y.; Zu, X. MXene-Activated Graphene Oxide Enhancing NO₂ Capture and Detection of Surface Acoustic Wave Sensors. *Sens. Actuators, B* **2024**, *401*, 135006.
- (176) Nirschl, M.; Schreiter, M.; Vörös, J. Comparison of FBAR and QCM-D Sensitivity Dependence on Adlayer Thickness and Viscosity. *Sens. Actuators, A* **2011**, *165* (2), 415–421.
- (177) Wang, X.; Zhang, T. An Overview: A Design Strategy for Dioxide Carbon QCM Gas Sensor Based on MOFs or COFs. *Adv. Sens. Res.* **2025**, *4* (4), 2400170.
- (178) Xuan, W.; Cole, M.; Gardner, J. W.; Thomas, S.; Villa-López, F.-H.; Wang, X.; Dong, S.; Luo, J. A Film Bulk Acoustic Resonator Oscillator Based Humidity Sensor with Graphene Oxide as the Sensitive Layer. *J. Micromech. Microeng.* **2017**, *27* (5), 055017.
- (179) Li, W.; Li, C.; Wan, Z.; Li, J. Photothermally Actuated Ti₃C₂T_x MXene-GO Resonator for Highly Sensitive Humidity Sensing. *Sens. Actuators, B* **2026**, *461*, 139983.
- (180) Lee, H. J.; Kim, B.-H.; Lee, S.; Kim, K.; Rho, J. Coupled-Mode Theory and Numerical Analysis of High-Q Reconfigurable Split-Ring Resonators (SRRs) for Graphene/rGO Conductivity-Based Microwave Sensing. *Adv. Theory Simul.* **2026**, *9* (1), No. e01559.
- (181) Kazemi, K. K.; Hosseini, E.; Hu, S.; Narang, R.; Li, S.; Arjmand, M.; Zarifi, M. H. MXene Membrane in Planar Microwave Resonant Structures for 5G Applications. *Appl. Mater. Today* **2022**, *26*, 101294.
- (182) Pal, D.; Dutta, J.; Mora, D. C.; Mu, J.; Wang, X.; Go, D.; Ross, A. E.; Kummel, A. C. Grain-Size Engineering in Vapor-Deposited ZIF-8 MOF: Low-Leakage Low-K Films with Enhanced Breakdown Field. *ACS Appl. Mater. Interfaces* **2026**, *18* (10), 15859–15866.
- (183) Kumar, A.; Wang, C.; Meng, F.-Y.; Zhou, Z.-L.; Zhao, M.; Yan, G.-F.; Kim, E.-S.; Kim, N.-Y. High-Sensitivity Quantified, Linear and Mediator-Free Resonator-Based Microwave Biosensor for Glucose Detection. *Sensors* **2020**, *20* (14), 4024.
- (184) Kumar, A.; Wang, C.; Meng, F.-Y.; Jiang, C.-P.; Yan, G.-F.; Zhao, M.; Jing, C.-Q.; Wang, L. Ultrafast Detection and Discrimination of Methanol Gas Using a Polyindole-Embedded Substrate Integrated Waveguide Microwave Sensor. *ACS Sens.* **2020**, *5* (12), 3939–3948.
- (185) Lieberzeit, P. *Piezoelectric Sensors*; Springer, 2024.
- (186) Wang, X.; Liang, J.-G.; Wu, J.-K.; Gu, X.-F.; Kim, N.-Y. Microwave Detection with Various Sensitive Materials for Humidity Sensing. *Sens. Actuators, B* **2022**, *351*, 130935.

- (187) Deng, G.; Zhang, Q.; Yu, Y.; Yang, J.; Li, Y.; Zhang, W.; Yin, Z. Microwave Humidity Sensor Based on Interdigital Capacitors with Compact Size and Enhanced Sensitivity. *Measurement* **2026**, 258, 119195.
- (188) Ma, Z.; Fu, M.; Gao, C.; Fan, S.; Chi, H.; Li, W.; Hou, D.; Cao, Y. Trenched Microwave Resonator Integrated with Porous PDMS for Detection and Classification of VOCs with Enhanced Performance. *J. Hazard. Mater.* **2024**, 472, 134553.
- (189) Chen, Q.; Tan, K.; Huang, X.-H.; Yao, Y.; Yang, G.; Liu, D. High-Sensitivity Chitin Nanofiber-Coated Series Piezoelectric Quartz Crystal Humidity Sensors. *IEEE Sens. J.* **2023**, 23 (10), 10316–10323.
- (190) Wei, Z.; Huang, J.-Q.; Huang, Q.-A. Enhanced Sensitivity of a Film Bulk Acoustic Resonator (FBAR) Humidity Sensor by a Nonlinear Parity-Time (PT) Symmetric System. *Sens. Actuators, B* **2026**, 449, 139072.
- (191) Zhang, D.; Kang, Z.; Liu, X.; Guo, J.; Yang, Y. Highly Sensitive Ammonia Sensor Based on PSS Doped ZIF-8-Derived Porous Carbon/Polyaniline Hybrid Film Coated on Quartz Crystal Microbalance. *Sens. Actuators, B* **2022**, 357, 131419.
- (192) Zhou, T.-L.; Wu, E.-K.; Wu, J.-K.; Qiang, T.; Gu, X.-F.; Yan, M.-Y.; von Grätowski, S.; Liang, J.-G. Boosting the Sensitivity and Stability of Microwave Gas Sensor for Acetone Detection Based on SnO₂/Co₃O₄ Nanofibers and Micro-Heater. *Sens. Actuators, B* **2025**, 440, 137938.
- (193) Bahoumina, P.; Hallil-Abbas, H.; Lachaud, J.-L.; Rebiere, D.; Dejous, C.; Abdelghani, A.; Frigui, K.; Bila, S.; Baillargeat, D.; Zhang, Q.; et al. VOCs Monitoring Using Differential Microwave Capacitive Resonant Transducer and Conductive Pedot: PSS-MWCNTs Nanocomposite Film for Environmental Applications. *IEEE Trans. Nanotechnol.* **2024**, 1.
- (194) Ali, L.; Wei, J.; Meng, F.-Y.; Qureshi, M. W.; Adhikari, K. K.; Li, M.-Y.; Liang, J.-G.; Wang, X.-L.; Ding, X.-M.; Kim, N.-Y.; et al. Sensitivity-Enhanced detection of acetone gas using MXene-Immobilized planar microwave sensor. *Sens. Actuators, B* **2023**, 392, 134048.
- (195) Atea, M.; Wei, H.; Andreescu, S. Sensors for Emerging Water Contaminants: Overcoming Roadblocks to Innovation. *Environ. Sci. Technol.* **2024**, 58 (6), 2636–2651.
- (196) Singh, J.; Gupta, S. K.; Vinita. Room Temperature Operating Formaldehyde Sensor Based on N-Type ZnO Functionalized Surface Acoustic Wave Resonator. *Sens. Actuators, A* **2024**, 365, 114818.
- (197) Ali, L.; Adhikari, K. K.; Wei, J.; Yi, Y.; Wang, G.; Wang, C. Sensitivity-Enhanced Detection of Ethanol Gas Using Moo₃ Nanoflake-Functionalized Laser-Induced Graphene in a Planar Microwave Resonator. *Microchim. Acta* **2025**, 192 (8), 516.
- (198) Wang, Z.; Chen, W.; Gu, S.; Wang, J.; Wang, Y. Discrimination of Wood Borers Infested Platycladus Orientalis Trunks Using Quartz Crystal Microbalance Gas Sensor Array. *Sens. Actuators, B* **2020**, 309, 127767.
- (199) Qin, P.; Day, B. A.; Okur, S.; Li, C.; Chandresh, A.; Wilmer, C. E.; Heinke, L. VOC Mixture Sensing with a MOF Film Sensor Array: Detection and Discrimination of Xylene Isomers and Their Ternary Blends. *ACS Sens.* **2022**, 7 (6), 1666–1675.
- (200) Okur, S.; Hashem, T.; Bogdanova, E.; Hodapp, P.; Heinke, L.; Bräse, S.; Wöll, C. Optimized Detection of Volatile Organic Compounds Utilizing Durable and Selective Arrays of Tailored UiO-66-X SURMOF Sensors. *ACS Sens.* **2024**, 9 (2), 622–630.
- (201) Malhotra, J. S.; Kubus, M.; Pedersen, K. S.; Andersen, S. I.; Sundberg, J. Room-Temperature Monitoring of CH₄ and CO₂ Using a Metal–Organic Framework-Based QCM Sensor Showing Inherent Analyte Discrimination. *ACS Sens.* **2023**, 8 (9), 3478–3486.
- (202) Zeng, G.; Wu, C.; Chang, Y.; Zhou, C.; Chen, B.; Zhang, M.; Li, J.; Duan, X.; Yang, Q.; Pang, W. Detection and Discrimination of Volatile Organic Compounds Using a Single Film Bulk Acoustic Wave Resonator with Temperature Modulation as a Multiparameter Virtual Sensor Array. *ACS Sens.* **2019**, 4 (6), 1524–1533.
- (203) Guo, Y.; Ma, Z.; Li, Y.; Pan, X.; Yan, Y.; Zhong, A.; Wang, D.; Ding, G.; Xue, S. QCM Sensor Based on Ferrocene-Containing COF for Ammonia Detection. *IEEE Trans. Electron. Devices* **2025**, 72 (12), 6998–7003.
- (204) Wang, X.; Xu, X.; Zhou, T.; Zhang, T. Nanoscale MOF-74-Based QCM Gas Sensor for CO₂ Detection at Room Temperature. *Sens. Actuators, B* **2024**, 413, 135874.
- (205) Tang, P.; Xu, M.; Wang, Y.; Liang, J.; Jiang, Y.; Yu, P.; Qiang, T. Integrated Passive Device-Based Gas Sensor Combined with PANI/SnO₂ Composites for Ammonia Detection. *ACS Appl. Polym. Mater.* **2023**, 5 (9), 7220–7228.
- (206) Yang, X.; Wang, C.; Liu, Y.; Sun, Z.; Wang, X.; Liu, F.; Lu, G. Microwave Sensor with Light-Assisted Enhancement Based on Ti₃C₂T_x MXene: Toward ppb-Level NO₂ Detection Limits. *ACS Sens.* **2025**, 10 (2), 1461–1469.
- (207) Choudhari, A.; Vignol, T.; Lachaud, J.-L.; Rube, M.; Sadli, I.; Sebeloue, M.; Delepee, R.; Tamarin, O.; Dejous, C. MIP Functionalized Love Wave Sensor for Detection of 4-AP Organic Compound in Turbid Solutions. In *2023 IEEE SENSORS*; IEEE, 2023, pp. 1–4.
- (208) Takuli, A. P. S.; Patel, R.; Majumder, T.; Adhikari, M. S.; Bansal, D. High-Q Shear Mode ZnO-Based Film Bulk Acoustic Resonators with Air Cavity for Liquid-Phase Sensing Applications. *Semiconductors* **2026**, 60 (4), 420–430.
- (209) Wang, Y.-X.; Fu, S.-F.; Xu, M.-X.; Tang, P.; Liang, J.-G.; Jiang, Y.-F.; Qiang, T. Integrated Passive Sensing Chip for Highly Sensitive and Reusable Detection of Differential-Charged Nanoplastics Concentration. *ACS Sens.* **2023**, 8 (10), 3862–3872.
- (210) Cui, W.; Abbasi, Z.; Ren, C. Real-Time Lead Detection Device Based on Nanomaterials Modified Microwave-Microfluidic Sensor. *Sens. Actuators, A* **2023**, 362, 114652.
- (211) Augustine, S.; Venkadesh, A.; Kaushal, S.; Lee, E.; Ajaj, M.; Lee, N.-E. Point-of-Care Testing: The Convergence of Innovation and Accessibility in Diagnostics. *Anal. Chem.* **2025**, 97 (18), 9569–9599.
- (212) Zhu, Y.; Xia, P.; Liu, J.; Fang, Z.; Du, L.; Zhao, Z. Polyimide-Based High-Performance Film Bulk Acoustic Resonator Humidity Sensor and Its Application in Real-Time Human Respiration Monitoring. *Micromachines* **2022**, 13 (5), 758.
- (213) Ali, L.; Wang, G.; Meng, F.; Ding, X.; Adhikari, K. K.; Wang, C. MXene-Coated Planar Microwave Resonator Sensor for Ultra-sensitive Humidity Monitoring. *IEEE Microw. Wirel. Technol. Lett.* **2023**, 33 (11), 1572–1575.
- (214) Wang, C.; Ding, Y.; Li, M.; Li, H.; Xu, S.; Li, C.; Qian, L.; Yang, B. Surface Acoustic Wave Sensor Based on Au/TiO₂/Pedot with Dual Response to Carbon Dioxide and Humidity. *Anal. Chim. Acta* **2022**, 1190, 339264.
- (215) Yu, H.; Wang, C.; Meng, F.; Xiao, J.; Liang, J.; Kim, H.; Bae, S.; Zou, D.; Kim, E.-S.; Kim, N.-Y.; et al. Microwave Humidity Sensor Based on Carbon Dots-Decorated MOF-Derived Porous Co₃O₄ for Breath Monitoring and Finger Moisture Detection. *Carbon* **2021**, 183, 578–589.
- (216) Luo, W.; Li, C.; Wang, L.; Qian, L.; Li, D.; Miao, L.; Xiong, Y.; Li, M.; Tian, Y.; Li, H. Microwave Humidity Sensor Based on Shorted Split Ring Resonator with Interdigital Capacitance and Al₂O₃ Nanoflakes/Chitosan Sensitive Film for Respiration Monitoring. *Sens. Actuators, B* **2024**, 413, 135869.
- (217) Wang, D.; Xu, M.; Zhang, D.; Mi, Q.; Gong, D.; Zhou, Y.; Wang, H.; Zhang, X.; Gong, K.; Zhang, R.; et al. Multifunctional QCM Sensor Based on UiO-66-NH₂/V₂CT_x MXene Films for Detection of Formaldehyde Gas and Respiratory Behavior. *Sens. Actuators, B* **2026**, 447, 138804.
- (218) Amann, A.; Costello, B. D. L.; Miekisch, W.; Schubert, J.; Buszewski, B.; Pleil, J.; Ratcliffe, N.; Risby, T. The Human Volatilome: Volatile Organic Compounds (VOCs) in Exhaled Breath, Skin Emanations, Urine, Feces and Saliva. *J. Breath Res.* **2014**, 8 (3), 034001.
- (219) Righettoni, M.; Tricoli, A.; Pratsinis, S. E. Si:WO₃ Sensors for Highly Selective Detection of Acetone for Easy Diagnosis of Diabetes by Breath Analysis. *Anal. Chem.* **2010**, 82 (9), 3581–3587.
- (220) Esmeryan, K. D.; Grakov, T.; Lazarov, Y.; Vergov, L. G.; Staykov, S. Measurement of the Blood Alcohol Concentration Via a

Metal-Phenolic Film-Coated Quartz Crystal Microbalance. *Measurement* **2025**, 253, 117795.

(221) Kus, F.; Altinkok, C.; Zayim, E.; Erdemir, S.; Tasaltin, C.; Gurol, I. Surface Acoustic Wave (SAW) Sensor for Volatile Organic Compounds (VOCs) Detection with Calix [4] Arene Functionalized Gold Nanorods (AuNRs) and Silver Nanocubes (AgNCs). *Sens. Actuators, B* **2021**, 330, 129402.

(222) Xue, S.; Jin, Q.; Wang, X.; Lu, G. Microwave Gas Sensor Based on Differential Planar Resonator Synergistically Loaded with Pd-Doped CdS₂ for Enhanced H₂S Detection. *IEEE Electron Device Lett.* **2025**, 46 (3), 480–483.

(223) Lim, H. J.; Saha, T.; Tey, B. T.; Tan, W. S.; Hassan, S. S.; Ooi, C. W. Quartz Crystal Microbalance-Based Biosensing of Hepatitis B Antigen Using a Molecularly Imprinted Polydopamine Film. *Talanta* **2022**, 249, 123659.

(224) Ji, J.; Pang, Y.; Li, D.; Huang, Z.; Zhang, Z.; Xue, N.; Xu, Y.; Mu, X. An Aptamer-Based Shear Horizontal Surface Acoustic Wave Biosensor with a Cvd-Grown Single-Layered Graphene Film for High-Sensitivity Detection of a Label-Free Endotoxin. *Microsyst. Nanoeng.* **2020**, 6 (1), 4.

(225) Ji, J.; Pang, Y.; Li, D.; Wang, X.; Xu, Y.; Mu, X. Single-Layered Graphene/Au-Nanoparticles-Based Love Wave Biosensor for Highly Sensitive and Specific Detection of Staphylococcus Aureus Gene Sequences. *ACS Appl. Mater. Interfaces* **2020**, 12 (11), 12417–12425.

(226) Rughoobur, G.; Sugime, H.; DeMiguel-Ramos, M.; Mirea, T.; Zheng, S.; Robertson, J.; Iborra, E.; Flewitt, A. J. Carbon Nanotube Isolation Layer Enhancing In-Liquid Quality-Factors of Thin Film Bulk Acoustic Wave Resonators for Gravimetric Sensing. *Sens. Actuators, B* **2018**, 261, 398–407.

(227) Amanati Shahri, A.; Omidvar, A. H.; Pamplona Rehder, G.; Serrano, A. L. C. A Microwave-Based Microfluidic Cell Detecting Biosensor for Biological Quantification Using the Metallic Nanowire-Filled Membrane Technology. *Sensors* **2022**, 22 (9), 3265.

(228) Zhuang, Z.; Xuan, X.; Li, H.; Jiang, D.; Li, M. A Wearable Antenna Sensor Based on Epda/SiO₂ Nanowalls for the Detection of Lactic Acid in Sweat. *Sens. Actuators, B* **2024**, 404, 135265.

(229) Meenal; Aghwariya, M. K.; Goel, T.; Patnaik, A. Advanced Materials for RF Sensors in Harsh Environments: A Comprehensive Review. *IEEE Sens. Rev.* **2025**, 2, 324–338.

(230) Pang, J.; Le, X.; Pang, K.; Dong, H.; Zhang, Q.; Xu, Z.; Gao, C.; Fu, Y.; Xie, J. Highly Precision Carbon Dioxide Acoustic Wave Sensor with Minimized Humidity Interference. *Sens. Actuators, B* **2021**, 338, 129824.

(231) Das, A.; Manjunatha, R.; Kumar, K. N.; De, D.; Bandyopadhyay, R. Fabrication of Surface Functionalized QCM Sensor for BTX Detection at Ambient Conditions. *Talanta* **2025**, 283, 127081.

(232) Ahmed, L. R.; Luder, J.; Chuang, C.-H.; El-Mahdy, A. F. Covalent-Organic-Framework-Modified Quartz Crystal Microbalance Sensor for Selective Detection of Hazardous Formic Acid. *ACS Appl. Mater. Interfaces* **2024**, 16 (23), 30408–30420.

(233) Lu, Y.; Wang, D.; Lv, W.; Xia, Y.; Ou, K.; Li, Y.; Du, Z.; He, Y.; Dai, J.; Wu, S.; et al. Surface Acoustic Wave Hydrogen Sulfide Gas Sensors Based on Porous SnO₂-SiO₂ Composite Films. *Sens. Actuators, B* **2024**, 417, 136117.

(234) Gao, L.; Xu, J.; Yang, X.; Xu, C.; Wang, X.; Jin, Q. Waveguide Microwave Gas Sensor Based on Monolithic In₂O₃ for Enhanced Ammonia Detection. *Talanta* **2025**, 293, 128122.

(235) Liu, X.; He, G.; Huang, X.; Zhuang, Y.; Hou, J. Detection of CO₂, H₂S, NH₃, and Gas Mixtures Using a Quartz Crystal Microbalance Sensor Array with Five Sensitive Materials. *Measurement* **2025**, 242, 115891.

(236) Pereira, P. F.; de Sousa Picciani, P. H.; Calado, V.; Tonon, R. V. Electrical Gas Sensors for Meat Freshness Assessment and Quality Monitoring: A Review. *Trends Food Sci. Technol.* **2021**, 118, 36–44.

(237) Bekhit, A. E.-D. A.; Holman, B. W.; Giteru, S. G.; Hopkins, D. L. Total Volatile Basic Nitrogen (TVB-N) and Its Role in Meat Spoilage: A Review. *Trends Food Sci. Technol.* **2021**, 109, 280–302.

(238) Cheng, G.; Liu, C.; Chen, S.; Li, Y.; Gao, L.; Ren, L.; Chen, W.; Wang, Z. A QCM Gas Sensor Based on 0.5 Co-Mn-BTC MOF Nanocomposites for Salmon Meat Freshness Detection. *Microchem. J.* **2025**, 217, 114992.

(239) Liu, C.; Chen, S.; Liu, S.; Chen, W.; Wang, Z. Highly Sensitive QCM Gas Sensor Based on ZIF-8-Derived Porous Carbon and Multi-Walled Carbon Nanotubes Decorated Nanocomposite for Salmon Meat Freshness Detection. *J. Food Compos. Anal.* **2024**, 133, 106392.

(240) Wang, Z.; Wang, X.; Qi, G.; Chen, S.; Liu, C.; Marrazza, G.; Gu, S.; Li, P.; Chen, W. Nanoporous Graphene Oxide/Molybdenum Disulfide Based on Quartz Crystal Microbalance Gas Sensor for Total Volatile Basic Nitrogen Detection in Salmon Meat. *J. Food Compos. Anal.* **2025**, 142, 107417.

(241) Chen, S.; Qi, G.; Zhang, L.; Duan, X.; Bai, M.; Hu, M.; Li, P.; Zhao, W.; Sun, X.; Guo, Y.; et al. Detection of Salmon Meat Freshness Using QCM Gas Sensor Array Combined with Physicochemical Method. *Microchem. J.* **2023**, 194, 109353.

(242) Yen, T.-Y.; Yao, D.-J. Detection of the Freshness of Kiwifruit with a TD-GC-MS and a Gas-Sensing Array Based on the Surface-Acoustic-Wave Technique. *IEEE Trans. NanoBiosci.* **2022**, 21 (3), 363–369.

(243) Tao, B.; Bai, J.; Wu, H.; Miao, F. Design and Research of Multi-Information Fusion RFID Sensor for Fruit and Vegetable Quality Detection. *Microchim. Acta* **2024**, 191 (9), 568.

(244) Wu, C.; Li, J. Portable FBAR Based E-nose for Cold Chain Real-Time Bananas Shelf Time Detection. *Nanotechnol. Precis. Eng.* **2023**, 6 (1), 013004.

(245) Ayala, C. E.; Vaughan, S. R.; Pérez, R. L.; Leonard, B. S.; King, B.; Jorgensen, K.; Warner, I. M. Influence of Humidity on Accuracy of QCM–Ir780-Based Gumbos Sensor Arrays. *Anal. Chim. Acta* **2023**, 1278, 341677.

(246) Bao, Z.; Alexandre, M.; Hasegawa, S.; Nakamoto, T. A Robust Odor Mixture Quantification Method Based on Active Sensing Using Both QCM Frequency Shifts and Resistance Changes of Multiple Harmonics. *IEEE Sens. J.* **2025**, 25 (23), 42482–42489.

(247) Hu, R.; Su, X.; Wang, Y.; Wang, Y.; Stevenson, A. C.; Li, G. A Hybrid Wireless Electrodeless QCM-D for the Discrimination of Volatile Organic Compounds. *IEEE Trans. Instrum. Meas.* **2023**, 73, 1–9.

(248) Sun, T.; Feng, B.; Huo, J.; Xiao, Y.; Wang, W.; Peng, J.; Li, Z.; Du, C.; Wang, W.; Zou, G.; et al. Artificial Intelligence Meets Flexible Sensors: Emerging Smart Flexible Sensing Systems Driven by Machine Learning and Artificial Synapses. *Nano Micro Lett.* **2024**, 16 (1), 14.

(249) Yan, J.; Kang, Y.; Wang, T.; Tian, Y.; Wang, C.; Ma, Z.; Song, Z. AI Empowers Intelligent Chemical Sensing Systems. *Innov. Inform.* **2025**, 1 (1), 100014.

(250) Adhikari, K. K.; Ali, L.; Wei, J.; Yi, Y.; Tang, X.; Li, Z.; Gao, J.; Mei, L.; Qu, Y.; Yin, J.; et al. Patterned Laser-Induced Graphene Enabling a High-Performance Gas Sensing Split-Ring Resonator. *Chem. Eng. J.* **2024**, 499, 155984.

(251) Niksan, O.; Wyatt, B. C.; Kazemi, K. K.; Anasori, B.; Zarifi, M. H. MXene Free Standing Films: Unlocking the Impact of Flake Sizes in Microwave Resonant Structures in Humid Environments. *Small* **2023**, 19 (37), 2300848.